

Essential Histology

A Concept-Based Guide for PGME

Understanding the structure. Seeing the pattern.
Distinguishing the alternatives. Retrieving the logic.

Definitive Edition

AREMS-HY Academic Series

Primary scientific foundation

Mescher AL. **Junqueira's Basic Histology: Text and Atlas**. 17th ed. McGraw Hill.

Official 16-Chapter Syllabus

Aligned with Junqueira 17th Chapters 1, 2, 3, 4, 5, 6, 9, 11, 12, 13, 14, 15, 16, 17, 18, 20

Built for the Afghanistan 1405 Specialty Examination

Chapter 1 — Histology and Its Methods of Study

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 1

Opening Question

How does a piece of fresh tissue — soft, transparent, and already beginning to digest itself — become a stained slide that can sit on a microscope stage for twenty years and still reveal a diagnosis?

Why This Matters

Every biopsy is really a chemistry experiment, not a photograph. The fixative, the solvent, the dye, and the microscope each reveal one specific molecular or physical property of the tissue while concealing everything else. A 45-year-old man presents with fatigue and a swollen liver; on routine H&E the hepatocytes look pale and swollen, but the diagnosis differs entirely depending on whether the accumulating substance is lipid (Oil Red O on frozen section), glycogen (PAS with diastase control), iron (Prussian blue), or amyloid (Congo red under polarized light). Wrong stain, missed diagnosis. The pathologist's first decision — which technique for which question — is the most consequential act of histology.

Learning Objectives

By the end of this chapter, you will be able to:

- 1. Explain** why living tissue must be processed through a chain of physical and chemical steps before it can be examined.
- 2. Distinguish** the role of each step in processing — fixation, dehydration, clearing, embedding, sectioning, staining — from the next.
- 3. Predict** the artifact produced when a step is skipped, shortened, or substituted.
- 4. Classify** stains by their chemical mechanism (charge attraction, oxidation-Schiff, metal reduction, lipid solubility, birefringence).
- 5. Relate** each stain in the standard panel to its molecular target and to the clinical question it answers.
- 6. Match** the correct microscopy modality to the type of contrast required (refractive index, fluorescence, electron scattering, surface topography).
- 7. Distinguish** frozen section from paraffin section by what each preserves, what each sacrifices, and what each is used for.
- 8. Interpret** an empty vacuole, a magenta band, a black network, or an apple-green glow as a specific molecular statement about the tissue.

Big Picture

This chapter is the lens through which every other chapter must be read. The hierarchy:

- **Problem** — tissue is transparent, fragile, and self-destructing within minutes.
 - **Solution chain** — fixation preserves → dehydration removes water → clearing makes the tissue paraffin-miscible → embedding gives it a rigid support → sectioning reduces it to a transparent slice → staining adds contrast → microscopy resolves the contrast.
 - **Master principle** — every step exists because the next step requires it; every stain reveals one specific molecular property and nothing else.
- The chapter proceeds: Build the Concept (the chain) → Structure (each step) → Function (each step solves one problem) → Structure → Function (why this chain works) → Classification (stains and microscopies by mechanism) → Compare & Distinguish → Recognition Logic → Clinical Correlation → Common Misconceptions → High-Yield → Integrated Summary → Mastery Check → Transition → Rapid Review.

Core Concept

Tissue is transparent and fragile; histology is the controlled transformation of tissue into a slice thin enough to transmit light and stained brightly enough to see what matters. Each technique reveals primarily one property; the art of diagnosis is choosing the technique that answers the question.

Build the Concept

Start with the problem. A cell removed from the body loses its oxygen supply within minutes. Mitochondria stop producing ATP; the sodium-potassium pump fails; the cytoplasm swells; lysosomal enzymes escape and digest the cell from within — a process called **autolysis**. Even before the student opens the textbook, the tissue has begun to change. The whole of histological technique is the controlled answer to this problem: preserve the structure closely enough to its living state that cells, matrix, boundaries, and relationships can still be interpreted, while making the tissue physically suitable for sectioning and providing contrast against an otherwise transparent background.

The chain that solves the problem has six logical links:

- 1. Fixation** — chemically cross-link the proteins so that autolysis stops and the structure is locked in place.
- 2. Dehydration** — remove the water, because paraffin (the embedding medium) is hydrophobic and cannot infiltrate a wet specimen.
- 3. Clearing** — replace the dehydrating alcohol with a solvent that is miscible both with alcohol and with paraffin.
- 4. Embedding** — infiltrate the tissue with molten paraffin and let it solidify into a block that can be cut.
- 5. Sectioning** — cut the block on a microtome into slices roughly 5 μm thick.

6. Staining — apply dyes that bind specific molecular targets, creating contrast.

Each step exists because the next needs it. Skip the dehydration and the paraffin will not infiltrate. Skip the clearing and the paraffin will not mix with the alcohol. Skip the embedding and the tissue will collapse under the microtome blade. Skip the staining and the section will be invisible.

Structure — the Six Steps in Detail

1. Fixation

Fixation is the chemical arrest of biological change. The most common fixative for routine light microscopy is **formalin** — a ~10% solution of formaldehyde in phosphate-buffered saline. Formaldehyde reacts with primary amine groups on proteins to form methylene bridges that cross-link adjacent proteins into an insoluble network. The cross-linking preserves cellular architecture, blocks autolysis, inactivates infectious agents, and makes the tissue firm enough to handle.

Glutaraldehyde is a stronger cross-linker used for electron microscopy, but it penetrates tissue slowly and is therefore often used as a secondary fixative after a primary aldehyde. **Osmium tetroxide** is used as a secondary fixative for EM; it reacts with unsaturated lipid bonds and stabilizes membranes, and because osmium is electron-dense it also contributes contrast in the electron beam.

A critical concept: **fixation is not neutral**. Formalin preserves morphology but destroys enzymatic activity and partially degrades nucleic acids. Glutaraldehyde preserves ultrastructure but makes immunolabelling harder. Osmium fixes lipid but dissolves into the lipid bilayer, which is exactly why the EM image shows membranes as black lines.

2. Dehydration

Water and paraffin are not miscible. The tissue must be passed through a graded series of ethanol (typically 70% → 80% → 95% → absolute) to remove the water. Each step concentrates the ethanol gradually so that the tissue does not shrink or distort from osmotic shock.

3. Clearing

Ethanol and paraffin are also not miscible. **Xylene** — a hydrocarbon solvent — is miscible with both. The tissue is passed through two or three changes of xylene to remove the ethanol and prepare the way for paraffin infiltration. Xylene is the clearing agent: it makes the tissue translucent because its refractive index is close to that of protein.

> **DO NOT CONFUSE**. Dehydration removes **water** (using ethanol). Clearing removes **ethanol** (using xylene). The student who reverses these terms has already lost the logic.

4. Embedding

The cleared tissue is placed in molten paraffin wax (typically ~60 °C) for several changes, then oriented in a mould and allowed to cool into a solid block. Paraffin gives the tissue a rigid support that can be cut on the microtome without distortion. Alternative embedding media include **resin** (for EM, thinner sections), **celloidin** (for large specimens), and **OCT compound** (for frozen sections, where water is not removed).

5. Sectioning

A microtome holds the paraffin block and advances it against a steel or glass blade in micrometre increments. Sections of **approximately 5 µm thickness** (range ~1-10 µm) are thin enough to be transparent when mounted on a glass slide. The section is floated on a water bath to flatten it, picked up on a slide, and dried.

6. Staining

The slide is now a thin, transparent, pale-yellow slice of tissue — essentially invisible. Staining adds contrast. The paraffin must first be removed (with xylene) and the tissue rehydrated (back through ethanol to water) before aqueous dyes can penetrate.

Function — Why Each Step Solves the Problem It Solves

Step	Problem it solves	Without it
Fixation	Tissue autolyses; structures collapse	Cells rupture; no diagnosis
Dehydration	Paraffin is hydrophobic; water blocks infiltration	Block will not cut
Clearing	Ethanol and paraffin are immiscible	Block will not infiltrate
Embedding	Tissue is too soft to section thinly	Sections tear or compress
Sectioning	Whole tissue is opaque and too thick	Microscope cannot focus through
Staining	Sections are nearly transparent	Structures are invisible

Structure → Function — Why This Chain Works

The chain works because each step transforms the tissue into the form the next step requires. The principle is sequential dependency: fixation prepares for dehydration, dehydration for clearing, clearing for embedding, embedding for sectioning, sectioning for staining, staining for microscopy. A break at any link breaks the chain. The diagnostic implication is equally important: if a step is poorly executed, the resulting artifact tells the student where to look. Shrinkage and vacuolation suggest poor fixation or rapid dehydration. Faded nuclei suggest a weak hematoxylin or an old solution. A black precipitate scattered through the section suggests oxidized hematoxylin or carbon from the microtome blade.

Classification — Stains by Chemical Mechanism

The stain menu looks long but reduces to five underlying mechanisms:

Mechanism 1 — Charge attraction. Basic (cationic) dyes bind acidic tissue components. Hematoxylin is basic; it binds the negatively charged phosphate groups of DNA and RNA, staining the nucleus blue. Acidic (anionic) dyes bind basic tissue components. Eosin is acidic; it binds the positively charged amino groups of cytoplasmic proteins, staining the cytoplasm pink. This is the basis of the **H&E** stain and accounts for most of what you see in routine sections.

Mechanism 2 — Oxidation followed by Schiff condensation. Periodic acid oxidizes the 1,2-glycol groups of carbohydrates to aldehydes; the Schiff reagent then reacts with those aldehydes to produce a magenta colour. **PAS** (Periodic Acid-Schiff) highlights carbohydrates — glycogen, mucins, basement membranes, fungal cell walls. Diastase (alpha-amylase) digestion removes glycogen, so a PAS-positive, diastase-sensitive structure is glycogen; a PAS-positive, diastase-resistant structure is mucin or basement membrane.

> **EXAM PEARL.** PAS highlights carbohydrate-rich structures and is not a collagen-specific stain. Basement membranes are PAS-positive because of their carbohydrate-rich components (laminin, perlecan, entactin associated with the type IV collagen network), not because PAS stains collagen directly.

Mechanism 3 — Metal reduction or substitution. Silver methods reduce silver ions to metallic silver at specific chemical sites. **Reticulin stains** deposit silver on the carbohydrate sheath of type III collagen, producing a black network in lymphoid stroma and around vessels. **Von Kossa** does not stain calcium ions directly; silver nitrate substitutes for the phosphate and carbonate anions bound to calcium in mineralised deposits, and the silver is then reduced to metallic silver, producing a black colour. Von Kossa is therefore an indirect indicator of mineralisation, not a stain for calcium as such.

Mechanism 4 — Lipid solubility. **Oil Red O** and the **Sudan dyes** are lipophilic dyes that partition preferentially into neutral lipid. The crucial processing requirement is that the lipid must still be present in the tissue at the time of staining — which means frozen section, because routine paraffin processing uses ethanol and xylene, both of which dissolve lipid. On a frozen section stained with Oil Red O, lipid appears bright red-orange. On a routine paraffin section, the same adipocyte appears as an empty vacuole. The cell is not empty; the lipid was washed away.

Mechanism 5 — Birefringence. **Congo red** binds amyloid in a regular, oriented manner, and when the section is viewed under crossed polarising filters the oriented dye produces a characteristic apple-green birefringence. Collagen is also birefringent, but white, not green. The diagnosis of amyloid therefore requires the combination of Congo red staining and apple-green birefringence.

Other stains worth knowing:

- **Trichrome stains** (Masson, Mallory): collagen blue or green, muscle red, nuclei black. Used to assess fibrosis.
- **Orcein / Verhoeff-van Gieson**: elastic fibres brown-black. Used to assess vessel-wall integrity.
- **Prussian blue (Perls)**: ferric iron + potassium ferrocyanide → ferric ferrocyanide, blue. Used to demonstrate haemosiderin and to diagnose iron overload.
- **Alcian blue**: acidic mucins (sulfated and carboxylated GAGs) blue at pH 2.5; used in combination with PAS to distinguish neutral from acidic mucins.
- **Feulgen**: acid hydrolysis removes purine bases from DNA, exposing aldehyde groups on the deoxyribose that react with Schiff reagent. Used in flow cytometry for DNA ploidy analysis.
- **Alizarin red**: chelates calcium directly; the appropriate stain when calcium itself (not just mineralisation) is in question.

Classification — Microscopy Modalities by Contrast Mechanism

The microscope is not a magnifier; it is a contrast generator. Each modality exploits a different physical property:

- **Bright-field light microscopy** — the routine. Resolution limited to $\sim 0.2 \mu\text{m}$ by the wavelength of visible light; below this, electron microscopy is required.
- **Phase-contrast microscopy** — converts refractive-index differences in unstained, living cells into brightness differences. Used for live cells, tissue culture, sperm motility.
- **Differential interference contrast (DIC, Nomarski)** — similar principle, gives a pseudo-3-D shadowed image.
- **Dark-field microscopy** — only scattered light reaches the objective; bright objects on a black background. Used for spirochaetes (syphilis serology) and for the demonstration of certain crystals.
- **Fluorescence microscopy** — fluorochromes (FITC, rhodamine, DAPI, GFP) absorb light at one wavelength and emit at a longer wavelength; the tissue appears bright on a dark background. Used in immunofluorescence.
- **Confocal microscopy** — uses a pinhole to reject out-of-focus light, allowing optical sectioning of thick specimens.
- **Polarising microscopy** — uses crossed polarising filters; birefringent structures (amyloid under Congo red, foreign material, crystals) appear bright on a dark background.
- **Transmission electron microscopy (TEM)** — electrons pass through an ultrathin section ($\sim 70 \text{ nm}$); electron-dense structures (osmium-stained membranes, heavy-metal-stained nucleic acids) scatter electrons and appear dark. Resolution $\sim 0.1 \text{ nm}$ because the de Broglie wavelength of the electron is far shorter than visible light.
- **Scanning electron microscopy (SEM)** — electrons scan the surface; secondary electrons are collected to produce a 3-D image of surface

- topography. Resolution ~1-10 nm.
- **Freeze-fracture EM** — frozen tissue is cleaved along the plane of weakest adhesion (usually the lipid bilayer); the exposed faces (E-face, the outer leaflet seen from within; P-face, the inner leaflet seen from within) are coated with platinum and examined by TEM. Reveals integral membrane proteins as particles within the bilayer.

Compare & Distinguish

Frozen section vs paraffin section

Feature	Frozen section	Paraffin section
Speed	Minutes	Days
Lipid preserved	Yes	No (dissolved by ethanol/xylene)
Morphology	Acceptable but inferior	Excellent
Special stains	Limited	All routine + special stains
Typical use	Intraoperative margin	Routine diagnosis

TEM vs SEM vs light microscopy

Feature	TEM	SEM	Light
Resolution	~0.1 nm	~1-10 nm	~0.2 μm
Sample type	Ultrathin section, internal	Surface, no sectioning	Section ~5 μm
Image	2-D internal	3-D surface	2-D stained

H&E vs special stains

Question	Stain	Why
General architecture	H&E	Nucleus blue, cytoplasm pink
Collagen in fibrosis	Trichrome	Collagen blue/green vs muscle red
Reticular stroma	Silver	Reticular fibres black
Elastic lamina	Orcein / Verhoeff	Elastic fibres brown-black
Lipid	Oil Red O on frozen	Lipid dissolved by paraffin processing
Amyloid	Congo red + polarised	Apple-green birefringence
Iron	Prussian blue	Ferric iron blue
Mineralisation	Von Kossa	Indirect via silver substitution
Glycogen	PAS $\pm$ diastase	Diastase-sensitive control

Recognition Logic — How to Read a Stain

> **Look for** the dominant colour. Blue means basic dye $\rightarrow$ acidic tissue component. Pink/red means acidic dye $\rightarrow$ basic tissue component. Black

means metal deposition (silver or osmium). Empty means the molecule was dissolved.

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> **Then confirm with** the expected location of that component. Blue should be in nuclei, ribosome-rich cytoplasm, or mucins; pink in cytoplasm and extracellular matrix; black in reticulin or mineral; empty in lipid vacuoles.

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> **Do not confuse** absence of colour with absence of structure. An empty vacuole in a paraffin section of liver can be lipid (which was dissolved) or glycogen (which was dissolved) or even a fixation artefact.

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> **The decisive feature is** the combination of colour + location + context. A blue dot inside a cell is not a nucleus if it sits in the cytoplasm; it is more likely a bacterium or a contaminating artefact.

Clinical Correlation

Clinical Case — Four Liver Biopsies, One H&E, Four Diagnoses

A 45-year-old man presents with fatigue, hepatomegaly, and abnormal liver function tests. On routine H&E, the liver biopsy shows pale, swollen hepatocytes — essentially the same picture regardless of the underlying cause. Four additional stains resolve four different diagnoses:

- **Oil Red O on frozen section** — bright red-orange droplets in hepatocytes. Diagnosis: **hepatic steatosis** (fatty liver).
- **PAS with diastase control** — magenta cytoplasm that disappears after diastase. Diagnosis: **glycogen storage disease**.
- **Prussian blue** — bright blue granules in hepatocytes and Kupffer cells. Diagnosis: **haemochromatosis** (iron overload).
- **Congo red under crossed polarisers** — apple-green birefringence in the perisinusoidal space. Diagnosis: **hepatic amyloidosis**.

Histological lesson. H&E shows that something is present; special stains identify what that something is. The choice of stain is a clinical decision, not a routine one.

Clinical Case — Frozen Section in the Operating Room

A surgeon is operating on a breast tumour and asks for a margin. The tissue is sent fresh from the operating theatre to the pathology laboratory, mounted in OCT, frozen in a cryostat at approximately -20°C , sectioned, and stained with H&E within minutes. The pathologist reports: invasive ductal carcinoma extends to the inked margin. The surgeon removes more tissue.

Histological lesson. Speed comes at the cost of morphology. Frozen sections are good enough to answer the binary question "is tumour at the margin?" but not good enough for full pathological typing. The lipid stain (Oil Red O) is also a frozen-section technique for exactly this reason.

Common Misconceptions

Misconception: Von Kossa stains calcium.

Why it seems plausible: the stain is associated with bones and calcified deposits. The correct understanding: Von Kossa demonstrates phosphate and carbonate anions bound to calcium in mineralised deposits by silver substitution; it is an indirect indicator of mineralisation, not a stain for calcium ion directly. When the clinical question is "is calcium present?", alizarin red is the more appropriate stain.

Misconception: Clearing removes water.

Why it seems plausible: the tissue does become translucent after clearing, which sounds like water removal. The correct understanding: dehydration removes water (with ethanol); clearing removes ethanol (with xylene) and makes the tissue miscible with paraffin. The two terms are not interchangeable.

Misconception: H&E shows everything.

Why it seems plausible: H&E is the routine stain and the most familiar image. The correct understanding: H&E reveals primarily nucleic acids (blue) and proteins (pink). It does not reveal carbohydrates, lipids, amyloid, iron, mineralisation, or specific antigens without further steps.

Misconception: Higher magnification means more resolution.

Why it seems plausible: "zooming in" feels like seeing more. The correct understanding: resolution is limited by wavelength ($\sim 0.2 \mu\text{m}$ for visible light); once the diffraction limit is reached, empty magnification produces a larger but no clearer image. To see finer detail, switch to a shorter-wavelength probe (electrons).

Misconception: PAS stains collagen.

Why it seems plausible: basement membranes, which contain type IV collagen, are strongly PAS-positive. The correct understanding: PAS is not a collagen-specific stain; basement membranes are PAS-positive because of their carbohydrate-rich components (laminin, perlecan, entactin), not because PAS reacts with collagen directly.

Misconception: Frozen sections are just faster paraffin sections.

Why it seems plausible: the difference appears to be processing time alone. The correct understanding: frozen sections preserve lipid and enzyme activity but produce inferior morphology; paraffin sections provide excellent morphology but destroy lipid and partially degrade nucleic acids. The two techniques are complementary, not interchangeable.

High-Yield Knowledge

Must Know

- The six-step chain: fixation → dehydration → clearing → embedding → sectioning → staining.

- H&E chemistry: hematoxylin (basic) → nuclei blue; eosin (acidic) → cytoplasm pink.
- PAS targets glycogen, mucin, basement membrane, fungi; diastase removes glycogen.
- Oil Red O requires frozen sections because routine processing dissolves lipid.
- Von Kossa is an indirect indicator of mineralisation, not a stain for calcium.
- Resolution of light microscopy is $\sim 0.2 \mu\text{m}$; resolution of TEM is $\sim 0.1 \text{ nm}$.
- Frozen section use: intraoperative diagnosis, lipid staining.
- Formalin is the routine LM fixative; glutaraldehyde + osmium are the EM fixatives.

Must Distinguish

- Dehydration (water → ethanol) vs clearing (ethanol → xylene).
- Frozen section (lipid preserved, morphology inferior) vs paraffin section (lipid lost, morphology excellent).
- PAS-positive + diastase-sensitive (glycogen) vs PAS-positive + diastase-resistant (mucin, basement membrane).
- Von Kossa (mineralisation, indirect) vs alizarin red (calcium, direct).
- Apple-green birefringence (amyloid) vs white birefringence (collagen).

Must Understand

- Stain selection is a clinical decision driven by the diagnostic question.
- Every stain reveals one specific molecular property and nothing else.
- Resolution is limited by wavelength, not by magnification.
- Frozen sections sacrifice morphology for speed and lipid preservation.

Integrated Summary

Histology is the controlled transformation of fresh tissue into a stained, transparent, mechanically stable slice thin enough for a microscope. The transformation follows a six-step chain (fixation → dehydration → clearing → embedding → sectioning → staining) in which each step exists because the next requires it. Stains reduce to five chemical mechanisms — charge attraction, oxidation–Schiff, metal reduction or substitution, lipid solubility, and birefringence — and each stain reveals one specific molecular property and nothing else. Microscopy reduces to a set of contrast-generation mechanisms, each exploiting a different physical property (refractive index, fluorescence, electron scattering, birefringence, surface scattering). The pathologist's first decision is which technique answers the diagnostic question; the second is how to interpret what that technique shows. A stained slide is a chemistry experiment, not a photograph.

Mastery Check

- 1. Reconstruct the chain** — explain why each step exists, what it solves, and what artefact its failure produces.
- 2. Predict the diagnosis** — a biopsy is sent to the laboratory with the request "amyloid?". Describe the procedure, the stain, the molecular target, and the diagnostic criterion (what positive looks like under the microscope).
- 3. Defend the choice** — the clinical question is "does this bone biopsy contain calcium?". Explain why Von Kossa alone is not the right answer and what stain should be added.
- 4. Resolve the paradox** — the same liver on H&E looks pale regardless of cause; explain how four different special stains produce four different diagnoses, and identify the molecular target of each.
- 5. Choose the technique** — the question is "is this tumour at the inked surgical margin?" Justify a frozen section; explain what is sacrificed and what is preserved.
- 6. Interpret an image** — a section shows a hepatocyte with a clear, round, centrally located vacuole that displaces the nucleus. Name three possible contents and the stain that would distinguish them.

Transition

You now know how tissue is prepared, stained, and examined. Step inside the cell itself — the machinery that every stain is lighting up, and the architectural decisions that allow a single cell to build proteins, generate force, and renew itself.

Rapid Review

- **Six-step chain** — fixation (formalin LM, glutaraldehyde + osmium EM) → dehydration (graded ethanol, removes water) → clearing (xylene, removes ethanol) → embedding (paraffin) → sectioning (~5 µm) → staining.
- **H&E** — hematoxylin basic → nucleic acids blue; eosin acidic → proteins pink.
- **PAS** — periodic acid + Schiff reagent → carbohydrates magenta; PAS+diastase-sensitive = glycogen; PAS+diastase-resistant = mucin/BM.
- **Silver (reticulin)** — type III collagen argyrophilic → black network.
- **Oil Red O / Sudan** — neutral lipid → red-orange; **frozen section only**.
- **Congo red + polarised** — amyloid β-sheet → apple-green birefringence (combination required).
- **Prussian blue (Perls)** — ferric iron → blue.
- **Von Kossa** — silver substitution for phosphate/carbonate in mineralised deposits → black; **indirect indicator of mineralisation, not calcium**.
- **Alizarin red** — direct calcium chelator.
- **Trichrome** — collagen blue/green, muscle red.
- **Orcein / Verhoeff** — elastic fibres brown-black.
- **Resolution** — light ~0.2 µm; TEM ~0.1 nm.

- **Frozen section** — intraoperative, lipid staining; morphology inferior.
- **Common trap** — clearing removes ethanol, not water; Von Kossa is not a calcium stain; PAS is not a collagen stain; magnification is not resolution.

Chapter 2 — The Cytoplasm

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 2

Opening Question

How does a cell build, fold, sort, and ship thousands of different proteins — and destroy the ones that misfolded — without mixing them up, while constantly renewing its own membrane and reshaping itself in response to its surroundings?

Why This Matters

Every protein that makes you work — every enzyme, every receptor, every hormone, every antibody — passes through the cytoplasm on its way from gene to function. Many of the most informative diagnostic features in pathology live in the cytoplasm: the eosinophilic granularity of an oncocyte (mitochondria-rich), the basophilic cytoplasm of a plasma cell (RER-rich), the clear cytoplasm of a clear-cell renal carcinoma (lipid and glycogen). Reading cytoplasm correctly means predicting function from appearance.

Learning Objectives

By the end of this chapter, you will be able to:

- 1. Explain** how the endomembrane system (RER → Golgi → vesicles → lysosomes / plasma membrane) coordinates the synthesis, modification, sorting, and degradation of proteins.
- 2. Distinguish** rough from smooth endoplasmic reticulum by structure, function, and disease association.
- 3. Predict** the histological appearance of a cell from the dominant organelle population (euchromatic nucleus + basophilic cytoplasm → protein-synthesising cell; eosinophilic cytoplasm → mitochondria-rich cell; clear cytoplasm → lipid or glycogen).
- 4. Classify** cytoskeletal filaments by diameter, subunit, polarity, associated motor, and example location.
- 5. Relate** each organelle to the disease produced when it fails (I-cell, α 1-antitrypsin, Kartagener, congenital disorders of glycosylation).
- 6. Identify** the cellular targets of common histological stains in cytoplasmic structures.

Big Picture

The cytoplasm is the cell's workshop. Membrane flow connects compartments, the cytoskeleton organises space and movement, and every organelle has a specific job.

- **Membrane flow** — RER (synthesis) → Golgi (modification, sorting) → vesicles (delivery) → lysosomes (degradation) or plasma membrane.

- **Cytoskeleton** — actin (6–8 nm, microfilaments, myosin motor) for surface contraction; intermediate filaments (~10 nm, no motor, tissue-type specific) for mechanical integrity; microtubules (~25 nm, kinesin and dynein motors) for long-range transport and chromosome separation.
- **Specialised compartments** — mitochondria (oxidative phosphorylation, own DNA), peroxisomes (β -oxidation of very-long-chain fatty acids, hydrogen peroxide detoxification), lysosomes (acid hydrolases), inclusions (stored lipid, glycogen, pigment).

Core Concept

A cell is a membrane-traffic network built on a cytoskeletal scaffold, organised into compartments that each do one job, connected by vesicles that ensure cargo arrives at the right address.

Build the Concept

Begin with the problem: the human genome encodes ~20,000 proteins; many of them must be synthesised, folded, modified (glycosylated, sulfated, phosphorylated), sorted to the right destination (lysosome, plasma membrane, secretion), and degraded when misfolded or no longer needed. The cytoplasm solves these problems by partitioning work into membrane-bound organelles, with the cytoskeleton providing the spatial framework and the vesicle transport system providing the delivery service.

The logical sequence to teach the cytoplasm:

1. **Plasma membrane** — the lipid bilayer that defines the cell. Receptors, channels, transporters.
2. **Rough endoplasmic reticulum (RER)** — ribosome-studded membrane continuous with the nuclear envelope; site of synthesis of membrane proteins, secreted proteins, and lysosomal enzymes.
3. **Smooth endoplasmic reticulum (SER)** — no ribosomes; site of lipid synthesis, steroid synthesis, detoxification, calcium storage.
4. **Golgi apparatus** — stack of flattened cisternae (cis → medial → trans); receives vesicles from the RER, modifies the cargo (glycosylation, sulfation, phosphorylation), sorts it, and dispatches it to lysosomes, the plasma membrane, or secretion.
5. **Lysosomes** — membrane-bound vesicles full of acid hydrolases (proteases, lipases, nucleases, glycosidases) that operate at pH ~4.5. Receive cargo from endocytosis, phagocytosis, and autophagy.
6. **Peroxisomes** — smaller, single-membrane organelles; contain oxidases that generate hydrogen peroxide and catalase that destroys it; site of β -oxidation of very-long-chain fatty acids and plasmalogen synthesis.
7. **Mitochondria** — double-membrane organelles with inner-membrane cristae; site of oxidative phosphorylation (ATP production); contain their own circular DNA (mtDNA) and ribosomes;

maternally inherited.

8. Cytoskeleton — actin filaments, intermediate filaments, microtubules.

9. Inclusions — stored lipid droplets, glycogen granules, pigment (lipofuscin, haemosiderin, melanin).

10. Cell-surface specialisations — microvilli (absorption), cilia (movement), stereocilia (sensory), basal specialisations (basal lamina, hemidesmosomes).

This is the order in which each organelle is taught below.

Structure

Plasma Membrane

A phospholipid bilayer with embedded proteins. The lipid bilayer is fluid; proteins diffuse laterally unless anchored. The outer leaflet bears glycolipids and glycoproteins whose carbohydrate moieties form the glycocalyx. Cholesterol stiffens the membrane. The bilayer is asymmetric — phosphatidylserine is kept on the inner leaflet in living cells (its appearance on the outer surface is an early signal of apoptosis).

Rough Endoplasmic Reticulum (RER)

A continuous network of flattened membrane sacs (cisternae) studded with ribosomes on the cytoplasmic face. The RER is continuous with the outer nuclear membrane; the perinuclear space is the lumen of the RER. Ribosomes on the RER translate proteins that carry an N-terminal **signal peptide**, which is recognised by the **signal recognition particle (SRP)**, which delivers the ribosome to the SRP receptor on the RER membrane. The protein is then threaded through a translocon into the RER lumen, where the signal peptide is cleaved, the protein folds (with the help of chaperones such as BiP), and disulfide bonds form. N-linked glycosylation begins in the RER with the addition of a preformed oligosaccharide to an asparagine residue.

Histologically: RER-rich cytoplasm is **basophilic** (blue on H&E) because the abundant ribosomal RNA binds hematoxylin. Examples: plasma cells, pancreatic acinar cells, hepatocytes, plasma cells.

Smooth Endoplasmic Reticulum (SER)

A continuous network of tubules without ribosomes. Functions:

- **Lipid and steroid synthesis** — fatty acids, phospholipids, steroids (in adrenal cortex, Leydig cells, theca interna).
- **Detoxification** — cytochrome P450 enzymes in hepatocytes oxidise xenobiotics.
- **Calcium storage** — the sarcoplasmic reticulum of muscle is a specialised SER; calcium release triggers contraction.
- **Glucose-6-phosphatase** — the final step of gluconeogenesis and glycogenolysis.

Histologically: SER is not visible by light microscopy. Its presence is inferred from eosinophilic cytoplasm in cells known to make steroids (adrenal cortex) or detoxify (hepatocytes). SER-rich cells lack basophilia.

Golgi Apparatus

A stack of flattened cisternae (typically 4–6) with a cis face (entry, near the RER) and a trans face (exit, near the plasma membrane). Proteins from the RER arrive in transport vesicles, fuse with the cis-Golgi, and progress through the stack as their carbohydrate chains are trimmed and elaborated (mannose trimming, N-acetylglucosamine addition, sialic acid capping in the trans-Golgi). From the trans-Golgi, proteins are sorted into different vesicle populations:

- **Secretory vesicles** — for constitutive or regulated secretion.
- **Lysosomal vesicles** — bearing mannose-6-phosphate tags (see I-cell disease below).
- **Membrane vesicles** — for delivery to the plasma membrane.

Histologically: the Golgi appears as a pale, often crescent-shaped negative image next to the nucleus in routine H&E; it is best demonstrated by EM.

Lysosomes

Membrane-bound spherical organelles (~0.2–0.5 µm) containing acid hydrolases — proteases (cathepsins), lipases, nucleases, glycosidases — that operate at pH ~4.5. The acidic pH is maintained by a vacuolar H⁺-ATPase in the lysosomal membrane. Lysosomes receive cargo from:

- **Endocytosis** — receptor-mediated endocytosis brings LDL, transferrin, hormones into the cell; the endosome eventually fuses with a lysosome.
- **Phagocytosis** — macrophages and neutrophils engulf bacteria and debris into phagosomes, which fuse with lysosomes to form phagolysosomes.
- **Autophagy** — the cell wraps its own organelles in a phagophore, which fuses with a lysosome to form an autolysosome.

Histologically: lysosomes are visible by light microscopy only when they contain identifiable cargo (e.g., haemosiderin in macrophages, lipofuscin in cardiac myocytes). The diagnostic feature of lysosomal storage diseases is the accumulation of undigested material inside lysosomes (e.g., glycogen in Pompe disease, sphingomyelin in Niemann-Pick).

Peroxisomes

Single-membrane organelles (~0.2–1 µm) that contain oxidases (which generate hydrogen peroxide) and catalase (which destroys it). Peroxisomes oxidise very-long-chain fatty acids (which mitochondria cannot handle), synthesise plasmalogens (a class of phospholipid), and participate in bile-acid synthesis and D-amino-acid detoxification. Peroxisomes form by growth and fission of pre-existing peroxisomes; they do not arise from the ER.

Histologically: invisible by routine light microscopy; demonstrable by EM or by immunohistochemistry for catalase.

Mitochondria

Double-membrane organelles with a smooth outer membrane and a highly folded inner membrane (the cristae). The space between the membranes is the intermembrane space; the space enclosed by the inner membrane is the matrix. The inner membrane contains the electron-transport chain and ATP synthase; the matrix contains the enzymes of the citric acid cycle, mtDNA, mitochondrial ribosomes, and tRNAs.

Mitochondria are dynamic: they undergo fission and fusion, are transported along microtubules by kinesin and dynein motors, and are degraded by mitophagy (a form of autophagy). A typical hepatocyte contains 1,000–2,000 mitochondria; a cardiac myocyte contains 5,000–10,000.

Mitochondria contain their own genome — a circular double-stranded DNA of ~16,500 bp encoding 13 proteins of the electron-transport chain, 22 tRNAs, and 2 rRNAs. Mitochondrial DNA is maternally inherited (sperm mitochondria are degraded after fertilisation) and accumulates mutations with age (the mitochondrial theory of ageing).

Histologically: mitochondria produce the eosinophilic cytoplasm of protein-rich cells (cardiac myocytes, hepatocytes, oncocytes). On EM, mitochondria are visible as oval double-membrane organelles with cristae.

Function — The Membrane-Flow Pathway

The endomembrane system operates as a one-way flow with recycling:

- 1. Synthesis** — RER-bound ribosomes translate proteins carrying an N-terminal signal peptide.
- 2. Folding and quality control** — chaperones (BiP, calnexin, calreticulin) assist folding; misfolded proteins are retrotranslocated to the cytoplasm and degraded by the proteasome (ER-associated degradation, ERAD).
- 3. Vesicular transport to Golgi** — COPII vesicles bud from ER exit sites and travel to the cis-Golgi.
- 4. Golgi processing and sorting** — glycosylation is modified; cargo is sorted in the trans-Golgi network.
- 5. Delivery** — secretory vesicles, lysosomal vesicles, or membrane vesicles travel to their destinations along microtubules and actin filaments.
- 6. Endocytosis and recycling** — plasma membrane receptors and cargo are internalised into early endosomes, sorted in the endosome, recycled to the plasma membrane, or sent to lysosomes for degradation.

Structure → Function — Why the Architecture Works

The endomembrane system works because it physically separates incompatible biochemical environments:

- The reducing environment of the cytoplasm is separated from the oxidising environment of the ER lumen (which allows disulfide bond formation).
- The acidic interior of the lysosome is separated from the neutral cytoplasm (acid hydrolases would digest the cell at neutral pH).
- The high membrane potential of the mitochondrial inner membrane (generated by pumping protons out of the matrix) is the energy source for ATP synthesis.

When the architecture fails, the disease is informative:

- **I-cell disease** (mucopolidosis II) — N-acetylglucosamine-1-phosphotransferase is deficient, so lysosomal enzymes are not tagged with mannose-6-phosphate in the cis-Golgi. The unmarked enzymes are secreted from the cell instead of being delivered to lysosomes, and undigested substrates accumulate in lysosomes. Clinical features: coarse facies, restricted joint movement, skeletal dysplasia, psychomotor retardation; death usually by age 8.
- **α 1-antitrypsin deficiency** — a misfolded α 1-antitrypsin protein accumulates in the RER of hepatocytes, producing PAS-positive, diastase-resistant cytoplasmic globules; the secreted deficiency causes panacinar emphysema (loss of elastic recoil in the lung).
- **Congenital disorders of glycosylation (CDG)** — defects in N-linked glycosylation in the ER/Golgi produce multisystem disease (failure to thrive, inverted nipples, abnormal fat distribution, cerebellar hypoplasia).
- **Kartagener syndrome** (primary ciliary dyskinesia) — defective dynein arms in cilia produce immotile cilia, leading to situs inversus, chronic sinusitis, bronchiectasis, and infertility.
- **Mitochondrial diseases** — MELAS (mitochondrial encephalopathy, lactic acidosis, stroke-like episodes), MERRF (myoclonic epilepsy with ragged-red fibres), Leber hereditary optic neuropathy, and others result from mtDNA mutations; muscle biopsy shows ragged-red fibres on Gomori trichrome (subsarcolemmal accumulation of abnormal mitochondria).

Classification — Cytoskeletal Filaments

The three filament types are distinguished by diameter, subunit, polarity, motor protein, and example location:

Filament	Diameter	Subunit	Polarity	Motor	Example location
Microfilament (actin)	~6–8 nm	G-actin	Polarised (+/–)	Myosin	Cell cortex, microvilli, muscle sarcomere
Intermediate filament	~10 nm	Tissue-specific (keratin, vimentin, desmin, neurofilament, GFAP, lamin)	Non-polar	None	Mechanical integrity

Filament	Diameter	Subunit	Polarity	Motor	Example location
Microtubule	~25 nm	α/β -tubulin heterodimer	Polarised (+/-)	Kinesin (anterograde), dynein (retrograde)	Mitotic spindle, axon, centrosome, cilium

Intermediate filaments are tissue-type specific and are used as immunohistochemical markers of cell origin:

- **Cytokeratins** — epithelial cells (cytokeratin 5/6 in basal epithelia, cytokeratin 8/18 in simple epithelia).
- **Vimentin** — mesenchymal cells (fibroblasts, endothelium, macrophages, lymphocytes).
- **Desmin** — muscle cells (skeletal, cardiac, smooth).
- **Neurofilaments** — neurons.
- **GFAP** (glial fibrillary acidic protein) — astrocytes.
- **Lamins A/B/C** — the nuclear lamina.

Function — Cytoskeleton

- **Actin filaments** — drive cell migration, cytokinesis, microvillar structure, muscle contraction (with myosin II), and cytoplasmic streaming. The actin cortex beneath the plasma membrane generates cell shape and tension.
- **Intermediate filaments** — provide mechanical resilience. They are the most stable of the three filament types and are not dynamic on the timescale of actin and microtubules.
- **Microtubules** — provide long-range tracks for intracellular transport (vesicles, organelles) and form the mitotic spindle, the axoneme of cilia and flagella, and the centrosome.

Structure → Function — Motor Proteins and Directionality

The polarity of actin filaments and microtubules allows motor proteins to move unidirectionally:

- **Myosin** moves along actin filaments toward the plus end (or toward the minus end for myosin VI).
- **Kinesin** moves along microtubules toward the plus end (anterograde, from the cell centre outward).
- **Dynein** moves along microtubules toward the minus end (retrograde, toward the cell centre).

This directionality is what allows a vesicle to be transported from the trans-Golgi to the plasma membrane (kinesin, anterograde) and back from the axon terminal to the cell body (dynein, retrograde). Defects in dynein cause Kartagener syndrome; defects in kinesin produce peripheral neuropathy.

Classification — Centrioles, Centrosome, and Basal Bodies

The **centrosome** is the principal microtubule-organising centre of the cell. It contains a pair of **centrioles** — cylindrical structures built from nine triplets of microtubules arranged in a pinwheel — surrounded by pericentriolar material that nucleates microtubules. During mitosis, the centrosome duplicates and the two centrosomes form the two poles of the mitotic spindle.

Basal bodies are structurally identical to centrioles but located at the base of cilia, where they nucleate the ciliary axoneme (the 9+2 microtubule arrangement that drives ciliary beating).

Cell-Surface Specialisations

- **Microvilli** — finger-like apical projections supported by a core of actin filaments; increase the apical surface area for absorption. Dense, regular microvilli form the **brush border** of the small intestine and the **striated border** of the proximal renal tubule.
- **Stereocilia** — long, actin-filled microvilli (not true cilia) found on the epididymis (absorption) and the hair cells of the inner ear (mechanosensation).
- **Cilia** — motile projections built on a 9+2 microtubule axoneme; covered in Chapter 14 in the context of the respiratory tract. The **primary cilium** (9+0, non-motile) is found on most vertebrate cell types and functions as a sensory antenna for fluid flow, morphogens (Sonic hedgehog), and receptor localisation.
- **Basal lamina / basement membrane** — the specialised ECM to which epithelial cells attach; described in detail in Chapter 4.

Inclusions

- **Lipid droplets** — storage form of triglyceride; visible in adipocytes, hepatocytes (in steatosis), steroid-producing cells.
- **Glycogen granules** — storage form of glucose; PAS-positive, diastase-sensitive; abundant in hepatocytes and skeletal muscle.
- **Lipofuscin** — yellow-brown pigment representing the residue of lysosomal digestion; accumulates in long-lived post-mitotic cells (cardiac myocytes, neurons) with age; called "wear-and-tear pigment."
- **Haemosiderin** — golden-brown aggregates of ferritin; storage form of iron; demonstrated by Prussian blue.
- **Melanin** — black-brown pigment produced by melanocytes from tyrosine via tyrosinase; transferred to keratinocytes in melanosomes.

Compare & Distinguish

RER vs SER

Feature	RER	SER
Ribosomes	Yes	No
Appearance	Basophilic cytoplasm	Not visible by LM
Function	Protein synthesis, N-glycosylation	Lipid/steroid synthesis, detoxification, Ca ²⁺ storage
Examples	Plasma cell, pancreatic acinar	Hepatocyte, adrenal cortex, Leydig cell

Lysosome vs Peroxisome

Feature	Lysosome	Peroxisome
pH inside	Acidic (~4.5)	Cytosolic (~7)
Key enzyme	Acid hydrolases	Catalase, oxidases
Origin	Golgi (via M6P pathway)	Growth and fission
Substrate	Macromolecules	Very-long-chain fatty acids, H ₂ O ₂

Microfilament vs Intermediate Filament vs Microtubule

See table above. The decisive features are diameter (6–8 / 10 / 25 nm), subunit (actin / tissue-specific / tubulin), and motor (myosin / none / kinesin and dynein).

Mitochondrion vs Peroxisome

Both are small organelles involved in oxidation. Mitochondria generate ATP via oxidative phosphorylation and contain mtDNA; peroxisomes generate and destroy hydrogen peroxide and contain no DNA.

Recognition Logic

- > **Look for** the dominant cytoplasm colour on H&E.
- > - **Basophilic (blue)** cytoplasm with an eccentric nucleus and a perinuclear pale zone (hof) → plasma cell (RER-rich).
- > - **Strongly basophilic** cytoplasm with apical eosinophilic granules and basal nuclei → pancreatic acinar cell.
- > - **Eosinophilic (pink) granular** cytoplasm → mitochondria-rich cell (oncocyte, cardiac myocyte, hepatocyte).
- > - **Clear cytoplasm** with peripheral nucleus → adipocyte (lipid dissolved).
- > - **Clear cytoplasm** with central nucleus and PAS-positive, diastase-sensitive material → glycogen-rich cell.
- >
- > **Then confirm with** nucleus shape, cell location, and tissue context.
- >
- > **Do not confuse** basophilia due to RER with basophilia due to DNA — the cytoplasm is stained, not the nucleus.
- >

> **The decisive feature is** the combination of cytoplasm colour, granularity, and tissue context. A basaloid cell with eosinophilic cytoplasm in a salivary gland is an oncocyte (mitochondria-rich); a basaloid cell with basophilic cytoplasm in a lymph node is a lymphocyte (RER-poor).

Clinical Correlation

Clinical Case — Liver Biopsy Showing Two Different "Pale Cytoplasm" Patterns

A 50-year-old alcoholic man and a 9-month-old infant with failure to thrive both have pale hepatocyte cytoplasm on H&E. The mechanisms are different.

- **Alcoholic steatosis** — triglycerides accumulate in lipid droplets that dissolve during paraffin processing, leaving large empty vacuoles. Confirmed by Oil Red O on frozen section.
- **Glycogen storage disease (type I, von Gierke)** — glucose-6-phosphatase deficiency causes glycogen accumulation in hepatocytes. Confirmed by PAS-positive, diastase-sensitive cytoplasmic staining.

Histological lesson. Pale hepatocyte cytoplasm is not a diagnosis; it is a question. The stain that distinguishes the causes is the diagnostic test.

Clinical Case — Plasma Cell Neoplasm

A 65-year-old man presents with bone pain, hypercalcaemia, and renal failure. Bone marrow biopsy shows sheets of plasma cells with clock-face heterochromatin, basophilic cytoplasm, and a perinuclear hof. Serum protein electrophoresis shows an IgG monoclonal spike.

Histological lesson. The basophilic cytoplasm reflects abundant RER producing immunoglobulin; the perinuclear hof is the prominent Golgi where the immunoglobulin is glycosylated and packaged. The combination is diagnostic of plasma cell differentiation — and the architecture (monotonous sheets rather than scattered reactive plasma cells) is diagnostic of neoplasia.

Clinical Case — Kartagener Syndrome

A child with recurrent sinusitis, bronchiectasis, and situs inversus. Nasal ciliary biopsy shows absent dynein arms by EM.

Histological lesson. Dynein is the motor that bends cilia; without it, cilia are immotile. Mucus cannot be cleared from the airways (bronchiectasis), sperm cannot swim (male infertility), and the nodal cilia that direct left-right asymmetry cannot function (random situs, ~50% situs inversus).

Common Misconceptions

Misconception: The Golgi makes proteins.

Why it seems plausible: the Golgi is the dispatch centre. The correct understanding: proteins are made on RER-bound ribosomes; the Golgi

modifies, sorts, and dispatches them. The Golgi does not perform translation.

Misconception: Lysosomes are the cell's "garbage disposers" and their contents are always harmful.

Why it seems plausible: lysosomes are full of hydrolytic enzymes. The correct understanding: lysosomes are essential for nutrient recycling (autophagy provides amino acids during fasting), for antigen presentation (via MHC class II loading), and for bone resorption (osteoclast lysosomes dissolve hydroxyapatite). Lysosomal enzyme deficiencies cause storage diseases, but lysosomes themselves are not pathological.

Misconception: All cytoplasmic granules are the same.

Why it seems plausible: secretory vesicles, lysosomes, and peroxisomes all appear as small dots. The correct understanding: secretory vesicles contain cargo for regulated or constitutive secretion; lysosomes contain acid hydrolases at pH 4.5; peroxisomes contain oxidases and catalase. The cell biology is different, and so is the disease when each fails.

Misconception: Intermediate filaments are present in all cells.

Why it seems plausible: the cytoskeleton seems universal. The correct understanding: intermediate filaments are tissue-type specific; lamins in the nucleus, keratins in epithelia, vimentin in mesenchymal cells, desmin in muscle, neurofilaments in neurons, GFAP in astrocytes. Immunohistochemistry against these filaments is used to identify the origin of poorly differentiated tumours.

Misconception: Mitochondria are static oval organelles.

Why it seems plausible: textbooks draw them as static beans. The correct understanding: mitochondria are dynamic — they undergo fission and fusion, are transported along microtubules, and are degraded by mitophagy. The mitochondrial network changes shape with metabolic demand.

High-Yield Knowledge

Must Know

- RER is basophilic cytoplasm; SER is not visible by LM.
- The Golgi is cis → medial → trans; lysosomal enzymes receive mannose-6-phosphate in the cis-Golgi.
- Lysosomes are acidic (pH ~4.5) due to vacuolar H⁺-ATPase.
- Peroxisomes contain catalase and oxidases; β-oxidise very-long-chain fatty acids.
- Mitochondria have their own DNA (mtDNA), maternal inheritance.
- Cytoskeleton: microfilaments (actin, 6–8 nm, myosin); intermediate filaments (~10 nm, no motor, tissue-specific); microtubules (~25 nm, kinesin anterograde, dynein retrograde).
- Intermediate filament markers: cytokeratin (epithelium), vimentin (mesenchyme), desmin (muscle), neurofilament (neurons), GFAP (astrocytes), lamin (nuclear envelope).

- I-cell disease = failure of M6P tagging → lysosomal enzymes secreted instead of delivered.

Must Distinguish

- RER vs SER (ribosomes, function, appearance).
- Lysosome vs peroxisome (acidic pH vs cytosolic pH; hydrolytic enzymes vs catalase/oxidases).
- Basophilia due to RER (cytoplasm) vs basophilia due to DNA (nucleus).
- Actin filament vs microtubule vs intermediate filament (diameter, subunit, motor).

Must Understand

- The endomembrane system is a one-way flow with recycling, not a static set of compartments.
- Vesicle targeting depends on the cytoskeleton (microtubule-based long-range transport, actin-based short-range transport).
- Mitochondrial diseases are maternally inherited and accumulate with age.

Integrated Summary

The cytoplasm is a membrane-traffic network built on a cytoskeletal scaffold. Proteins are synthesised on RER-bound ribosomes, folded and N-glycosylated in the RER lumen, transported to the Golgi by COPII vesicles, modified and sorted in the cis → medial → trans stack, and dispatched by vesicles to lysosomes, the plasma membrane, or secretion. Lysosomes are acidic degradative organelles; peroxisomes are oxidative organelles without DNA or ribosomes. Mitochondria are dynamic double-membrane organelles with their own genome that generate ATP by oxidative phosphorylation. The cytoskeleton has three filament types — actin (~6–8 nm), intermediate (~10 nm), and microtubules (~25 nm) — each with a characteristic motor and function. Defects in the endomembrane system and cytoskeleton produce I-cell disease, α 1-antitrypsin deficiency, CDG, Kartagener syndrome, and mitochondrial disease — clinical correlations that make the cell biology memorable.

Mastery Check

- 1. Trace a secreted protein** from the gene to the extracellular space, naming each organelle, modification, and transport step.
- 2. Explain** how the I-cell disease defect produces a clinical disease, and what cellular finding would confirm the diagnosis.
- 3. Predict the H&E appearance** of a plasma cell, an oncocyte, an adipocyte, and a cardiac myocyte, and explain each appearance in terms of the dominant organelle.
- 4. Classify** the three cytoskeletal filaments by diameter, subunit, motor, polarity, and example location.
- 5. Compare and contrast** lysosomes and peroxisomes.
- 6. Explain** why dynein defects cause both infertility and situs inversus.

7. Reconstruct the ERAD quality-control pathway and explain why a misfolded $\alpha 1$ -antitrypsin accumulates in hepatocytes but a misfolded cytosolic protein does not.

Transition

You have now seen the cell's machinery. The next chapter moves inside the command centre — the nucleus, where the genome is stored, protected, transcribed, and replicated.

Rapid Review

- **RER** — ribosome-studded, basophilic cytoplasm, protein synthesis and N-glycosylation; plasma cell, pancreatic acinar.
- **SER** — no ribosomes, lipid/steroid synthesis, detoxification, Ca^{2+} storage; hepatocyte, adrenal cortex.
- **Golgi** — cis $\rightarrow$ trans, glycosylation modification, sorting.
- **Lysosome** — acidic (pH 4.5) acid hydrolases; receives cargo from endocytosis, phagocytosis, autophagy; mannose-6-phosphate targeting.
- **Peroxisome** — oxidases + catalase; β -oxidation very-long-chain fatty acids.
- **Mitochondrion** — double membrane, cristae, mtDNA, maternal, ATP via oxidative phosphorylation.
- **Cytoskeleton** — actin (6–8 nm, myosin), intermediate (10 nm, no motor, tissue-specific), microtubule (25 nm, kinesin/dynein).
- **Intermediate filament markers** — cytokeratin (epithelium), vimentin (mesenchyme), desmin (muscle), neurofilament (neurons), GFAP (astrocytes), lamin (nuclear envelope).
- **Disease associations** — I-cell (M6P), $\alpha 1$ -antitrypsin (RER misfolding), CDG (glycosylation), Kartagener (dynein), mitochondrial diseases (mtDNA, maternal).
- **Recognition** — basophilic cytoplasm = RER-rich; eosinophilic granular = mitochondria-rich; clear vacuolated = lipid or glycogen.

Chapter 3 — The Nucleus

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 3

Opening Question

How does approximately two metres of DNA fit inside a nucleus roughly six micrometres across — and how does the cell keep that DNA organised, protected, transcribed, and replicated without tangling itself into uselessness?

Why This Matters

The nucleus is the cell's command centre. Every decision about whether a cell will divide, what proteins it will make, when it will die, and how it will respond to its environment originates in the nucleus. Nuclear pathology — mutations, translocations, aneuploidy, viral integration — underlies most human disease. Histologically, the appearance of the nucleus is one of the most informative features of any cell: size, shape, chromatin pattern, and nucleolar prominence tell the pathologist whether the cell is resting, active, premalignant, or malignant.

Learning Objectives

- 1. Describe** the structural components of the interphase nucleus.
- 2. Explain** how nuclear import and export are mediated by NLSs, NESs, importins, exportins, and Ran-GTP.
- 3. Distinguish** heterochromatin from euchromatin by structure, transcriptional activity, and microscopic appearance.
- 4. Describe** the structure and function of the nucleolus.
- 5. Explain** the cell-cycle phases (G₁, S, G₂, M) and the role of cyclins, CDKs, and the restriction point.
- 6. Distinguish** apoptosis, necrosis, and autophagy by morphology and by clinical context.
- 7. Predict** the nuclear appearance of metabolically active, quiescent, malignant, and apoptotic cells.

Big Picture

- **Envelope** — double membrane with pores, continuous with the RER.
- **Lamina** — intermediate filament meshwork (lamins A/B/C) supporting the envelope.
- **Chromatin** — DNA + histone, organised as euchromatin (active, pale) and heterochromatin (inactive, dark).
- **Nucleolus** — site of rRNA synthesis and ribosomal subunit assembly; non-membrane-bound.
- **Cell cycle** — G₁ → S → G₂ → M, driven by cyclin-CDK pairs.
- **Death** — apoptosis (programmed, non-inflammatory), necrosis (uncontrolled, inflammatory), autophagy (self-digestion).

Core Concept

The nucleus is a regulated gateway that decides which molecules enter and leave, which genes are transcribed, and how the genome is protected. A pale, euchromatic nucleus with a prominent nucleolus indicates active transcription and ribosome production; a small, dark, heterochromatic nucleus indicates quiescence.

Structure

Nuclear Envelope

A double membrane: the outer membrane is continuous with the RER and bears ribosomes; the inner membrane is lined by the nuclear lamina. The perinuclear space between the two membranes is continuous with the RER lumen. The envelope breaks down during mitosis and reassembles afterwards.

Nuclear Pore Complexes (NPCs)

Large protein complexes (~30 different nucleoporins, ~1000 protein molecules per pore) that span both membranes. Small molecules pass freely; proteins larger than ~40 kDa require an NLS.

Nuclear Lamina

A meshwork of intermediate filaments (lamins A, B, C) beneath the inner membrane. Provides mechanical support, anchors NPCs, and binds heterochromatin at the nuclear periphery (lamina-associated domains, LADs). Mutations in LMNA cause progeria (premature aging), Emery-Dreifuss muscular dystrophy, and cardiomyopathies.

Chromatin

DNA packaged with histones into nucleosomes (147 bp DNA + histone octamer H2A, H2B, H3, H4), then folded into higher-order structures.

- Heterochromatin — densely packed, transcriptionally inactive, dark-staining. Two kinds:
 - Constitutive — permanently silenced (centromeres, telomeres, inactive X = Barr body).
 - Facultative — silenced in this cell type, can be activated in another.
- Euchromatin — loosely packed, transcriptionally active, pale-staining.

Nucleolus

Non-membrane-bound; composed of fibrillar centres (RNA Pol I transcription), dense fibrillar component (rRNA processing), and granular component (subunit assembly). Size reflects rRNA synthesis rate — large in protein-synthesising cells, small in quiescent cells. Disassembles during mitosis.

Nuclear Import and Export in Detail

The trafficking between nucleus and cytoplasm is more sophisticated than a simple "DNA in, RNA out." Most regulatory proteins that act in the nucleus — transcription factors, DNA repair enzymes, signalling molecules — are synthesised in the cytoplasm and must be imported when needed. Most RNAs are synthesised in the nucleus and exported to the cytoplasm for translation. The NLS-NES system is the trafficking machinery.

An NLS is typically a short stretch of basic residues (lysine, arginine). Importin binds the cargo in the cytoplasm, carries it through the NPC, and releases it in the nucleoplasm upon Ran-GTP binding. An NES is a stretch of hydrophobic residues recognised by exportin, which carries cargo from the nucleus to the cytoplasm in a Ran-GTP-dependent manner. The Ran GTPase provides directionality: Ran-GEF in the nucleus keeps Ran-GTP high there; Ran-GAP in the cytoplasm keeps Ran-GDP high there. Importins release cargo when they encounter Ran-GTP (nucleus); exportins bind cargo when they encounter Ran-GTP and release it when GTP is hydrolysed (cytoplasm).

This trafficking is the basis for several important cellular phenomena. Transcription factors such as NF- κ B are held in the cytoplasm by I κ B; phosphorylation of I κ B causes its degradation, exposes the NF- κ B NLS, and allows nuclear import and target-gene activation. Steroid hormone receptors are cytoplasmic; hormone binding causes conformational change, NLS exposure, nuclear import, and transcriptional activation. Many viruses (HIV, herpes, adenovirus) deliver their genomes to the nucleus through NPCs.

Chromatin Higher-Order Structure

DNA wraps around histone octamers to form nucleosomes (147 bp). Nucleosomes are connected by linker DNA and linker histones (H1) and folded into higher-order structures. The 30-nm fibre is the next level, formed by nucleosome coiling. The 30-nm fibre is further looped and attached to a protein scaffold (the nuclear matrix), forming the familiar metaphase chromosome. This hierarchical organisation compacts ~2 metres of DNA into a ~6 μ m nucleus while still allowing access for transcription and replication.

Histone modifications — acetylation, methylation, phosphorylation, ubiquitination — form the histone code that regulates chromatin accessibility. Acetylation of H3 and H4 by HATs (histone acetyltransferases) opens chromatin and promotes transcription; deacetylation by HDACs (histone deacetylases) closes chromatin and silences transcription. Methylation can be activating or repressing depending on the residue. Bromodomain proteins recognise acetylated lysines; chromodomain proteins recognise methylated lysines.

Nucleolus in Detail

The nucleolus assembles around nucleolar organiser regions (NORs) — tandem repeats of rRNA genes on the acrocentric chromosomes (13, 14, 15, 21, 22 in humans). RNA polymerase I transcribes the 45S pre-rRNA in the fibrillar centres. The 45S is processed in the dense fibrillar component

to 28S, 18S, and 5.8S rRNAs. These rRNAs are assembled with ribosomal proteins (imported from the cytoplasm) in the granular component to form the 40S and 60S subunits. The subunits are exported through NPCs and join in the cytoplasm to form functional 80S ribosomes.

Stress (heat shock, oxidative damage, viral infection, certain chemotherapeutics) causes nucleolar segregation — separation of fibrillar and granular components. The nucleolus also serves as a stress sensor that regulates p53 stability through the MDM2-ARF pathway.

Function

The nucleus stores the genome, regulates transcription, replicates DNA in S phase, and protects DNA from cytoplasmic nucleases. The NPC controls bidirectional nucleocytoplasmic transport. The nucleolus assembles ribosomal subunits. The nuclear lamina provides mechanical support and organises heterochromatin.

Structure → Function

The double membrane allows the nucleus to maintain a different biochemical environment from the cytoplasm (e.g., distinct redox state, ion concentrations). The NPC allows selective, energy-dependent traffic. The lamina supports the envelope and tethers heterochromatin at the nuclear periphery. The nucleolus concentrates rRNA transcription and ribosome assembly. Histone modifications regulate chromatin accessibility in a heritable way (the epigenome).

Classification — Cell Cycle

The cell cycle proceeds through:

- **G₁** — growth; duration variable; sensitive to extracellular signals; **restriction point** in late G₁ commits the cell to division. Before the restriction point, the cell can exit to G₀ (quiescence); after, it proceeds to S phase regardless.
- **S** — DNA replication; each chromosome duplicated to form two sister chromatids.
- **G₂** — preparation for mitosis; DNA repair checkpoint; synthesis of mitotic proteins.
- **M** — mitosis (prophase, prometaphase, metaphase, anaphase, telophase, cytokinesis). Driven by cyclin B-CDK1 (MPF, maturation-promoting factor).

CDKs are activated by binding cyclins. Different cyclin-CDK pairs act at different phases: cyclin D-CDK4/6 (G₁), cyclin E-CDK2 (G₁/S), cyclin A-CDK2 (S), cyclin A-CDK1 (G₂/M), cyclin B-CDK1 (M). CDKs are inhibited by CKIs (p21, p27). Loss of restriction-point control (e.g., via Rb mutation or p16 loss) is a hallmark of cancer.

The mitotic spindle is built from microtubules nucleated by centrosomes at the two poles. Kinetochore microtubules attach to chromosomes at

kinetochores; polar microtubules interdigitate at the spindle midzone; astral microtubules anchor the spindle poles. Kinesin and dynein motors move chromosomes. Mitotic figures (recognisable as condensed chromosomes in cells lacking a nuclear envelope) are diagnostic of proliferation in pathology.

Meiosis produces gametes through two sequential divisions without intervening DNA synthesis, with crossing-over in prophase I and independent assortment at metaphase I. The result is four haploid cells with novel allele combinations.

Classification — Cell Death

- **Apoptosis** — programmed, non-inflammatory; cell shrinkage; chromatin condensation (pyknosis); nuclear fragmentation (karyorrhexis); formation of apoptotic bodies; phagocytosis without inflammation. Mediated by caspases (cysteine proteases cleaving at aspartate). Intrinsic (mitochondrial) pathway: cellular stress → cytochrome c release → apoptosome → caspase-9. Extrinsic (death receptor) pathway: FasL/Fas or TNF/TNFR1 → caspase-8. Both converge on executioner caspases (3, 6, 7).
- **Necrosis** — uncontrolled; cell swelling (oncosis); membrane rupture; karyolysis; inflammation. Types: coagulative (most solid organs, ischemia); liquefactive (brain, abscesses); caseous (TB); fat (pancreatitis); fibrinoid (vasculitis); gangrenous (limb ischemia ± infection).
- **Autophagy** — regulated self-digestion; phagophore → autophagosome → autolysosome; ATG genes; protective during nutrient deprivation (provides amino acids, energy); quality control (removes damaged organelles, misfolded proteins); defense (engulfing pathogens). LC3 is the key autophagosome marker. Insufficient autophagy contributes to neurodegeneration; excessive to cell death.

Compare & Distinguish

Euchromatin vs Heterochromatin

Feature	Euchromatin	Heterochromatin
Density	Loose	Dense
Transcription	Active	Inactive
Appearance	Pale	Dark, peripheral, around nucleolus
Histone acetylation	High	Low
Example	Active genes	Centromeres, Barr body

Apoptosis vs Necrosis

Feature	Apoptosis	Necrosis
Cell size	Shrinks	Swells

Feature	Apoptosis	Necrosis
Plasma membrane	Intact	Ruptures
Nucleus	Pyknosis, karyorrhexis	Karyolysis
Inflammation	No	Yes
Context	Programmed	Acute injury
Pathway	Caspases	None (passive)

Constitutive vs Facultative Heterochromatin

Feature	Constitutive	Facultative
Permanence	Always silent	Can be reactivated
Examples	Centromeres, telomeres, Barr body	Tissue-specific silenced genes

Function

The nucleus stores the genome, regulates transcription, replicates DNA in S phase, and protects DNA from cytoplasmic nucleases. The NPC controls bidirectional nucleocytoplasmic transport. The nucleolus assembles ribosomal subunits.

Structure → Function

The double membrane allows the nucleus to maintain a different biochemical environment from the cytoplasm (e.g., higher calcium, different redox state). The lamina supports the envelope and organises heterochromatin. NPCs allow selective traffic. The nucleolus concentrates rRNA transcription and ribosomal subunit assembly.

Classification — Nuclear Import and Export

- **NLS** (nuclear localisation signal) — short basic-rich sequence recognised by **importin**; importin-cargo translocates through NPC; Ran-GTP in nucleus causes cargo release.
- **NES** (nuclear export signal) — hydrophobic sequence recognised by **exportin**; exportin-cargo-Ran-GTP exits to cytoplasm; Ran-GTP hydrolysis releases cargo.
- Ran-GTP is high in the nucleus (Ran-GEF), Ran-GDP high in cytoplasm (Ran-GAP).

Classification — Cell Cycle

- **G₁** — growth, variable duration, sensitive to extracellular signals; **restriction point** in late G₁ commits the cell to division.
- **S** — DNA replication.
- **G₂** — preparation for mitosis, DNA repair.

- **M** — mitosis (prophase, prometaphase, metaphase, anaphase, telophase, cytokinesis), driven by cyclin B-CDK1.
Cyclin-CDK pairs: cyclin D-CDK4/6 (G₁), cyclin E-CDK2 (G₁/S), cyclin A-CDK2 (S), cyclin A-CDK1 and cyclin B-CDK1 (G₂/M). Inhibitors: p21, p27.

Classification — Cell Death

- **Apoptosis** — programmed; cell shrinkage, chromatin condensation (pyknosis), nuclear fragmentation (karyorrhexis), apoptotic bodies, phagocytosis without inflammation; mediated by caspases; intrinsic (mitochondrial) and extrinsic (death receptor) pathways.
- **Necrosis** — uncontrolled; cell swelling (oncosis), membrane rupture, karyolysis, inflammation. Types: coagulative (most solid organs), liquefactive (brain, abscess), caseous (TB), fat (pancreatitis), fibrinoid (vasculitis), gangrenous (limb).
- **Autophagy** — regulated self-digestion; phagophore → autophagosome → autolysosome; ATG genes; protective during nutrient deprivation, contributes to disease when dysregulated.

Compare & Distinguish

Euchromatin vs Heterochromatin

Feature	Euchromatin	Heterochromatin
Density	Loose	Dense
Transcription	Active	Inactive
Appearance	Pale	Dark, peripheral, around nucleolus

Apoptosis vs Necrosis

Feature	Apoptosis	Necrosis
Cell size	Shrinks	Swells
Nucleus	Pyknosis, karyorrhexis	Karyolysis
Plasma membrane	Intact	Ruptures
Inflammation	None	Yes
Context	Programmed	Acute injury

Recognition Logic

- **Small, dark, dense nucleus, no visible nucleolus** → quiescent (small lymphocyte).
- **Large, pale, open nucleus with prominent nucleolus** → active (plasma cell, hepatocyte).
- **Coarse clumped heterochromatin in "clock-face" pattern + perinuclear hof** → plasma cell.
- **Multilobed nucleus** → neutrophil (3–5 lobes), eosinophil (2 lobes).

- **Enlarged irregular nucleus with multiple prominent nucleoli and atypical mitoses** → malignant cell.
- **Condensed, fragmented, dark nucleus in eosinophilic cytoplasm** → apoptotic cell.

Clinical Correlation

Case — Melanoma In-Transit Metastasis

A 65-year-old man with a pigmented skin lesion. Biopsy shows nests of large cells with abundant cytoplasm, large nuclei with open chromatin, multiple prominent nucleoli, and atypical mitoses. The decisive features for malignancy are the nuclear enlargement, prominent nucleoli, and atypical mitoses — not just the pigment.

Case — Viral Cytopathic Effect

Hepatocytes with "ground-glass" nuclei are characteristic of HBV infection. CMV produces large "owl's eye" intranuclear inclusions with a clear halo. HSV produces eosinophilic Cowdry type A inclusions. Papillomavirus produces koilocytes with perinuclear halos and raisinoid nuclei.

Common Misconceptions

- Euchromatin = active gene and heterochromatin = inactive gene. Approximately true, but constitutive heterochromatin contains few genes regardless, and some facultative heterochromatin can be reactivated.
- All dark nuclei are heterochromatic. Apoptotic nuclei are densely stained but fragmented.
- The nuclear envelope is just a barrier. It is a regulated gateway — NPCs are selective, energy-consuming transporters.
- Necrosis is always pathological. Programmed necrosis (necroptosis) is a regulated form.

High-Yield Knowledge

Must Know

- Nuclear envelope = double membrane continuous with RER + NPCs + lamina.
- Euchromatin (pale, active) vs heterochromatin (dark, inactive).
- Nucleolus size reflects rRNA synthesis.
- Cell cycle: $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$; restriction point in late G_1 .
- Apoptosis is non-inflammatory; necrosis is inflammatory.

Must Distinguish

- Apoptosis vs necrosis vs autophagy.
- Constitutive vs facultative heterochromatin.

Must Understand

- Nuclear import/export depends on NLS/NES and Ran-GTP gradient.
- Loss of restriction-point control is a hallmark of cancer.

Integrated Summary

The nucleus is a regulated gateway whose envelope, pores, lamina, chromatin, and nucleolus together control transcription, replication, repair, and cell-cycle progression. Chromatin pattern (euchromatin vs heterochromatin) reflects transcriptional activity; the nucleolus is the visible signature of ribosome biogenesis. The cell cycle progresses through $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$, with the restriction point as the commitment decision. Cells die by apoptosis (programmed, non-inflammatory), necrosis (uncontrolled, inflammatory), or autophagy (regulated self-digestion).

Mastery Check

1. Reconstruct the Ran-GTP gradient and explain how it provides directionality for nuclear import vs export.
2. Explain why a plasma cell has an open nucleus with a prominent nucleolus, while a small lymphocyte has a dense nucleus without a nucleolus.
3. Compare the morphological features of apoptosis and necrosis, and explain the clinical significance of each.
4. Predict the nuclear appearance of: cardiac myocyte, small lymphocyte, plasma cell, malignant epithelial cell, apoptotic cell.

Advanced Concepts

Recognition Micro-Features

At higher magnification and EM, additional nuclear features become diagnostic:

- **Heterochromatin pattern** — facultative heterochromatin (X chromosome inactivation = Barr body, attached to nuclear envelope); constitutive heterochromatin (centromeres, telomeres). Barr body count = number of X chromosomes – 1; useful in sex-chromosome aneuploidies (Turner 0, Klinefelter 1).
- **Nuclear bodies** — Cajal bodies (coiled bodies, snRNP assembly, coilin scaffold); PML bodies (PML protein scaffold, role in DNA damage response); nuclear speckles (splicing factor storage).
- **Nuclear lamina** — lamin A/C meshwork underlying the inner nuclear membrane; mutations in LMNA cause progeria (Hutchinson-Gilford) and laminopathies (Emery-Dreifuss muscular dystrophy, dilated cardiomyopathy).
- **Nuclear pores** — ~100 nm, eight-fold symmetric; basket structure on nuclear side; FG-repeats in nucleoporins mediate selective transport of molecules >40 kDa (Ran-GTP gradient drives directionality).

Cell Cycle Regulatory Detail

The cell cycle is governed by cyclin-dependent kinases (CDKs) whose activity oscillates with cyclin levels:

- **G1/S transition** — cyclin D/CDK4-6 phosphorylates Rb → releases E2F → transcription of S-phase genes; cyclin E/CDK2 drives S-phase entry. Restriction point = commitment to S-phase.
- **S-phase** — cyclin A/CDK2 drives DNA replication.
- **G2/M transition** — cyclin B/CDK1 (MPF) drives mitotic entry; CDK1 activation = the commitment to mitosis.
- **Mitotic exit** — APC/C (anaphase-promoting complex) degrades cyclin B and securin → sister chromatid separation and mitotic exit.

Checkpoints monitor fidelity: G1 checkpoint (DNA damage → p53 → p21 → CDK2 inhibition → G1 arrest), intra-S checkpoint, G2/M checkpoint (DNA damage → ATR → Chk1 → CDK1 inhibition), spindle assembly checkpoint (kinetochore-microtubule attachment → separase activation).

p53 (the "guardian of the genome") is the most commonly mutated gene in human cancer (~50% of all cancers); p53 loss removes the G1 DNA-damage checkpoint, allowing cells with damaged DNA to proliferate. Li-Fraumeni syndrome (germline p53 mutation) predisposes to multiple cancers.

Apoptosis vs Necrosis vs Autophagy

Feature	Apoptosis	Necrosis	Autophagy
Trigger	Programmed, physiological/pathological	Pathological, acute injury	Stress, starvation
Cell size	Shrinks	Swells	Variable
Nucleus	Pyknosis, karyorrhexis	Karyolysis	Intact
Membrane	Blebs, apoptotic bodies	Rupture	Intact
Inflammatory	No	Yes	No
Mechanism	Caspase cascade	ATP failure → membrane rupture	Lysosomal degradation
Marker	Annexin V+, TUNEL+	HMGB1 release	LC3-II, p62

Apoptosis pathways:

- **Intrinsic (mitochondrial)** — stress (DNA damage, growth factor withdrawal) → Bax/Bak → cytochrome c release → apoptosome (Apaf-1 + caspase-9) → caspase-3/-7 activation.
- **Extrinsic (death receptor)** — FasL/FasR, TRAIL/TRAIL-R, TNF/TNFR1 → FADD → caspase-8 → caspase-3/-7.
- **Common pathway** — caspase-3/-7 execute: DNA fragmentation (CAD), cytoskeletal breakdown, blebbing, phagocytic uptake.

Necrosis is uncontrolled cell death from membrane rupture (inflammation, ATP depletion, complement). Coagulative necrosis (preserved architecture, e.g., MI), liquefactive necrosis (CNS, abscess), caseous necrosis (TB), fat necrosis (pancreatitis, breast trauma), gangrenous necrosis (limb ischaemia ± infection).

Autophagy ("self-eating") wraps organelles in autophagosomes that fuse with lysosomes; degrades long-lived proteins and damaged organelles; protective during nutrient stress; implicated in neurodegeneration, cancer, infection.

Transition

You now understand the command centre. The next chapter moves outward to the first tissue type built from cells — the epithelium, the boundary layer that separates the body from the outside world.

Rapid Review

- Interphase nucleus = envelope (double membrane + NPCs) + lamina (lamins) + chromatin (euchromatin/heterochromatin) + nucleolus (rRNA transcription).
- Nuclear import via NLS + importin + Ran-GTP; export via NES + exportin + Ran-GTP.
- Heterochromatin = condensed, inactive, dark. Euchromatin = open, active, pale.
- Nucleolus = fibrillar centre + dense fibrillar + granular component; size = rRNA synthesis rate.
- Cell cycle: $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$; cyclin-CDK pairs; restriction point in late G_1 .
- Apoptosis (programmed, non-inflammatory, caspase-mediated) vs necrosis (uncontrolled, inflammatory) vs autophagy (regulated self-digestion).

Chapter 4 — Epithelial Tissue

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 4

Opening Question

How does the body maintain a sharp internal/external boundary — the inside of the gut, the wall of a blood vessel, the lining of a renal tubule — while still permitting selective absorption, secretion, and protection?

Why This Matters

Epithelium is the body's first line of defence and principal site of exchange. Skin, gut, kidney tubules, glandular secretory cells, blood vessel linings — all are epithelium. Roughly 60% of all human cancers are carcinomas, malignant tumours of epithelial origin.

Learning Objectives

1. List the six defining features of epithelium and explain why each contributes to function.
2. Classify epithelia by number of cell layers and shape of surface cells.
3. Describe the apical, lateral, and basal surfaces of an epithelial cell.
4. Identify the major cell-cell junctions (tight, adherens, desmosome, gap, hemidesmosome) by structure, function, and location.
5. Describe the structure and functions of the basement membrane.
6. Distinguish exocrine from endocrine glands and classify exocrine glands.
7. Recognise surface specialisations (microvilli, stereocilia, cilia, keratin) and predict function.

Big Picture

- **Polarity** — every epithelial cell has an apical surface (lumen) and a basal surface (basement membrane).
- **Cell-cell junctions** — tight (seal) → adherens (adhesion, actin) → desmosome (adhesion, intermediate filament) → gap (communication) → hemidesmosome (basal, to BM, intermediate filament).
- **Specialisations** — microvilli (absorption), cilia (movement), stereocilia (sensory/absorption), keratin (protection).
- **Glands** — exocrine (duct to surface) vs endocrine (no duct, hormone to blood); classified by morphology and secretion.

Core Concept

Epithelium is a polarised sheet of cells attached to a basement membrane, with apical specialisations for specific functions, lateral junctions for adhesion and communication, and basal specialisations for attachment and vectorial transport.

Structure

Defining Features

1. Cellularity — almost entirely cells, little ECM.
2. Polarity — free (apical) and basal surfaces with different lipids and proteins.
3. Attachment to a basement membrane — every epithelial cell rests on a basal lamina.
4. Avascularity — no blood vessels; nutrients diffuse from underlying CT.
5. Innervation — rich free nerve endings.
6. High regenerative capacity — constant shedding and replacement by stem cells.

Why Polarity Matters

Polarity is not a cosmetic feature. The apical and basolateral membranes have different lipid compositions, different protein compositions, and different cytoskeletal connections. Tight junctions separate them, fencing membrane proteins to one domain. This allows the apical membrane to face the lumen (low-pH gastric juice, hyperosmolar renal medulla, abrasive food) while the basolateral membrane faces the connective tissue (physiological pH, isotonic fluid, nurturing capillaries). The polarised distribution is what makes vectorial transport possible.

Why Avascularity Matters

Avascularity limits thickness but creates an opportunity. Because the epithelium itself has no vessels, anything that crosses the epithelium must pass through cells (transcellular) or between cells (paracellular). Tight junctions regulate the paracellular pathway; transporters regulate the transcellular pathway. An avascular tissue is also a tissue that can be rapidly replaced — stem cells in the basal layer divide, progeny migrate upward, terminally differentiated cells are shed. The gut epithelium turns over every 3–5 days; the epidermis every 4–6 weeks.

Why Junctions Matter

The junctional complex (apical to basal: tight → adherens → desmosome → gap) is not redundancy; it is layered function. Tight junctions seal and fence. Adherens junctions provide a continuous adhesion belt that holds the sheet together mechanically. Desmosomes provide spot strong adhesion that resists shear (especially in stratified squamous epithelia). Gap junctions allow cell-cell communication. Hemidesmosomes anchor the basal surface to the BM. Loss of any junction produces a specific disease: pemphigus (desmosome), pemphigoid (hemidesmosome), diarrhoea due to defective tight junctions.

Apical Surface Specialisations

- **Microvilli** — actin-cored projections that increase absorptive surface area; dense regular microvilli form the brush border (intestine) or

- striated border (proximal renal tubule).
- **Stereocilia** — long actin-filled microvilli (not true cilia); epididymis (absorption), hair cells of inner ear (mechanosensation).
- **Cilia** — 9+2 microtubule axoneme, motile; respiratory tract, fallopian tube. Primary cilia (9+0) are sensory antennae.
- **Keratin** — intracellular fibrous protein in keratinised stratified squamous epithelium.

Lateral Junctions (apical to basal)

- **Tight junction (zonula occludens)** — claudins + occludins, seal paracellular space, fence (separate apical from basolateral membrane proteins).
- **Adherens junction (zonula adherens)** — E-cadherin + catenins + vinculin, linked to actin, adhesion belt.
- **Desmosome (macula adherens)** — desmoglein/desmocollin (desmosomal cadherins) + plakoglobin + desmoplakin, linked to keratin intermediate filaments, spot adhesion.
- **Gap junction** — connexons (connexin), allow ion and small-molecule passage between cells.
- **Hemidesmosome** — basal; integrin $\alpha 6 \beta 4$ + laminin-332 + BP180/230 + plectin, linked to keratin, anchors cell to basement membrane.

Junction Detail

Tight junction claudins (the principal barrier proteins) form paired strands that bind across the intercellular space, sealing the paracellular pathway. ZO-1/ZO-2/ZO-3 anchor the strands to the actin cytoskeleton. Different claudins confer different paracellular selectivity: claudin-2 in the proximal renal tubule is leaky (high paracellular Na^+ flux); claudin-16 in the thick ascending limb mediates paracellular Mg^{2+} reabsorption. Mutations in claudin-14 cause familial hypomagnesaemia.

Adherens junction β -catenin links E-cadherin to the actin cytoskeleton; α -catenin links the complex to actin. Adherens junctions mediate contact inhibition of proliferation; β -catenin released from the junction can enter the nucleus and activate Wnt target genes when the junction is destabilised. APC mutations in colorectal cancer disrupt β -catenin regulation.

Desmosome plakoglobin and desmoplakin connect the cytoplasmic tail of desmogleins/desmocollins to keratin IFs. Desmosomes are abundant in stratified squamous epithelia (stratum spinosum) and cardiac muscle (intercalated disc). Pemphigus vulgaris = autoantibodies against desmoglein 3 $\rightarrow$ intraepidermal split (acantholysis).

Gap junction connexons formed by connexins (Cx26, Cx32, Cx43). In heart, gap junctions allow action potentials to spread cell-to-cell. Mutations in Cx26 cause the most common form of autosomal recessive deafness (Cx26 is required for cochlear K^+ recycling).

Hemidesmosome integrin $\alpha 6 \beta 4$ anchors the cell to laminin-332 in the lamina lucida; BP180 (collagen XVII) and BP230 connect the integrin to

keratin IFs. Bullous pemphigoid = autoantibodies against BP180 → subepidermal split.

Basement Membrane

LM term: basement membrane = basal lamina (epithelial product) + reticular lamina (fibroblast product).

EM term: basal lamina = lamina lucida (electron-lucent, integrin/laminin extracellular domains) + lamina densa (electron-dense, type IV collagen network + laminin network + nidogen/entactin bridges + perlecan heparan sulfate proteoglycan).

Below: lamina fibroreticularis (type III collagen) with anchoring fibrils (type VII collagen).

Functions: structural support, selective filter (glomerular filtration barrier), barrier to tumour invasion (in situ vs invasive carcinoma), signals to epithelial cell (integrin-mediated).

Basement Membrane Detail

Components:

- **Type IV collagen** — forms a sheet network via NC1 domain interactions, rather than fibrils; provides the scaffold. The $\alpha3\alpha4\alpha5$ isoforms are specialised in the glomerular basement membrane.
- **Laminin** — forms a separate network; binds integrins ($\alpha6\beta4$ in hemidesmosomes, $\alpha3\beta1$ in focal adhesions) and type IV collagen; provides adhesion and signalling.
- **Nidogen/entactin** — bridges laminin and type IV collagen networks.
- **Perlecan** (heparan sulfate proteoglycan) — hydrated gel; provides the size- and charge-selective filter (glomerulus).
- **Type VII collagen** — anchoring fibrils extending from lamina densa into the CT.

Clinical correlations: Alport syndrome (mutations in COL4A3/4/5) → hereditary nephritis with hearing loss; Goodpasture syndrome (anti-GBM $\alpha3$ antibodies) → pulmonary haemorrhage + glomerulonephritis. Epidermolysis bullosa: junctional (laminin-332), dystrophic (type VII collagen), simplex (keratin K5/K14).

Classification

By number of layers × shape of surface cell:

- **Simple** (one layer) — transport (absorption, secretion, filtration).
- **Stratified** (≥ 2 layers) — protection.
- **Pseudostratified** (one layer, all touch BM, nuclei at different heights) — often ciliated.

Shapes: squamous (flat), cuboidal (height = width, central round nucleus), columnar (tall, basal oval nucleus), transitional (urothelium, dome-shaped at surface).

Type	Example	Function
Simple squamous	Endothelium, mesothelium, alveoli	Diffusion
Simple cuboidal	Renal tubules, thyroid follicles, small ducts	Secretion/absorption
Simple columnar	Stomach, intestine, gallbladder	Secretion/absorption
Pseudostratified ciliated columnar	Respiratory tract, epididymis	Propulsion/absorption
Stratified squamous (keratinized)	Epidermis	Protection
Stratified squamous (non-keratinized)	Esophagus, vagina, oral cavity	Protection
Stratified cuboidal	Large gland ducts	Protection
Stratified columnar	Conjunctiva	Protection/secretion
Transitional (urothelium)	Urinary tract	Stretch + barrier

Function

Epithelia cover, line, secrete, absorb, filter, and protect. The avascularity and tight junctions create a barrier; polarity creates directional transport; the basement membrane anchors and filters; regeneration handles mechanical/chemical assault.

Structure → Function

- **Polarity** allows directional transport (Na⁺-glucose cotransporter apical, GLUT2 basolateral, Na⁺/K⁺-ATPase basolateral).
- **Tight junctions** create the paracellular barrier and maintain polarity by fencing membrane proteins.
- **Basement membrane** is a size- and charge-selective filter (glomerular filtration barrier retains albumin).
- **Avascularity** keeps epithelium thin enough for diffusion.
- **Desmosomes** and **hemidesmosomes** provide mechanical resistance (epidermis, cardiac muscle).

Glands

- **Exocrine** — duct to surface; classified by morphology (tubular/acinar/tubuloacinar), branching (simple/compound), secretion (serous/mucous/mixed), mode (merocrine/apocrine/holocrine).
- **Endocrine** — no duct; secretion to blood; arranged in cords/follicles surrounded by fenestrated capillaries.

Exocrine mode of secretion:

- Merocrine (eccrine) — exocytosis; most glands.
- Apocrine — apical cytoplasm shed; mammary, axillary sweat, prostate.
- Holocrine — whole cell disintegrates; sebaceous.

Compare & Distinguish

Microvilli vs Cilia

Feature	Microvilli	Cilia
Core	Actin	9+2 microtubules
Length	~1 µm	~5–10 µm
Motile	No	Yes
Function	Increase surface area	Move fluid

Stereocilia vs True Cilia

Stereocilia are long actin-filled microvilli; not motile; not true cilia.

Simple Columnar vs Pseudostratified Columnar

Both have apical-basal polarity. Simple: all nuclei basal, single row. Pseudostratified: all cells touch BM but nuclei at different heights; often ciliated.

Recognition Logic

- **Single layer of flat cells** → simple squamous (endothelium).
- **Single layer of cuboidal cells with central round nuclei** → simple cuboidal (renal tubule, thyroid follicle).
- **Single layer of tall cells with apical brush border + goblet cells** → simple columnar (intestine).
- **Single layer, nuclei at varying heights, apical cilia + goblet cells** → pseudostratified ciliated columnar (respiratory tract).
- **Multiple layers, surface flat, may be keratinised** → stratified squamous.
- **Multiple layers, surface dome-shaped, in distensible organ** → transitional (urothelium).

Clinical Correlation

Case — Barrett's Esophagus

A 35-year-old woman with chronic heartburn. Biopsy of the distal esophagus shows simple columnar epithelium with goblet cells where stratified squamous should be. Diagnosis: Barrett's esophagus. The columnar epithelium with goblet cells is intestinal metaplasia caused by chronic acid reflux. The clinical significance is an increased risk of esophageal adenocarcinoma.

Lesson. Metaplasia is a histological adaptation to chronic stress; it protects but predisposes to malignancy.

Case — Pemphigus Vulgaris vs Bullous Pemphigoid

Both produce blistering skin disease. In **pemphigus vulgaris**, autoantibodies against desmoglein disrupt desmosomes → intraepidermal split (acantholysis) → superficial blister. In **bullous pemphigoid**, autoantibodies against BP180 disrupt hemidesmosomes → subepidermal

split → tense blister.

Lesson. The level of the split equals the molecule targeted. Desmosome = intraepidermal blister; hemidesmosome = subepidermal blister.

Common Misconceptions

- Stratified means many cells of the same shape. Surface-cell shape gives the name; basal cells in stratified squamous are cuboidal/columnar.
- Pseudostratified means multilayered. All cells touch BM; only the nuclei are at different heights.
- Microvilli and cilia are the same thing. Actin core vs microtubule core; absorption vs movement.
- The basement membrane is one thing. Multiple sublayers; signalling + anchoring + filtering function.
- Transitional epithelium is stratified squamous. It is a distinct type found only in the urinary tract.

High-Yield Knowledge

Must Know

- Simple = transport; stratified = protection; pseudostratified = propulsion.
- Apical → basal junctions: tight → adherens → desmosome → gap → hemidesmosome.
- Basement membrane = basal lamina (epithelial) + reticular lamina (CT); EM: lamina lucida + lamina densa.
- Microvilli = actin core, absorption; cilia = 9+2 microtubule, movement.

Must Distinguish

- Simple columnar vs pseudostratified columnar (nuclear position).
- Microvilli vs stereocilia vs cilia.
- Merocrine vs apocrine vs holocrine secretion.
- Desmosome vs hemidesmosome (cell-cell vs cell-BM).

Must Understand

- Tight junctions seal and fence.
- Basement membrane breach = in situ → invasive carcinoma.
- Polarity enables directional transport.

Integrated Summary

Epithelium is a polarised sheet of cells attached to a basement membrane, with apical specialisations for specific functions, lateral junctions for adhesion and communication, and basal specialisations for attachment and vectorial transport. The six defining features (cellularity, polarity, BM attachment, avascularity, innervation, regeneration) make epithelium ideal for covering, lining, and secreting. The breach of the basement membrane is the critical step between carcinoma in situ and invasive carcinoma.

Mastery Check

1. Compare and contrast the five junctions by structure, cytoskeletal linkage, and function.
2. Explain the role of polarity in glucose absorption by the small intestine.
3. Describe how the basement membrane breach converts in situ to invasive carcinoma.
4. Distinguish the histological appearance of microvilli, stereocilia, and cilia.
5. Reconstruct the levels at which pemphigus and pemphigoid split the epithelium and explain why.

Advanced Concepts

Recognition Micro-Features

At higher magnification, the type of epithelium is diagnostic:

- **Basement membrane thickness** — ~50 nm in routine H&E, but thicker in PAS stain; thickening of BM is a feature of diabetes (capillary BM), bullous pemphigoid (subepidermal split above thickened BM), and membranous glomerulonephritis (subepithelial immune deposits).
- **Goblet cells** — distended apical cytoplasm with clear/pale mucin (mucinogen), basal compressed nucleus. Number increases from duodenum to colon (most numerous in colon). Goblet cell metaplasia in respiratory epithelium → chronic bronchitis, smoker's airway; goblet cells in Barrett's esophagus define intestinal metaplasia.
- **Brush border vs striated border** — intestinal microvilli form the brush border (well-defined, ~1-2 µm thick, eosinophilic, contains brush-border enzymes); kidney proximal tubule has a similar but taller striated border.
- **Cilia** — apical eosinophilic fringe in ciliated epithelium; 9+2 axoneme, basal bodies; in the respiratory tract, ciliary dyskinesia (primary ciliary dyskinesia, Kartagener syndrome) → immotile cilia → recurrent sinopulmonary infections, situs inversus.
- **Stratified squamous keratinised vs non-keratinised** — keratinised has anucleate surface layer (skin, keratinised portion of anal canal above pectinate line); non-keratinised retains nuclei in surface layer (esophagus, vagina, oral cavity).

Epithelial Renewal

Epithelia are among the most rapidly renewing tissues in the body:

- **Intestinal epithelium** — 3-5 days from crypt stem cell to villus tip shedding.
- **Skin epidermis** — 4-6 weeks from basal layer to stratum corneum desquamation.
- **Respiratory epithelium** — turnover ~1 month.

- **Liver hepatocytes** — long-lived but can regenerate: 70% hepatectomy → full mass restoration in ~2 weeks via hepatocyte proliferation.

Stem cells reside in specific niches:

- **Intestinal crypts** — Lgr5+ stem cells at the crypt base; Paneth cells provide Wnt, EGF, Notch signals.
- **Epidermal basal layer** — interfollicular stem cells; hair follicle bulge stem cells (K15+) contribute to wound re-epithelialisation.
- **Corneal limbus** — limbal stem cells; deficiency → corneal neovascularisation and opacification.

Common Misconceptions Extended

- All epithelia are avascular. Stratified squamous of the cornea is avascular; most epithelia rest on a vascularised basement membrane.
- Transitional epithelium is "transitional" because it changes shape. It is transitional because it was thought to be intermediate between stratified squamous and stratified columnar; it is specialised urothelium, not a "transition" between other types.
- Pseudostratified means multiple layers. A single layer with nuclei at different heights (every cell touches the basement membrane).
- Microvilli = cilia. Microvilli = membrane protrusions (actin core); cilia = membrane-bound organelles (microtubule core). Different ultrastructure, different motility, different functions.
- Squamous cells are only on surfaces. Mesothelium is also simple squamous (peritoneum, pleura, pericardium); mesothelial cells are squamous but have microvilli on their free surface (unique).

Transition

Epithelium is the body's boundary. Connective tissue, the subject of the next chapter, is the body's structural and metabolic support — the tissue that binds, supports, defends, and stores.

Rapid Review

- Epithelium = polarised sheet on BM, avascular, with high regeneration.
- Classification: simple/stratified/pseudostratified × squamous/cuboidal/columnar (+ transitional in urinary tract).
- Apical: microvilli (actin, absorption), cilia (9+2 microtubule, movement), stereocilia (long microvilli), keratin (protection).
- Lateral junctions: tight (claudins, seal + fence), adherens (E-cadherin + actin), desmosome (desmoglein + keratin IF), gap (connexin), hemidesmosome (integrin $\alpha 6\beta 4$ + laminin-332 + keratin IF).
- Basement membrane: BM = basal lamina + reticular lamina; EM: lamina lucida + lamina densa + lamina fibroreticularis.
- Glands: exocrine (duct) vs endocrine (bloodstream); merocrine vs apocrine vs holocrine.

Chapter 5 — Connective Tissue

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 5

Opening Question

How does the body build tissues that bind organs together, support them, defend them, store energy, and transport nutrients and waste — and how does it do this with relatively few cells, by making the cells' products (collagen, elastin, ground substance) the functional element?

Why This Matters

Connective tissue is the most abundant and most diverse of the four basic tissue types. It binds organs (loose CT), supports epithelia (BM + underlying CT), transmits forces (dense regular CT of tendons), and circulates cells and molecules (blood). Many diseases directly affect connective tissue — fibrosis, scurvy, Marfan syndrome, Ehlers-Danlos, osteogenesis imperfecta.

> **Note on scope.** Per the official 16-chapter syllabus, cartilage (Junqueira Ch 7) and bone (Junqueira Ch 8) are not in the core. Cartilage is type II collagen-rich specialised CT; bone is type I collagen + hydroxyapatite specialised CT. They share the same cell-matrix principle taught here.

Learning Objectives

1. List the structural components of connective tissue and explain the function of each.
2. Describe the structure, synthesis, and function of collagen types I-IV.
3. Explain the composition and function of the ground substance.
4. Identify the major connective-tissue cell types and their functions.
5. Classify connective tissues by composition and architecture.
6. Distinguish connective tissue proper from blood, cartilage, and bone.
7. Recognise the histological appearance of the major connective-tissue types.

Big Picture

- **Three components** — cells + fibers + ground substance. Matrix, not cells, is the functional element.
- **Fibers** — collagen (type I, II, III, IV by location), elastic (elastin + fibrillin-1), reticular (type III + glycoprotein coating).
- **Ground substance** — GAGs + proteoglycans + glycoproteins; hydrated gel that resists compression.
- **Cells** — fibroblast (builder), macrophage (defense), mast cell (allergy), plasma cell (antibody), adipocyte, mesenchymal stem cell.

Core Concept

Connective tissue is the most abundant and most diverse tissue type, defined by an abundant extracellular matrix produced and maintained by relatively few cells. The matrix has three components — fibers, ground substance, and water — and its composition determines every disease.

Structure

Fibers

Collagen — most abundant protein in the body (~30% of total protein). Twenty-eight types identified; four clinically most important:

Type	Distribution	Function	Disease
I	Dermis, bone, tendon, dentin, capsule, sclera	Tensile strength	OI, EDS (arthrochalasia)
II	Hyaline/elastic cartilage matrix, vitreous, nucleus pulposus	Compression resistance	Achondrogenesis
III	Reticular: lymphoid stroma, around vessels, granulation tissue, uterus	Delicate stroma	Vascular EDS
IV	Basal lamina (sheet-forming, not fibrillar)	Filtration, BM scaffold	Goodpasture, Alport

Collagen I synthesis — gene → preprocollagen (RER) → hydroxylation of proline/lysine (requires vitamin C as cofactor for prolyl/lysyl hydroxylase) → glycosylation → triple helix → procollagen → exocytosis → cleavage by procollagen peptidases → tropocollagen → cross-linking by **lysyl oxidase** (copper-dependent) → collagen fibril → collagen fiber.

Light microscopy: collagen I is pink (eosinophilic). Masson trichrome: collagen blue/green. Reticular (type III) is not visible by H&E; demonstrated by silver (argyrophilic). Type IV is in the BM, demonstrated by PAS or silver.

Elastic fibers — elastin core (desmosine + isodesmosine cross-links unique to elastin) + fibrillin-1 microfibril scaffold. Stained by orcein, Verhoeff-van Gieson, aldehyde fuchsin. Found in skin (dermis), large elastic arteries, lung, ligamenta flava, vocal ligaments.

Marfan syndrome: FBN1 mutation → fibrillin-1 defect → defective elastic recoil + dysregulated TGF- β signalling → aortic aneurysm, lens dislocation, skeletal overgrowth.

Reticular fibers — type III collagen + glycoprotein/proteoglycan coating. Argyrophilic. Found in lymphoid organs (lymph node, spleen, thymus, tonsil), liver stroma, around vessels, in early wound granulation tissue.

Ground Substance

GAGs (glycosaminoglycans) — long unbranched disaccharide chains, usually sulfated or carboxylated (negative charge). Major GAGs: hyaluronic acid (hyaluronan, non-sulfated, not protein-bound, high MW), chondroitin sulfate, dermatan sulfate, keratan sulfate, heparan sulfate.

Proteoglycans — GAGs attached to a core protein. Examples: aggrecan (cartilage, binds hyaluronic acid → huge aggregates), versican, decorin, biglycan, perlecan (BM).

Glycoproteins — fibronectin (binds integrins + collagen), laminin (binds integrins + type IV collagen; found in BM).

Function: hydrated gel resists compression; binds and stores growth factors (TGF- β , FGF); medium for diffusion of nutrients/waste/signals.

Ground Substance Detail

Hyaluronic acid (hyaluronan) is the simplest GAG — non-sulfated, not covalently linked to protein, very high MW (up to millions of daltons). It is synthesised by hyaluronan synthases on the plasma membrane and extruded directly into the ECM. Hyaluronan is highly hydrated; in CT it creates the gel that allows diffusion. In cartilage (out of core) it forms the backbone of large aggregates with aggrecan.

The sulfated GAGs (chondroitin sulfate, dermatan sulfate, keratan sulfate, heparan sulfate) are attached to core proteins to form proteoglycans. Aggrecan binds hyaluronic acid via link proteins to form huge aggregates that hydrate cartilage matrix. Decorin and biglycan bind TGF- β and regulate its availability. Perlecan in the BM is essential for the size- and charge-selective filtration in the glomerulus; mutations cause Pierson syndrome.

Fibronectin is a dimeric glycoprotein that binds integrins ($\alpha 5 \beta 1$ in fibroblasts) and collagen; it is essential for cell migration during wound healing. Laminin-1 is the principal laminin of epithelial BMs; laminin-332 (laminin-5) is specific to hemidesmosomes. Tenascin and thrombospondin are matricellular proteins that regulate cell-matrix interactions during development and wound healing.

Cells

- **Fibroblast** — principal cell; spindle-shaped, euchromatic nucleus, abundant RER (basophilic); synthesises collagen, elastin, proteoglycans, glycoproteins. Quiescent form (fibrocyte) has less RER and more heterochromatin.
- **Macrophage** — monocyte-derived; phagocytic, antigen-presenting, cytokine-secreting; large vacuolated cytoplasm, kidney-shaped nucleus; CD68+ by IHC.
- **Mast cell** — round, central nucleus, basophilic metachromatic granules (toluidine blue purple); granules contain heparin, histamine, tryptase, chymase; degranulates on IgE crosslinking → type I hypersensitivity.
- **Plasma cell** — oval, eccentric "clock-face" heterochromatic nucleus, basophilic cytoplasm (RER for Ig), perinuclear hof (Golgi).

- **Lymphocyte** — small, round, dense heterochromatic nucleus, scant cytoplasm.
- **Adipocyte** — see Chapter 6.
- **Mesenchymal stem cell** — multipotent, perivascular; can become fibroblast, adipocyte, chondrocyte, osteoblast.

Cell Detail

Fibroblasts are activated by TGF- β released from platelets and macrophages in wound healing; activated fibroblasts (myofibroblasts) express α -smooth muscle actin and contract to close wounds. Excessive activation $\rightarrow$ fibrosis (scleroderma, keloid, pulmonary fibrosis, hepatic cirrhosis).

Macrophages polarise into M1 (classically activated by IFN- γ + LPS) or M2 (alternatively activated by IL-4/IL-13). M1 macrophages are pro-inflammatory, microbicidal; M2 macrophages are anti-inflammatory, tissue-repairing. Tumour-associated macrophages are often M2-polarised and support tumour growth. CD68 is the pan-macrophage marker; CD163 marks M2-polarised macrophages.

Mast cells are tissue-resident (mucosal mast cells vs connective-tissue mast cells, with different granule contents and cytokine profiles). Upon activation, they release preformed mediators (histamine, heparin, tryptase, chymase) within minutes, and synthesised mediators (leukotriene C4, prostaglandin D2, IL-4, IL-5) over hours. Mastocytosis is a clonal proliferation of mast cells with elevated serum tryptase.

Plasma cells secrete large amounts of immunoglobulin (up to thousands of antibodies per second per cell). They have a well-developed RER (basophilic cytoplasm) and a large Golgi (perinuclear hof, where the immunoglobulin is glycosylated). Plasma cells do not divide; they are terminally differentiated. Multiple myeloma is a malignant proliferation of plasma cells in the bone marrow with monoclonal immunoglobulin (M-protein) in serum and free light chains (Bence-Jones proteins) in urine.

Mesenchymal stem cells (MSCs) reside in perivascular niches in many tissues. They are multipotent (can become fibroblasts, adipocytes, chondrocytes, osteoblasts) and are being investigated for regenerative medicine applications. They are identified by surface markers (CD73, CD90, CD105) and the ability to differentiate into the three lineages in vitro.

Function

Connective tissue binds, supports, defends, stores energy, and transports. The matrix is load-bearing, space-filling, and signal-carrying. Its molecular composition determines every disease of the tissue.

Structure $\rightarrow$ Function

- **Type I collagen** in parallel arrays = tendon (one-direction tension); in meshwork = dermis (multidirectional tension).

- **Elastic fibers** = recoil after stretch (skin, large arteries).
- **Reticular fibers** = delicate stromal scaffold (lymphoid organs, liver).
- **Hyaluronic acid hydration** = compression resistance (cartilage).
- **GAG negative charge** = binds cations and water; resists compression.
- **Heparan sulfate in BM** = size- and charge-selective filter.

Classification

Connective Tissue Proper

- **Loose (areolar)** — abundant ground substance, moderate fibers (mostly type I), variety of cells; found in subcutaneous tissue, lamina propria, around vessels.
- **Dense irregular** — thick interwoven collagen bundles, few cells; dermis, organ capsules, periosteum, dura.
- **Dense regular** — parallel collagen bundles, compressed fibroblasts; tendons, ligaments, aponeuroses.
- **Elastic** — dominated by elastic fibers; ligamenta flava, vocal ligaments, elastic arteries.
- **Reticular** — reticular fibers dominant; lymphoid stroma.
- **Mucous** — abundant ground substance, few fibers; Wharton's jelly of umbilical cord.

Wound Healing — A Worked Example of CT Dynamics

Wound healing illustrates the CT principle: cells + matrix + signals in dynamic balance.

1. **Haemostasis** (minutes) — platelet plug + fibrin clot; platelets release TGF- β , PDGF, FGF.
2. **Inflammation** (hours to days) — neutrophils (24–48 h), then macrophages (day 2–3); macrophages phagocytose debris and release cytokines.
3. **Proliferation/granulation tissue** (days to weeks) — fibroblasts migrate into the wound along fibrin scaffold, proliferate, and produce type III collagen (granulation tissue is rich in type III). Macrophages continue to drive the process via VEGF (angiogenesis) and TGF- β (fibroblast activation). Myofibroblasts contract the wound.
4. **Remodelling** (weeks to months) — type III collagen is replaced by type I; wound contracts; tensile strength increases but never reaches 100% of unwounded tissue (~80%).

Excessive healing → keloid (scar extends beyond original wound, exuberant collagen deposition) or hypertrophic scar (raised but within wound boundaries). Defective healing → chronic wound (diabetic foot ulcer, pressure ulcer). The molecular basis of fibrosis is sustained TGF- β signalling and myofibroblast persistence.

Specialized Connective Tissues (out of core, brief mention)

Cartilage (J7) and bone (J8) are specialised connective tissues with type II / type I collagen respectively and are not covered in core per the official 16-chapter syllabus. The cell-matrix principle is the same: cells produce

matrix, matrix determines function.

Blood is fluid CT (Chapter 9).

Compare & Distinguish

Collagen I vs III vs IV

Type	Form	Stain	Disease
I	Fibril	Eosin (H&E), blue/green (trichrome)	OI, EDS
III	Fibril	Silver (argyrophilic)	Vascular EDS
IV	Sheet network	PAS, silver	Goodpasture, Alport

Mast Cell vs Plasma Cell

Feature	Mast cell	Plasma cell
Granules	Basophilic, metachromatic	None (RER)
Nucleus	Central, round	Eccentric, clock-face
Cytoplasm	Granular	Basophilic with perinuclear hof
Function	Allergy (histamine)	Antibody secretion

Recognition Logic

- **Pink wavy collagen bundles in parallel, compressed fibroblasts between** → dense regular CT (tendon).
- **Thick interwoven pink collagen, few cells** → dense irregular CT (dermis).
- **Loose meshwork of pink fibers with abundant cells** → loose CT.
- **Black network on silver stain** → reticular CT (lymph node stroma).
- **Metachromatic purple granules** → mast cells.
- **Clock-face nucleus with basophilic cytoplasm + perinuclear hof** → plasma cell.

Clinical Correlation

Case — Ehlers-Danlos Syndrome (Vascular Type)

A 25-year-old woman presents with easy bruising, joint hypermobility, and translucent skin. She has a family history of arterial rupture. Diagnosis: vascular EDS. The mutation is in COL3A1 (type III collagen). Histologically, the dermis shows sparse, disorganised collagen bundles and reduced reticular network.

Lesson. The disease names the fiber type: vascular EDS = type III; classical EDS with hypermobile joints = type I/V; arthrochalasia EDS = type I.

Case — Scurvy

A 60-year-old man with alcoholism and poor diet presents with bleeding gums, perifollicular haemorrhages, and poor wound healing. Diagnosis: vitamin C deficiency → impaired hydroxylation of proline and lysine in collagen → defective collagen cross-linking.

Lesson. Vitamin C is the cofactor for prolyl and lysyl hydroxylase; without it, collagen is synthesised but improperly folded.

Case — Marfan Syndrome

A 20-year-old man with tall stature, arachnodactyly, and lens dislocation dies suddenly of aortic dissection. Histology of the aorta shows fragmented, disorganised elastic fibers. Diagnosis: Marfan syndrome (FBN1 mutation → defective fibrillin-1).

Lesson. The disease is in the elastic scaffold, not the collagen.

Common Misconceptions

- Collagen is one protein. Twenty-eight types; I, II, III, IV most common.
- Elastic fibers are made of elastin only. Elastin core + fibrillin-1 microfibril scaffold.
- Reticular fibers are different from collagen. Type III collagen with glycoprotein/proteoglycan coating.
- Fibroblasts are the only cells in CT. Macrophages, mast cells, plasma cells, adipocytes, mesenchymal stem cells are also resident.

High-Yield Knowledge

Must Know

- Three components of CT: cells + fibers + ground substance.
- Collagen I (tensile), III (reticular), IV (BM network).
- Vitamin C required for collagen hydroxylation.
- Copper required for lysyl oxidase.
- Fibrillin-1 scaffold + elastin core in elastic fibers; FBN1 mutation in Marfan.

Must Distinguish

- Collagen I vs III vs IV (form, location, disease).
- Mast cell vs plasma cell (granules, nucleus position).
- Dense regular vs dense irregular CT (fiber orientation).

Must Understand

- Matrix composition determines disease.
- GAG negative charge binds water; compression resistance.
- Type IV collagen is sheet-forming (BM), not fibrillar.

Integrated Summary

Connective tissue is the most abundant and most diverse of the four basic tissue types, defined by an abundant extracellular matrix. The matrix has

three components (fibers, ground substance, water); its composition determines every disease. Collagen I is the principal fibrillar collagen (tendon, dermis, bone); type III is reticular; type IV is sheet-forming in the BM. Elastic fibers are elastin core + fibrillin-1 scaffold. Ground substance is GAGs + proteoglycans + glycoproteins in a hydrated gel. CT cells include fibroblasts, macrophages, mast cells, plasma cells, adipocytes, and mesenchymal stem cells. Cartilage and bone are specialised connective tissues out of core per the official 16-chapter syllabus.

Mastery Check

1. Reconstruct type I collagen synthesis from gene to fiber, identifying key enzymes, cofactors, and clinical correlations.
2. Compare and contrast collagen I, III, and IV by form, location, and disease.
3. Explain why scurvy is a disease of collagen and what the clinical features are.
4. Distinguish loose CT, dense regular CT, dense irregular CT, and reticular CT histologically.
5. Why does the matrix, not the cells, determine CT function and disease?

Advanced Concepts

Collagen Synthesis Pathway Detail

Collagen synthesis is a multistep process that explains collagen diseases:

1. **Pre-procollagen synthesis** (RER) — signal peptide directs nascent polypeptide into ER; pro- α chains assemble into triple helix (procollagen) with N- and C-terminal propeptides.
 2. **Hydroxylation** (RER, requires vitamin C) — proline and lysine residues hydroxylated by prolyl/lysyl hydroxylases; vitamin C deficiency $\rightarrow$ hydroxyproline deficiency $\rightarrow$ triple helix unstable $\rightarrow$ scurvy (bleeding gums, poor wound healing, perifollicular haemorrhages).
 3. **Glycosylation** (RER, Golgi) — hydroxylysine residues glycosylated.
 4. **Triple helix formation** (Golgi) — procollagen triple helix with retained propeptides.
 5. **Exocytosis** — procollagen secreted.
 6. **Cleavage of propeptides** (extracellular) — procollagen peptidases cleave N- and C-terminal propeptides $\rightarrow$ insoluble tropocollagen.
 7. **Cross-linking** (extracellular, lysyl oxidase, requires Cu^{2+}) — lysine and hydroxylysine residues oxidatively deaminated $\rightarrow$ allysine $\rightarrow$ cross-links with adjacent tropocollagen molecules $\rightarrow$ collagen fibril.
- Menkes disease (Cu^{2+} transport defect) $\rightarrow$ defective cross-linking $\rightarrow$ kinky hair, brittle bones.
 - Ehlers-Danlos (various, e.g., lysyl hydroxylase deficiency in type VI) $\rightarrow$ defective cross-linking $\rightarrow$ hyperextensible skin, hypermobile joints.

8. Fibril assembly (extracellular) — staggered tropocollagen molecules → 67 nm D-banding periodicity.

Elastic Fibre Detail

Elastic fibres are composed of an elastin core surrounded by a fibrillin-1 microfibril sheath. The tropoelastin molecules are cross-linked by lysyl oxidase (same enzyme as collagen). The cross-links are desmosine and isodesmosine — unique to elastin.

- Marfan syndrome — FBN1 (fibrillin-1) mutation; fibrillin-1 microfibrils defective → aortic dissection, ectopia lentis, skeletal overgrowth. Fibrillin-1 also sequesters TGF- β ; defective fibrillin → excess TGF- β signalling → additional pathology.
- Williams syndrome — elastin gene deletion (7q11.23); supravalvular aortic stenosis, "elfin" facies, hypercalcaemia, friendly personality.
- Cutis laxa — elastin gene mutation or acquired (post-inflammatory); loose, redundant skin.

Ground Substance Detail Extended

GAG turnover is mediated by specific enzymes:

- Hyaluronidase degrades hyaluronic acid (target of bacterial hyaluronidase, which spreads infection through CT).
- Sulfatases degrade sulfated GAGs; mucopolysaccharidoses (Hurler, Hunter, Morquio) are deficiencies in specific lysosomal sulfatases or hydrolases → GAG accumulation → coarse facies, hepatosplenomegaly, skeletal dysplasia, intellectual disability.

Decorin, biglycan, lumican, and fibromodulin are small leucine-rich proteoglycans that bind TGF- β and regulate collagen fibril assembly. Decorin deficiency → irregular collagen fibrils and skin fragility.

Fibronectin has two forms: plasma fibronectin (circulating, in fibrin clots, supports wound healing) and cellular fibronectin (insoluble, in ECM). Fibronectin binds integrins (RGD sequence), collagen, fibrin, and heparan sulfate; it is essential for cell migration in embryogenesis and wound healing.

Transition

Connective tissue in general, and the loose connective tissue that supports every epithelium in particular, is the substrate on which the next chapter's focus — adipose tissue — is built. Adipose is a specialised form of connective tissue with a remarkable recent history of rediscovery as an endocrine organ.

Rapid Review

- CT = cells + fibers + ground substance.
- Fibers: type I collagen (tensile, eosinophilic), elastic (recoil, orcein/Verhoeff), reticular (type III, argyrophilic, lymphoid stroma).

- Ground substance: GAGs + proteoglycans (aggrecan) + glycoproteins (fibronectin, laminin).
- Cells: fibroblast (matrix production), macrophage (phagocytosis, CD68+), mast cell (allergy, metachromatic), plasma cell (clock-face nucleus, perinuclear hof, antibody secretion).
- Connective tissue proper: loose, dense irregular, dense regular, elastic, reticular, mucous.
- Specialised: cartilage (J7) and bone (J8) out of scope per official syllabus.

Chapter 6 — Adipose Tissue

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 6

Opening Question

How does fat do more than store energy — and why does its location in the body decide metabolic risk?

Why This Matters

Adipose tissue is at the centre of modern medicine's understanding of obesity, type 2 diabetes, cardiovascular disease, and chronic inflammation. Adipocytes are not just passive storage cells; they secrete hormones (leptin, adiponectin), cytokines (TNF- α , IL-6), and other signalling molecules that affect appetite, insulin sensitivity, vascular tone, and coagulation.

Learning Objectives

1. Compare white and brown adipose tissue by structure, function, and distribution.
2. Describe the development of adipocytes from mesenchymal stem cells.
3. Explain the molecular mechanism of adipogenesis (PPAR- γ , C/EBP) and lipid storage.
4. Describe the endocrine functions of adipose tissue (leptin, adiponectin).
5. Distinguish white from brown adipocytes histologically.
6. Recognise the concept of beige/brite adipocytes.
7. Identify adipose tissue and recognise common pathologies (lipoma, liposarcoma).

Big Picture

- **White adipose** — unilocular, signet-ring, leptin/adiponectin endocrine, energy storage, subcutaneous + visceral depots.
- **Brown adipose** — multilocular, mitochondria-rich, UCP1, non-shivering thermogenesis, neonates + adult neck/supraclavicular.
- **Beige/brite** — inducible UCP1-expressing cells in white fat.
- **Adipogenesis** — MSC $\rightarrow$ preadipocyte $\rightarrow$ mature adipocyte (PPAR- γ + C/EBP).

Core Concept

Adipose tissue is a specialised connective tissue where lipid storage dominates, but endocrine signalling (leptin, adiponectin) and thermogenesis (UCP1) make it a metabolic organ.

Structure

White Adipocytes

Large cells (~50–150 μm) with a single, large lipid droplet that pushes cytoplasm and nucleus to the periphery → signet-ring appearance. Thin rim of cytoplasm with SER (lipid synthesis), mitochondria (β -oxidation), Golgi, and a small amount of RER. The lipid droplet is not inert: bounded by a phospholipid monolayer coated with PAT-family proteins (perilipin, ADRP, TIP47); perilipin blocks lipase access in the basal state and is phosphorylated upon β -adrenergic stimulation → lipolysis.

Surrounded by basal lamina and reticular-fiber meshwork.

White Adipocyte Detail

The lipid droplet forms at the ER, where phospholipids and triglycerides are synthesised. Lipid droplet-associated proteins (PLIN1/perilipin, PLIN2/adipose differentiation-related protein, PLIN3/TIP47, PLIN4, PLIN5) regulate lipase access and droplet dynamics. In the basal state, perilipin coats the droplet and prevents hormone-sensitive lipase (HSL) and adipose triglyceride lipase (ATGL) from accessing the triglyceride. β -adrenergic stimulation → cAMP → PKA → perilipin phosphorylation → HSL and ATGL gain access → lipolysis → free fatty acids + glycerol released for use by other tissues.

The white adipocyte is the body's principal energy reservoir. In a lean adult male, adipose stores ~100,000–150,000 kcal as triglyceride (sufficient for ~30–40 days of fasting). The caloric density of triglyceride (~9 kcal/g) is far higher than glycogen (~4 kcal/g) or protein (~4 kcal/g), making adipose the optimal long-term storage. The body stores adipose in subcutaneous depots (insulation, cushioning) and visceral depots (more metabolically active and more strongly associated with metabolic disease).

Brown Adipocytes

Smaller (~20–40 μm) with multiple small lipid droplets (multilocular) and abundant mitochondria (brown colour from cytochrome). The molecular basis of brown-fat thermogenesis is uncoupling protein 1 (UCP1, thermogenin) in the inner mitochondrial membrane: it allows protons to re-enter the matrix without passing through ATP synthase, dissipating the proton gradient as heat.

Cold exposure → sympathetic norepinephrine → β -adrenergic receptors on brown adipocytes → cAMP → PKA → lipolysis → free fatty acids → β -oxidation → protons pumped by ETC → UCP1 dissipates gradient as heat.

Highly vascularised and richly innervated by sympathetic nerves.

Distribution

- **White** — subcutaneous (insulation, cushioning, energy storage); visceral (omental, mesenteric, retroperitoneal, perirenal). Visceral fat is metabolically more active and more strongly associated with metabolic syndrome.

- **Brown** — neonates: interscapular, around neck, around kidneys, around great vessels. Adults: detectable in neck, supraclavicular, perirenal, suprarenal on PET scans (avid ^{18}F -FDG uptake).
- **Beige** — scattered within white adipose depots; induced by chronic cold, β -adrenergic stimulation, or thiazolidinediones (PPAR- γ agonists).

Beige (Brite) Adipocytes

White adipocytes induced to express UCP1 and take on brown-fat phenotype. Important therapeutic target for obesity and metabolic disease.

Function

Energy Storage and Mobilisation

Triglycerides stored in the droplet can be mobilised as free fatty acids and glycerol during fasting, exercise, stress. Hormone-sensitive lipase and adipose triglyceride lipase (ATGL) are the principal lipases; perilipin regulates access.

Endocrine Function

- **Leptin** — 16 kDa, secreted in proportion to fat mass; acts on hypothalamus to suppress appetite and increase energy expenditure. Leptin deficiency or resistance causes severe obesity. *ob/ob* mouse (leptin-deficient) and *db/db* mouse (leptin-receptor-deficient) are classic monogenic obesity models. Discovered by Friedman in 1994; established adipose as an endocrine organ.
- **Adiponectin** — 30 kDa, secreted in inverse proportion to fat mass; increases insulin sensitivity, promotes fatty acid oxidation, anti-inflammatory, anti-atherogenic. Low in obesity, type 2 diabetes.
- **Resistin** — promotes insulin resistance (more clearly in rodents).
- **Inflammatory cytokines** — TNF- α , IL-6, IL-1 β , MCP-1 (mainly from infiltrating macrophages in obese fat).
- **Angiotensinogen, PAI-1, estrogen** (aromatase from androgens), IGF-1, visfatin, apelin.

Thermogenesis (Brown and Beige)

UCP1 uncouples oxidative phosphorylation to produce heat (non-shivering thermogenesis).

Structure → Function

- **Single large lipid droplet** in white adipocytes = efficient storage (maximum volume per cell).
- **Multilocular droplets + many mitochondria** in brown adipocytes = rapid mobilisation for thermogenesis.
- **PAT-family coating** prevents lipase access until hormonally required.
- **Signet-ring appearance** = maximum lipid storage with minimal cytoplasmic space.

Classification

By cell type: white, brown, beige. By distribution: subcutaneous, visceral. By clinical status: hypertrophic (cell size up), hyperplastic (cell number up), inflamed (crown-like structures of macrophages around dying adipocytes in obesity).

Development

Adipocytes develop from **mesenchymal stem cells** (MSCs), the same progenitors as fibroblasts, chondrocytes, osteoblasts. Adipogenesis is driven by a cascade of transcription factors:

- **PPAR- γ** (peroxisome proliferator-activated receptor gamma) — master regulator of adipogenesis; necessary and sufficient for differentiation. Thiazolidinedione drugs (pioglitazone, rosiglitazone) are PPAR- γ agonists used in type 2 diabetes.
- **C/EBP family** (α , β , δ) — act with PPAR- γ to drive terminal differentiation.
- **KLF, SREBP, ADD1** contribute.

Terminal differentiated adipocyte expresses lipoprotein lipase (LPL, fatty acid uptake), fatty acid synthase, acetyl-CoA carboxylase, hormone-sensitive lipase, ATGL, GLUT4 (insulin-stimulated glucose uptake), perilipin.

Recognition Logic

- **Large empty-appearing cells, signet-ring nuclei, in chicken-wire meshwork** → white adipose tissue.
- **Smaller cells with multiple small vacuoles + abundant pink cytoplasm, lobular with capillaries between cells** → brown adipose tissue.

Clinical Correlation

Case — Lipoma

A 55-year-old man presents with a soft, mobile, painless subcutaneous nodule on the back. Excision shows a well-circumscribed mass of mature adipocytes with peripheral nuclei. Diagnosis: lipoma (the most common benign soft-tissue tumour).

Case — Liposarcoma

A 60-year-old man presents with a deep, growing mass in the thigh. Biopsy shows atypical adipocytes with lipoblasts (signet-ring with scalloped hyperchromatic nucleus), atypical mitoses, and a delicate chicken-wire capillary network. Diagnosis: liposarcoma (most common soft-tissue sarcoma in adults).

Lesson. Lipoma = mature adipocytes; liposarcoma = lipoblasts with atypia and mitoses.

Case — Obesity-Induced Adipose Inflammation

A 50-year-old woman with BMI 38 has fasting glucose 180, BP 150/95, triglycerides 280. Omental fat biopsy shows enlarged adipocytes and macrophages forming ring-like structures (crown-like structures) around individual adipocytes.

Lesson. Hypertrophic adipocytes outpace angiogenesis → hypoxia → chemokine secretion (MCP-1) → macrophage recruitment → inflammatory cytokine secretion (TNF- α , IL-6) → insulin resistance. This is the cellular basis of metabolic syndrome.

Common Misconceptions

- Adipose is just fat storage. It is a major endocrine organ.
- All adipocytes are the same. White, brown, beige are distinct cell types.
- Adipocytes never divide. They can hypertrophy (cell size up) and hyperplasia (cell number up).
- Brown fat is only in babies. Adults have detectable brown fat in the neck and supraclavicular region.
- Leptin would cure obesity. Leptin works only in rare leptin deficiency; most obesity has leptin resistance.

High-Yield Knowledge

Must Know

- White: unilocular, signet-ring, leptin/adiponectin.
- Brown: multilocular, UCP1, thermogenesis.
- Beige: inducible in white fat.
- PPAR- γ is the master adipogenic regulator.
- Leptin deficiency → severe obesity (treatable with recombinant leptin).

Must Distinguish

- White vs brown vs beige (droplet morphology, mitochondria, UCP1).
- Lipoma vs liposarcoma (mature adipocytes vs lipoblasts with atypia).
- Subcutaneous vs visceral adipose (metabolic risk).

Must Understand

- Hypertrophy + hyperplasia of adipocytes in obesity.
- Macrophage crown-like structures in obese adipose.
- Visceral adipose is more metabolically active and more pro-inflammatory.

Integrated Summary

Adipose is a specialised connective tissue with three cell types: white (unilocular, energy storage, endocrine via leptin/adiponectin); brown (multilocular, mitochondria-rich, UCP1-mediated non-shivering thermogenesis); and beige (inducible UCP1-expressing cells in white fat). Adipogenesis from MSCs is driven by PPAR- γ and C/EBP. Adipose tissue

dysfunction — hypertrophy, hyperplasia, macrophage infiltration, chronic inflammation — underlies obesity, insulin resistance, and metabolic syndrome.

Mastery Check

1. Compare white, brown, and beige adipocytes by structure, function, and distribution.
2. Why is PPAR- γ called the master regulator of adipogenesis?
3. Reconstruct the leptin-hypothalamic feedback loop and explain why leptin therapy fails in most obese patients.
4. How does obesity cause adipose inflammation at the cellular level?
5. Distinguish the histological appearance of white fat, brown fat, lipoma, and liposarcoma.

Advanced Concepts

Adipose as an Endocrine Organ Extended

White adipose is one of the largest endocrine organs in the body. Principal adipokines:

- **Leptin** (16 kDa) — produced in proportion to fat mass; signals satiety via hypothalamic LepRb receptor → POMC/AgRP regulation → appetite suppression and increased energy expenditure.
- **Adiponectin** (30 kDa) — paradoxically produced more by lean than obese adipose; insulin-sensitising (AMPK activation in muscle and liver); anti-atherogenic; low levels in metabolic syndrome.
- **Resistin** — links adipose to insulin resistance (resistin name reflects this); more clearly demonstrated in rodents than humans.
- **Inflammatory cytokines** — TNF- α , IL-6, MCP-1 (CCL2); recruit macrophages; contribute to chronic low-grade inflammation in obesity.

Adipose macrophages (ATM) form crown-like structures around dead adipocytes in obese adipose. In lean adipose, ATMs are M2-polarised (anti-inflammatory, wound healing). In obese adipose, ATMs become M1-polarised (pro-inflammatory) and drive chronic inflammation, contributing to insulin resistance.

Adipose tissue dysfunction is a central feature of metabolic syndrome: hypertrophic obesity, hypoxia, macrophage infiltration, ER stress, and altered adipokine secretion all converge on insulin resistance. Bariatric surgery reverses much of this dysfunction, often before significant weight loss (via altered gut hormones GLP-1, PYY).

Thermogenesis Detail

Brown adipose tissue is metabolically active, sympathetic-innervated, and UCP1-dependent. Cold exposure activates the sympathetic nervous system → NE release on brown adipocytes → β 3-adrenergic receptor → cAMP → PKA → lipolysis and UCP1 activation → mitochondrial proton leak → heat. Newborns have substantial BAT (interscapular, perirenal) for non-shivering thermogenesis because they cannot shiver. Adult humans

retain BAT deposits (supraclavicular, neck, mediastinal) detectable by PET-CT.

Beige adipocytes arise within white fat depots upon chronic cold or β -adrenergic stimulation. They express UCP1 and have a thermogenic capacity intermediate between white and brown adipocytes. Recruitment of beige fat is being investigated as an obesity/metabolic disease therapy.

Adipocyte Plasticity and Pathology

Adipocytes are highly plastic:

- **Hypertrophy** — increase in cell size; associated with metabolic disease.
- **Hyperplasia** — increase in cell number; healthier; recruits new adipocytes from progenitors.
- **Dedifferentiation** — under extreme conditions, mature adipocytes can lose their lipid and become fibroblast-like.

Major pathological correlates:

- **Obesity** — BMI ≥ 30 ; visceral adipose accumulation $\rightarrow$ metabolic syndrome, type 2 diabetes, NAFLD, atherosclerosis.
- **Lipodystrophy** — selective loss of adipose; may be genetic (e.g., LMNA mutations in familial partial lipodystrophy) or acquired (HIV antiretroviral therapy).
- **Lipoma** — most common benign soft-tissue tumour; mature adipocytes.
- **Liposarcoma** — most common soft-tissue sarcoma in adults; lipoblasts with atypia and mitoses; well-differentiated, myxoid/round cell, pleomorphic subtypes.
- **Hibernoma** — rare benign tumour of brown adipose.

Transition

Storage, signalling, and thermogenesis are the work of adipose. The next chapter turns to the tissue that is the body's communication and command system — nerve tissue.

Rapid Review

- White adipose: unilocular, signet-ring, peripheral nucleus, energy storage, endocrine (leptin, adiponectin). Subcutaneous + visceral.
- Brown adipose: multilocular, mitochondria-rich, UCP1, non-shivering thermogenesis. Neonates + adult neck/supraclavicular.
- Beige: inducible UCP1-expressing cells in white fat.
- Adipogenesis: MSC $\rightarrow$ preadipocyte $\rightarrow$ mature adipocyte, driven by PPAR- γ + C/EBP.
- Adipokines: leptin (anorexigenic), adiponectin (insulin-sensitising), resistin, inflammatory cytokines.
- Obesity: hypertrophy + hyperplasia, macrophage infiltration (crown-like structures), chronic inflammation, insulin resistance.

Chapter 7 — Nerve Tissue

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 9

Opening Question

How does a cell that lasts a lifetime wire a body and still support itself, when its longest process may extend a metre from the cell body?

Why This Matters

The nervous system is the body's command and information system. Every conscious experience, every movement, every autonomic function, and every hormone secretion is in some way controlled by neurons. Many of the most common human diseases — stroke, multiple sclerosis, Parkinson's disease, Alzheimer's disease, traumatic injury — directly affect nerve tissue.

Learning Objectives

1. Describe the structural and functional features of neurons.
2. Identify the principal neuronal cell types (multipolar, bipolar, pseudounipolar, anaxonic).
3. List the major glial cell types (CNS and PNS) and their functions.
4. Explain the structure and function of myelin in CNS vs PNS.
5. Describe the structure of the synapse.
6. Distinguish gray matter from white matter.
7. Recognise neurons and major glial cells in routine sections.

Big Picture

- **Neuron** — long-lived, polarised, excitable cell. Soma (Nissl/RER, nucleus, nucleolus) → dendrites (input) → axon (output) → synaptic terminals.
- **Glia** — astrocytes, oligodendrocytes, microglia, ependymal cells (CNS); Schwann cells, satellite cells (PNS).
- **Myelin** — oligodendrocyte (CNS, many internodes) vs Schwann cell (PNS, one internode). Saltatory conduction at nodes of Ranvier.
- **Synapse** — chemical (predominant) vs electrical (gap junction).

Core Concept

A neuron is a long-lived, polarised, electrically excitable cell supported by glial cells. Myelin, formed by oligodendrocytes in the CNS and Schwann cells in the PNS, insulates axons and speeds conduction via saltatory jumping between nodes of Ranvier.

Structure

Neuron

- **Soma (perikaryon)** — contains nucleus, prominent nucleolus, Nissl substance (RER in parallel arrays), Golgi, mitochondria, lysosomes. Protein and neurotransmitter synthesis occurs here.
- **Dendrites** — branched processes receiving synaptic input. Membrane contains neurotransmitter receptors.
- **Axon** — single long process from the axon hillock, conducts action potentials to terminals. Contains neurofilaments and microtubules for transport.
- **Synaptic terminals** — specialised endings containing neurotransmitter-filled synaptic vesicles and the molecular machinery for transmitter release.

Axonal Transport

- Anterograde (soma → terminal) by kinesin along microtubules; fast (~200 mm/day) for membrane-bound organelles, slow (~0.1–2 mm/day) for soluble proteins and cytoskeleton.
- Retrograde (terminal → soma) by dynein. Carries recycling materials and also pathogens (rabies, tetanus, herpesvirus).

Neuronal Morphology

- Multipolar — many dendrites + one axon; most common; CNS, motor neurons.
- Bipolar — one dendrite + one axon at opposite poles; retina, olfactory, vestibulocochlear ganglia.
- Pseudounipolar — single process bifurcating into peripheral (to receptor) + central (to CNS) branches; dorsal root ganglia.
- Anaxonic — small, no distinguishable axon; CNS.

Functional: sensory (afferent), motor (efferent), interneuron.

Synapse

Chemical synapse (predominant): presynaptic terminal with vesicles separated from postsynaptic cell by ~20–30 nm cleft. Action potential opens voltage-gated Ca^{2+} channels → Ca^{2+} influx → vesicle fusion → transmitter release → receptor binding → postsynaptic potential (excitatory or inhibitory) → termination by reuptake, enzymatic degradation, or diffusion.

Electrical synapse: pre- and postsynaptic cells connected by gap junctions; fast, synchronous.

Axon hillock has the highest density of voltage-gated Na^+ channels and serves as the trigger zone.

Glial Cells

CNS:

- **Astrocytes** — most numerous; perivascular feet; functions: support, nutrition, K^+ buffering, glutamate uptake at synapses, blood-brain barrier formation, glial scar after injury.

- **Oligodendrocytes** — form myelin around CNS axons; one oligodendrocyte myelinates multiple internodes (up to ~50).
- **Microglia** — resident CNS macrophages, derived from yolk-sac progenitors; phagocytic, antigen-presenting; activated in injury/disease.
- **Ependymal cells** — line ventricles and central canal; produce CSF (choroid plexus).

PNS:

- **Schwann cells** — form myelin around PNS axons (one Schwann cell = one internode) or ensheath unmyelinated axons.
- **Satellite cells** — surround neuronal cell bodies in peripheral ganglia.

Glial Detail

Astrocytes have many processes; their perivascular feet ensheath capillaries and contribute to the blood-brain barrier (BBB) — a continuous endothelium with tight junctions, surrounded by BM and astrocyte foot processes. Astrocytic functions: uptake of glutamate and GABA at synapses (preventing excitotoxicity); K^+ buffering (siphoning excess K^+ away from active neurons); glycogen storage and lactate shuttling to neurons; water homeostasis (aquaporin-4 on foot processes); scar formation after injury (reactive gliosis). GFAP (glial fibrillary acidic protein) is the principal intermediate filament of astrocytes.

Oligodendrocytes in the CNS myelinate multiple internodes (up to ~50) by extending processes that wrap around axon segments. Oligodendrocyte precursor cells (OPCs) are present in the adult CNS and can remyelinate after demyelinating injury, but this capacity declines with age and is insufficient in multiple sclerosis. The CNS lacks the basal lamina + Schwann cell tube (neurilemma) that guides PNS regeneration; this is the structural basis of poor CNS recovery.

Microglia are the resident macrophages of the CNS, derived from yolk-sac progenitors that colonise the brain during embryonic development (not from bone marrow). In the resting state, microglia are highly ramified, surveying the parenchyma. Upon activation (injury, infection, neurodegeneration), they retract processes, become amoeboid, phagocytose debris, and secrete cytokines. Microglial activation is implicated in Alzheimer's disease, Parkinson's disease, and multiple sclerosis.

Ependymal cells are a simple cuboidal epithelium lining the ventricles and central canal. Modified ependymal cells (tanocytes) line the third ventricle and mediate transport between CSF and brain. The choroid plexus (specialised ependymal cells + fenestrated capillaries) produces CSF (~500 mL/day); tight junctions between choroid plexus cells form the blood-CSF barrier.

Schwann cells in the PNS myelinate one internode each (or ensheath multiple unmyelinated axons in a Remak bundle). Unlike oligodendrocytes, every Schwann cell has a basal lamina on its outer surface; this basal lamina forms the neurilemma — the tube that guides regenerating axons

after peripheral nerve injury.

Satellite cells surround neuronal cell bodies in peripheral ganglia (dorsal root ganglia, autonomic ganglia). They regulate the microenvironment and are derived from neural crest.

Myelin

Multilayered membrane wrapping. Major proteins: MBP, PLP (CNS); P0, PMP22, MBP (PNS). Major lipids: cholesterol, sphingomyelin, galactocerebroside.

Node of Ranvier — gap (~1 μm) between adjacent myelin segments; high density of voltage-gated Na^+ channels. Saltatory conduction: action potential jumps from node to node.

Schmidt-Lanterman incisures (PNS only) — cytoplasmic pockets within myelin.

CNS vs PNS myelin origin:

Feature	CNS	PNS
Cell	Oligodendrocyte	Schwann cell
Internodes per cell	Many	One
Neurilemma (basal lamina + Schwann cytoplasm)	No	Yes (allows regeneration)
Basal lamina around myelin	No	Yes
Disease association	MS	Guillain-Barré, Charcot-Marie-Tooth

Function

Neurons receive, integrate, transmit signals. Myelin increases conduction velocity via saltatory conduction. Astrocytes regulate the microenvironment. Microglia are the immune cells of the CNS. Ependymal cells produce CSF. Schwann cells support and myelinate PNS axons; their basal lamina guides regeneration.

Structure → Function

- **Nissl substance** (parallel RER) supports high protein synthesis for axonal transport.
- **Node of Ranvier** with high Na^+ channel density → saltatory conduction (fast).
- **Neurilemma** in PNS (basal lamina + Schwann cytoplasm) → tube for axonal regeneration; absent in CNS → poor regeneration.
- **Astrocyte perivascular feet** → blood-brain barrier component.
- **Microglia quiescence in normal brain → activation in injury:** surveillance and immune defense.

Classification — Synapse Types

- **Chemical** — most synapses; use neurotransmitter; directional.
- **Electrical** — gap junction; bidirectional, fast, synchronous; smooth muscle, some CNS circuits.

Recognition Logic

- **Large cell with pale nucleus, prominent nucleolus, basophilic Nissl substance, processes** → neuron soma.
- **Elongated cell with small dense nucleus, in parallel arrays around axons** → Schwann cell (PNS nerve).
- **Star-shaped cell with processes contacting vessels** → astrocyte (GFAP+).
- **Small dense cell in CNS parenchyma** → microglia or oligodendrocyte.
- **Long parallel eosinophilic fibers with elongated Schwann-cell or fibroblast nuclei** → peripheral nerve in longitudinal section.
- **Cross-section of a nerve with multiple small dots (myelinated axons) within a denser sheath** → peripheral nerve in cross-section; perineurium visible.

Clinical Correlation

Case — Multiple Sclerosis

A 35-year-old woman with progressive leg weakness, diplopia, and sensory loss. MRI shows periventricular hyperintensities. CSF shows elevated IgG and oligoclonal bands. Brain biopsy shows plaques of demyelination with preserved axons, perivascular lymphocytic infiltrate, reactive astrocytic gliosis.

Lesson. MS is autoimmune CNS demyelination. The absence of neurilemma (basal lamina) prevents effective CNS axonal regeneration after demyelination.

Case — Guillain-Barré Syndrome

A 30-year-old man with ascending paralysis after a gastrointestinal infection. CSF shows elevated protein with normal cell count (albuminocytologic dissociation). Nerve biopsy shows segmental demyelination with macrophage infiltration.

Lesson. GBS is autoimmune PNS demyelination. The presence of neurilemma allows PNS regeneration after the acute attack.

Case — Wallerian Degeneration

After a peripheral nerve transection, the distal segment degenerates (axon fragmentation, myelin ovoids, macrophage clearance), but the basal lamina tube remains. Schwann cells dedifferentiate and guide the regenerating axon back to its target.

Lesson. PNS regeneration depends on the basal lamina and Schwann cells; CNS lacks this scaffold and cannot regenerate effectively.

Common Misconceptions

- Neurons are the most numerous cells in the CNS. Glia outnumber neurons by $\sim 10\times$.
- All neurons are multipolar. Many sensory neurons are pseudounipolar; retinal/olfactory are bipolar.
- Myelin is part of the axon. Myelin is formed by oligodendrocytes or Schwann cells.
- Glial cells are passive support. They regulate microenvironment, form BBB, myelinate, phagocytose, contribute to disease.
- Action potentials travel at the same speed in all axons. Conduction velocity depends on diameter and myelination.

High-Yield Knowledge

Must Know

- Neuron: soma (Nissl), dendrites (input), axon (output), synapse.
- Synapse: chemical (predominant) vs electrical (gap junction).
- Myelin: oligodendrocyte (CNS, many internodes) vs Schwann cell (PNS, one internode).
- Neurilemma (PNS) guides regeneration; absent in CNS.
- Saltatory conduction at nodes of Ranvier.

Must Distinguish

- CNS vs PNS myelin origin.
- Astrocyte vs oligodendrocyte vs microglia.
- Axon vs dendrite (axon hillock, no Nissl in initial segment; dendrites have Nissl).

Must Understand

- MS = CNS demyelination; GBS = PNS demyelination.
- BBB = continuous endothelium + tight junctions + basal lamina + astrocyte foot processes.

Integrated Summary

Nerve tissue is built from neurons (signaling) and glia (support). Neurons are polarised cells with a soma containing Nissl substance, dendrites for input, an axon for output, and synaptic terminals for communication. Synapses are predominantly chemical (vesicle release $\rightarrow$ receptor binding). Myelin is formed by oligodendrocytes in the CNS (one cell, many internodes) and Schwann cells in the PNS (one cell, one internode). The node of Ranvier is the site of saltatory conduction. CNS lacks the neurilemma that enables PNS regeneration; this is the structural basis of the poor recovery from CNS compared to PNS injury.

Mastery Check

1. Reconstruct the events at a chemical synapse from action-potential arrival to postsynaptic response.
2. Compare the formation of myelin by oligodendrocytes and Schwann cells.
3. Why is Wallerian regeneration possible in the PNS but not in the CNS?
4. Predict the histological appearance of a peripheral nerve in longitudinal and cross-section.
5. Explain the molecular basis of multiple sclerosis and Guillain-Barré syndrome.

Advanced Concepts

Synapse Detail

Synapses are specialised junctions where one neuron communicates with another (or with an effector). Two major classes:

- **Electrical synapses** — gap junctions allow direct ionic current flow; fast, bidirectional; found in cardiac muscle, smooth muscle, some CNS neurons (esp. inhibitory interneurons in retina, thalamic reticular nucleus, hippocampal interneurons).
- **Chemical synapses** — neurotransmitter in presynaptic vesicles released into synaptic cleft; bind postsynaptic receptors; slower but plastic.

Chemical synapse structure: presynaptic terminal (active zones, mitochondria, synaptic vesicles), synaptic cleft (~20–30 nm, with adhesion molecules like neurexin/neuroligin), postsynaptic density (receptors, scaffolding proteins). Synaptic vesicles cluster at active zones (docked) and are primed for release. Ca^{2+} entry through voltage-gated Ca^{2+} channels triggers SNARE-mediated vesicle fusion (synaptobrevin/VAMP, syntaxin, SNAP-25). Botulinum toxin cleaves SNARE proteins → flaccid paralysis.

Neurotransmitter types:

- **Amino acids** — glutamate (principal excitatory), GABA (principal inhibitory), glycine (inhibitory, spinal cord).
- **Monoamines** — dopamine, norepinephrine, epinephrine, serotonin, histamine.
- **Acetylcholine** — neuromuscular junction, autonomic preganglionic, parasympathetic postganglionic.
- **Neuropeptides** — substance P, enkephalins, endorphins, neuropeptide Y, somatostatin; stored in dense-core vesicles, slower action.

Postsynaptic potentials:

- **EPSP** (excitatory postsynaptic potential) — depolarisation, opens Na^+ channels.
- **IPSP** (inhibitory postsynaptic potential) — hyperpolarisation, opens Cl^- or K^+ channels.

Myelination and Saltatory Conduction

Myelin is a multilamellar lipid-rich sheath (mostly lipid, some protein) wrapped around axons by oligodendrocytes (CNS) or Schwann cells (PNS). It increases conduction velocity by:

- Reducing membrane capacitance (less charge stored across the bilayer).
- Increasing membrane resistance (less current leak).

Conduction is saltatory: action potentials "jump" between nodes of Ranvier (gaps in myelin where Na^+ channels are concentrated). Saltatory conduction is much faster than continuous conduction (up to ~ 120 m/s in large myelinated axons vs ~ 1 m/s in small unmyelinated).

Demyelinating disease:

- **Multiple sclerosis** — autoimmune destruction of CNS myelin (oligodendrocytes); periventricular plaques on MRI; symptoms depend on plaque location (optic neuritis, hemiparesis, ataxia).
- **Guillain-Barré syndrome** — autoimmune PNS demyelination (Schwann cells); ascending paralysis.
- **Charcot-Marie-Tooth** — hereditary demyelinating neuropathy (PMP22, MPZ mutations); foot drop, stork-leg deformity.

Neuronal Cell Biology

Neurons are post-mitotic; they cannot divide. They have high metabolic demands (high O_2 consumption, glucose uptake, mitochondrial density). The neuronal cytoskeleton is specialised:

- **Neurofilaments** — intermediate filaments; accumulate in ageing neurons (neurofibrillary tangles in Alzheimer disease).
- **Microtubules** — track for fast axonal transport (kinesin anterograde, dynein retrograde).
- **Actin** — dendritic spines, growth cones.

Axonal transport:

- **Anterograde** (cell body $\rightarrow$ terminal) — kinesin; fast transport of vesicles, slow transport of cytoskeletal proteins.
- **Retrograde** (terminal $\rightarrow$ cell body) — dynein; carries trophic signals (NGF), viruses (rabies, HSV), toxins (tetanus).

Neuronal injury: chromatolysis (cell body swells, Nissl substance disperses, nucleus moves peripherally) after axonal injury reflects the metabolic shift to regeneration. Central neurons rarely regenerate (no Schwann cell tube, glial scar, myelin inhibitors); peripheral neurons regenerate along the Schwann cell tube (~ 1 mm/day).

Transition

Nerve tissue is the body's signalling system. The next chapter turns to the system that supplies every tissue — the circulatory system.

Rapid Review

- Neuron: soma (Nissl), dendrites, axon (axon hillock = trigger zone), synaptic terminals.

- Glia: CNS (astrocytes, oligodendrocytes, microglia, ependymal cells); PNS (Schwann cells, satellite cells).
- Myelin: oligodendrocytes (CNS, many internodes) vs Schwann cells (PNS, one internode, neurilemma present).
- Synapse: chemical (predominant) vs electrical (gap junction).
- Saltatory conduction at nodes of Ranvier; BBB = endothelium + tight + BM + astrocyte feet.
- Wallerian regeneration: PNS yes (basal lamina tube), CNS no (no neurilemma).

Chapter 8 — The Circulatory System

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 11

Opening Question

How does one structural plan — three concentric layers with varying composition — produce an aorta that stretches, a capillary that leaks on purpose, and a vein that stores blood?

Why This Matters

The circulatory system is the body's delivery service. It supplies every tissue with oxygen, nutrients, hormones, and immune cells; it removes metabolic waste; it maintains homeostasis. Most common human diseases — atherosclerosis, hypertension, myocardial infarction, stroke — directly affect the circulatory system. The endothelium is now recognised as a major endocrine and signalling organ whose dysfunction contributes to atherosclerosis.

Learning Objectives

1. Describe the three layers (tunics) of a typical blood vessel.
2. Identify the major vessel types (elastic artery, muscular artery, arteriole, capillary, venule, vein) by structure and function.
3. Describe the structure and function of the endothelium.
4. Explain the structural basis of vascular function — how architecture predicts pressure, flow, and resistance.
5. Describe the histological features of capillaries and the structural basis of capillary exchange.
6. Recognise the histological signatures of atherosclerosis, vasculitis.
7. Describe the layers and conduction system of the heart.

Big Picture

- **Three layers** — intima (endothelium + subendothelial CT + internal elastic lamina in arteries), media (smooth muscle + elastic/collagen), adventitia (CT with vasa vasorum in large vessels).
- **Vessel hierarchy** — elastic arteries → muscular arteries → arterioles → capillaries → venules → veins.
- **Capillary types** — continuous (most tissues, BBB), fenestrated (endocrine, intestinal villi, renal glomerulus), sinusoidal (liver, spleen, bone marrow).
- **Heart** — endocardium + myocardium + epicardium; fibrous skeleton; valves; conduction system.

Core Concept

Every blood vessel is built from three concentric layers (intima, media, adventitia); the relative thickness and composition of these layers defines the vessel type and predicts its mechanical and physiological function. Endothelium is a major endocrine and signalling organ.

Structure

The Three Layers

- **Tunica intima** — innermost; endothelium (single layer of flattened cells on basal lamina) + subendothelial loose CT + internal elastic lamina in arteries (fenestrated sheet of elastin between intima and media).
- **Tunica media** — concentric layers of smooth muscle with elastic and collagen fibers. External elastic lamina in arteries between media and adventitia.
- **Tunica adventitia** — outermost CT (fibroblasts, collagen, elastic fibers) anchoring the vessel; contains vasa vasorum (vessels supplying large-vessel walls) and nerve fibers.

Elastic Arteries (aorta, pulmonary, common carotid)

Largest arteries; receive blood directly from the heart; must accommodate the systolic pressure wave. Tunica media dominated by multiple concentric fenestrated elastic laminae (40–70 in newborn aorta) with smooth muscle cells sandwiched between them. Abundant elastin allows stretch during systole and recoil during diastole — the **Windkessel effect** that smooths pulsatile cardiac output.

Capillary Detail

The endothelium is a remarkable organ. A single adult has $\sim 10^{13}$ endothelial cells weighing ~ 1 kg and covering ~ 4000 m². The endothelium produces nitric oxide (NO, vasodilation via cGMP), endothelin (vasoconstriction), prostacyclin (vasodilation + anti-platelet), von Willebrand factor (coagulation, especially under stress), tissue plasminogen activator (fibrinolysis), thrombomodulin (anticoagulation), and various growth factors (VEGF, FGF). Endothelial dysfunction (loss of NO, increased endothelin, increased adhesion molecules) is the central event in atherosclerosis.

Endothelial heterogeneity by organ is substantial. Continuous capillaries with tight junctions (BBB) in the CNS require astrocyte signals. Fenestrated capillaries in the endocrine glands allow rapid hormone exchange. Sinusoidal capillaries in the liver allow massive bidirectional exchange. The glomerular fenestrae lack diaphragms and are surrounded by a thick GBM and podocyte foot processes; this triple filter (endothelium + GBM + slit diaphragm) is highly size- and charge-selective.

Lymphatic capillaries are blind-ended vessels with overlapping endothelial cells (function as primary valves) and anchoring filaments; they drain interstitial fluid back to the bloodstream. Larger lymphatics have smooth muscle and valves; lymph node filtration occurs in the medullary sinuses.

Muscular Arteries (most named arteries)

Medium-sized. Prominent wavy internal elastic lamina (key LM landmark). Media dominated by smooth muscle; relatively little elastic tissue. Smooth muscle regulates vessel caliber (vasoconstriction, vasodilation).

Arterioles

Small (lumen < 0.5 mm). 1-5 layers of smooth muscle in the media. **Resistance vessels:** their smooth muscle tone is the principal determinant of peripheral resistance and blood pressure. Metarterioles connect directly to capillaries with precapillary sphincters regulating capillary filling.

Capillaries

Exchange vessels. Wall = endothelium + basal lamina + pericytes. Three types:

- **Continuous** — uninterrupted endothelium with tight junctions; continuous basal lamina; muscle, CT, lung, CNS. Permit water, ions, small molecules; not large proteins.
- **Fenestrated** — endothelial cells with fenestrae (~60-80 nm), often closed by a thin diaphragm; endocrine glands, intestinal villi, renal glomeruli, choroid plexus. Rapid exchange of fluids and small molecules.
- **Sinusoidal (sinusoids)** — large, irregular, tortuous; discontinuous endothelium; discontinuous basal lamina; large lumen. Liver, spleen, bone marrow, adrenal cortex. Pass large molecules and even cells.

Total surface area of capillaries ~600 m²; total length ~40,000 km.

Venules and Veins

Post-capillary venules — endothelium + pericytes; principal site of leukocyte emigration in inflammation. Muscular venules — 1-2 layers of smooth muscle.

Veins — low-pressure; thin media; thick adventitia (relative); many valves (especially below heart) preventing backflow. Larger lumen than corresponding artery.

Function

Arteries conduct blood at high pressure; muscular arteries distribute; arterioles regulate resistance; capillaries exchange; veins return blood and store volume. The endothelium regulates vascular tone (NO, endothelin), coagulation (vWF, tPA, thrombomodulin), inflammation (selectins, ICAMs, VCAMs), and permeability.

Structure → Function

- **Elastic arteries' many elastic laminae** → Windkessel effect (stretch + recoil → smooth flow).
- **Muscular arteries' thick smooth muscle media** → precise regulation of regional flow.

- **Arterioles' 1-2 muscle layers** → resistance vessels; small change in radius → large change in flow (Poiseuille's law).
- **Capillary types** match function: continuous BBB (tight), fenestrated endocrine (rapid), sinusoidal liver (bulk exchange).
- **Veins' valves** → prevent backflow in the legs.
- **Endothelium's nitric oxide** → vasodilation (target of nitroglycerin, sildenafil potentiates downstream pathway).

Compare & Distinguish

Artery vs Vein

Feature	Artery	Vein
Wall	Thick	Thin
Lumen	Smaller, round	Larger, often collapsed
Media	Thick smooth muscle	Thin
Adventitia	Thinner	Thicker relative
Internal elastic lamina	Prominent	Not prominent
Valves	No	Yes (especially below heart)

Elastic vs Muscular Artery

Feature	Elastic	Muscular
Size	Largest	Medium
Media	Many elastic laminae	Smooth muscle dominant
Function	Windkessel	Distribution

Capillary Types

Type	Endothelium	Diaphragm	Example
Continuous	Unbroken	—	Muscle, CNS
Fenestrated	Pores	+	Endocrine, villi, glomerulus
Sinusoidal	Discontinuous	—	Liver, spleen, marrow

Recognition Logic

- **Large vessel, many concentric elastic laminae** → elastic artery.
- **Medium vessel, prominent wavy internal elastic lamina, thick smooth muscle media** → muscular artery.
- **Small vessel, 1-2 muscle layers** → arteriole.
- **Single endothelium + pericytes on basal lamina, no media** → capillary.
- **Thin media, thick adventitia, valves** → vein.

The Heart

Three layers:

- **Endocardium** — endothelium + subendothelial CT + subendocardial layer with Purkinje fibres.
- **Myocardium** — cardiac muscle (see muscle tissue chapter in out-of-scope appendix for full coverage); the principal mass.
- **Epicardium** — visceral pericardium; mesothelium on CT; contains coronary vessels and nerves.

Fibrous skeleton — dense CT rings (annuli fibrosi) around valve orifices; provides attachment for muscle and electrical insulation between atria and ventricles.

Valves — flaps of dense CT covered by endothelium on both sides. Chordae tendineae of AV valves attach to papillary muscles.

Conduction system — SA node, AV node, bundle of His, bundle branches, Purkinje fibres (modified cardiac muscle cells; large, pale, abundant glycogen, few myofibrils; rapidly conduct action potentials).

Clinical Correlation

Case — Atherosclerosis

A 65-year-old man with hypertension, diabetes, and hypercholesterolemia dies of acute MI. Coronary arteries show intimal plaques with fibrous cap over necrotic lipid-rich core. Plaque rupture → thrombosis → myocardial infarction.

Lesson. Endothelial injury (HTN, smoking, dyslipidemia, hyperglycemia) → LDL infiltration → monocyte recruitment → foam cell formation → smooth muscle migration → plaque growth → rupture → thrombosis. The intima is the primary site of disease.

Case — Varicose Veins

A 50-year-old woman with bilateral leg heaviness and visible tortuous veins. Histology shows dilated veins with valve incompetence.

Lesson. Valve failure → venous stasis → dilation → further valve failure (vicious cycle). Histologically, vein wall is thinned and irregular.

Common Misconceptions

- All arteries have thick muscular media. Elastic arteries have media dominated by elastic laminae.
- Capillaries have no smooth muscle. True, but pericytes wrap around the endothelium.
- Veins are just arteries with thinner walls. Veins have different layer organisation; many have valves; endothelium is a major endocrine organ, not just a passive lining.

High-Yield Knowledge

Must Know

- Three layers: intima (endothelium + IEL), media (smooth muscle), adventitia (CT).
- Elastic artery → Windkessel; muscular artery → distribution; arteriole → resistance; capillary → exchange; vein → return/valves.
- Capillary types: continuous (BBB), fenestrated (endocrine, glomerulus), sinusoidal (liver, marrow).
- Endothelium produces NO (vasodilation), endothelin (vasoconstriction), vWF, tPA.

Must Distinguish

- Elastic vs muscular artery (media composition).
- Capillary types by endothelium.
- Artery vs vein (wall thickness, lumen, valves).

Must Understand

- Arteriolar radius is the principal determinant of peripheral resistance.
- Endothelial dysfunction is central to atherosclerosis.

Integrated Summary

The circulatory system is a closed network organised into a vessel hierarchy — elastic arteries (Windkessel), muscular arteries (distribution), arterioles (resistance), capillaries (exchange: continuous, fenestrated, sinusoidal), venules (leukocyte emigration), veins (return, valves). All vessels have three layers (intima, media, adventitia); the relative composition defines function. The endothelium is a major endocrine and signalling organ whose dysfunction is central to atherosclerosis.

Mastery Check

1. Compare elastic and muscular arteries by structure and function.
2. Explain the Windkessel effect.
3. Why does arteriolar tone have a disproportionate effect on blood pressure?
4. Compare the three capillary types by structure, location, and function.
5. Reconstruct the pathogenesis of atherosclerosis from endothelial injury to plaque rupture.
6. Compare the structure of an artery and a vein of similar size.

Advanced Concepts

Endothelial Heterogeneity and Specialised Circulations

Endothelium is highly heterogeneous, specialised to local function:

- **Brain (continuous with tight junctions)** — BBB; pericytes, astrocyte foot processes; prevents most molecules >400 Da from crossing.
- **Endocrine glands (fenestrated)** — rapid hormone exchange; pituitary, adrenal, thyroid.

- **Liver (sinusoidal, discontinuous)** — large fenestrations, no diaphragm; allows bidirectional exchange of macromolecules.
- **Kidney glomerulus (fenestrated, no diaphragm)** — combined with podocyte slit diaphragm and GBM to form a triple filter.
- **Spleen (sinusoidal, open circulation)** — RBCs squeeze through; macrophages in sinuses phagocytose aged RBCs.
- **Lung (continuous)** — extremely thin for gas exchange; pneumocyte on air side.

Lymphatics:

- **Initial lymphatics** — blind-ended capillaries with overlapping endothelial cells (function as primary valves), no BM, no pericytes; entry of interstitial fluid driven by tissue pressure.
- **Collecting lymphatics** — have smooth muscle and valves; lymphangions contract to propel lymph.
- **Lymph nodes** — along the lymphatic pathway; filter lymph (see Chapter 11).

Vascular Pathology Correlates

- **Atherosclerosis** — intimal lipid accumulation + inflammation; plaques form in high-pressure, turbulent-flow regions (branch points); risk factors: hypertension, hyperlipidaemia, smoking, diabetes.
- **Aneurysm** — focal dilation of artery (>50% of normal); risk of rupture; abdominal aortic aneurysm is most common.
- **Vasculitis** — inflammatory destruction of vessel wall; examples: giant cell arteritis (temporal artery, elderly, headache, jaw claudication), Takayasu arteritis (aorta, young women), Kawasaki disease (coronary arteries, children), polyarteritis nodosa (medium arteries, ANCA-negative).
- **Capillary leak** — endothelial barrier breakdown → tissue oedema; seen in sepsis, burns, anaphylaxis.
- **Diabetic microangiopathy** — capillary BM thickening; retinopathy, nephropathy, neuropathy.

Cardiac Muscle vs Smooth Muscle vs Skeletal Muscle (recognition primer)

Although muscle tissue is out of core scope, the heart and vessels contain contractile cells worth contrasting:

Feature	Skeletal	Cardiac	Smooth
Striations	Yes	Yes	No
Nuclei	Multinucleate (peripheral)	1-2 central	1 central
Intercalated discs	No	Yes	No
Cell shape	Long cylinder	Branched cylinder	Spindle/fusiform
Ca ²⁺ source	SR only	SR + extracellular	Extracellular + SR
Gap junctions	No	Yes (intercalated disc)	Yes (single-unit)
Control	Voluntary	Involuntary	Involuntary

Feature	Skeletal	Cardiac	Smooth
Regeneration	Limited (satellite cells)	Minimal	Good

Smooth muscle in the tunica media of arteries is the principal effector of vascular tone (α 1-adrenergic $\rightarrow$ contraction; β 2-adrenergic $\rightarrow$ relaxation).

Transition

The circulatory system carries the formed elements of blood. The next chapter is about those formed elements and what they tell us in a single drop of blood on a slide.

Rapid Review

- Vessels: elastic artery (Windkessel) $\rightarrow$ muscular artery (distribution) $\rightarrow$ arteriole (resistance) $\rightarrow$ capillary (exchange: continuous, fenestrated, sinusoidal) $\rightarrow$ venule $\rightarrow$ vein (return, valves).
- Three layers: intima (endothelium + IEL), media (smooth muscle), adventitia (CT with vasa vasorum).
- Endothelium: endocrine organ (NO, endothelin, vWF, tPA).
- Heart: endocardium, myocardium, epicardium; fibrous skeleton; valves; conduction system (SA, AV, His, Purkinje).

Chapter 9 — Blood

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 12

Opening Question

How can you tell a bacterial infection from a parasitic one from a single blood smear — and how does a fluid tissue remain contained and flowing inside a closed circuit?

Why This Matters

The complete blood count and the peripheral-blood smear are among the most commonly ordered diagnostic tests in medicine. Anaemia, infection, leukaemia, thrombocytopenia — all are diagnosed and monitored by blood-cell morphology. The student who can read a blood smear is the student who can begin to understand haematology.

Learning Objectives

1. Describe the composition of blood: plasma and formed elements.
2. Identify the five leukocytes and the other formed elements on a Wright-stained smear.
3. Explain the function of each leukocyte.
4. Describe red-cell morphology and recognise pathological forms (spherocytes, sickle cells, target cells, schistocytes).
5. Describe platelet structure and function in primary haemostasis.
6. Recognise the morphological signatures of common blood pathologies.

Big Picture

- **Plasma** — water + proteins (albumin, globulins, fibrinogen) + ions + nutrients + waste + hormones.
- **Formed elements** — RBCs (~99%), WBCs, platelets. Haematocrit = % RBC volume (~45% men, ~40% women).
- **RBC** — anucleate, biconcave disc (~7.5 μm), haemoglobin, ~120-day life span, recycled by spleen.
- **Platelets** — anucleate megakaryocyte fragments, primary haemostasis.
- **WBCs** — neutrophils (acute bacterial), eosinophils (allergy, parasites), basophils (immediate hypersensitivity), lymphocytes (adaptive immunity), monocytes (chronic inflammation, macrophage precursors).

Core Concept

Blood is a fluid connective tissue in which cells (RBCs, WBCs, platelets) are suspended in plasma. The peripheral-blood smear is the most accessible histological preparation in medicine.

Structure

Red Blood Cells (Erythrocytes)

$\sim 5 \times 10^6/\mu\text{L}$. Anucleate, biconcave disc ($\sim 7.5 \mu\text{m}$ diameter). Packed with haemoglobin (~ 250 million molecules per cell). No organelles; the absence of a nucleus and mitochondria maximises space for O_2 transport and prevents O_2 consumption.

Shape on smear: pale, pink-orange (eosinophilic) discs with central pallor ($\sim 1/3$ cell diameter). Biconcave shape gives flexibility for capillary transit.

RBC Membrane and Cytoskeleton

The RBC membrane is a lipid bilayer supported by a spectrin-based cytoskeleton. Spectrin α and β chains form tetramers that connect junctional complexes (containing actin, protein 4.1, band 3, glycophorin). This meshwork gives the RBC its biconcave shape and the deformability needed to traverse capillaries narrower than its own diameter. Mutations in spectrin, ankyrin, band 3, or protein 4.2 cause hereditary spherocytosis (round, fragile RBCs $\rightarrow$ splenic sequestration $\rightarrow$ haemolytic anaemia).

The ABO blood group antigens are carbohydrate modifications on glycophorin and band 3; the Rh antigen is a transmembrane protein. Mismatched transfusion $\rightarrow$ IgM anti-A/anti-B $\rightarrow$ complement-mediated intravascular haemolysis. Rh incompatibility in pregnancy $\rightarrow$ IgG anti-D $\rightarrow$ haemolytic disease of the newborn.

G6PD deficiency (X-linked recessive) is the most common enzymatic disorder of RBCs; oxidative stress (fava beans, sulfa drugs, infections) precipitates haemoglobin oxidation and Heinz body formation $\rightarrow$ bite cells $\rightarrow$ haemolytic anaemia. Sickle cell disease (HbS point mutation) $\rightarrow$ polymerisation under low O_2 $\rightarrow$ sickle cells $\rightarrow$ vaso-occlusion and haemolysis.

Haemoglobin: four globin chains ($2\alpha + 2\beta$ in adult HbA) + four heme groups with iron that binds O_2 . Cooperative binding (sigmoidal O_2 -dissociation curve).

Life span: ~ 120 days. Aged RBCs phagocytosed by splenic macrophages. Iron recycled; heme $\rightarrow$ bilirubin (conjugated in liver, excreted in bile); globin $\rightarrow$ amino acids.

Pathological forms (anisocytosis, poikilocytosis):

Form	Description	Disease
Spherocyte	Round, small, no central pallor	Hereditary spherocytosis, AIHA
Elliptocyte	Oval	Hereditary elliptocytosis
Sickle cell	Crescent	Sickle-cell anaemia
Target cell	Bull's-eye	Thalassemia, liver disease, Fe deficiency
Schistocyte	Fragment	Microangiopathic haemolytic anaemia
Burr cell (echinocyte)	Spiky	Uraemia

Form	Description	Disease
Acanthocyte	Irregular spikes	Liver disease, abetalipoproteinaemia

Platelets

Small, anucleate cytoplasmic fragments (~2-4 μm) derived from bone-marrow megakaryocytes. ~250,000-400,000/ μL . Life span ~8-10 days. On smear: small, granular, bluish-purple, often in clusters.

Function — primary haemostasis: endothelial injury → platelet adhesion to subendothelial collagen via vWF and GPIb → activation → granule release (ADP, serotonin, TXA_2) → surface phosphatidylserine exposure (clotting-factor assembly) → aggregation → primary haemostatic plug.

White Blood Cells (Leukocytes)

Classified by specific cytoplasmic granules into granulocytes (neutrophils, eosinophils, basophils) and agranulocytes (lymphocytes, monocytes).

Neutrophils (50-70%)

Multi-lobed nucleus (3-5 lobes connected by thin chromatin strands). Pale pink cytoplasm with fine, neutral-staining specific granules. Azurophilic (primary) granules contain lysosomal enzymes; specific granules contain defensins, myeloperoxidase, lactoferrin.

Function: highly phagocytic, first responders in bacterial infection. Predominant cells in pus.

Hypersegmented (>5 lobes) → megaloblastic anaemia (B_{12} or folate deficiency). Band (band-shaped nucleus) → "left shift" in severe infection.

Eosinophils (1-4%)

Bilobed nucleus (occasionally unsegmented). Bright red-orange (eosinophilic) granules filling the cytoplasm; granules contain major basic protein (toxic to parasites), eosinophil cationic protein, eosinophil peroxidase.

Function: allergic reactions, parasitic (helminth) defence.

Eosinophilia: allergic diseases (asthma, allergic rhinitis, atopic dermatitis), parasites, eosinophilic granulomatosis with polyangiitis.

Basophils (0.5-1%)

Bilobed or S-shaped nucleus often obscured by large, dark purple-blue basophilic granules. Granules contain histamine, heparin, and proteases.

Function: bind IgE on $\text{Fc}\epsilon\text{RI}$; degranulate on antigen cross-linking → type I (immediate) hypersensitivity (anaphylaxis, urticaria, allergic rhinitis).

Lymphocytes (20-40%)

Round, dense, heterochromatic nucleus occupying most of the cell. Thin rim of pale blue cytoplasm.

Small (6-9 μm) are most abundant in peripheral blood. Large lymphocytes (10-15 μm) are rare in blood but common in lymphoid organs.

Three lineages: B (bone marrow maturation, plasma cell → antibody), T (thymic maturation, cellular immunity — cytotoxic and helper), NK (innate, virus/tumour killing without prior sensitisation).

Monocytes (2–8%)

Largest leukocytes (12–20 μm). Indented or kidney-shaped nucleus; abundant pale gray-blue cytoplasm with fine azurophilic granules.

Circulate 1–3 days, then migrate into tissue and differentiate into macrophages or dendritic cells. Once in tissue, take on resident phagocyte morphology (Kupffer cells in liver, microglia in brain, alveolar macrophages in lung, osteoclasts in bone).

Monocytosis: chronic infections (TB, endocarditis), autoimmune disease, certain leukaemias.

Function

RBCs transport O_2 and CO_2 . Platelets initiate primary haemostasis. Neutrophils phagocytose bacteria. Eosinophils combat parasites and contribute to allergy. Basophils and mast cells mediate immediate hypersensitivity. Lymphocytes mediate adaptive immunity. Monocytes become tissue macrophages for chronic inflammation and antigen presentation.

Structure → Function

- **Anucleate biconcave RBC** → maximum space for haemoglobin, maximum flexibility for capillary transit.
- **Multi-lobed neutrophil nucleus** → allows nuclear deformation through narrow tissue spaces.
- **Granule colour of granulocytes** = stain target: red (eosinophil, basic proteins), dark blue (basophil, heparin), neutral (neutrophil, fine enzymes).
- **Bilobed eosinophil nucleus** + large eosinophilic granules = distinctive parasite defence phenotype.
- **Round dense lymphocyte nucleus** = mostly heterochromatin (quiescent, awaiting antigen); activated lymphocytes enlarge with more euchromatin.

Compare & Distinguish

Granulocytes by Granule Colour

Cell	Granule colour	Function
Neutrophil	Pale neutral	Bacterial phagocytosis
Eosinophil	Bright red-orange	Parasites, allergy
Basophil	Dark blue-purple	Immediate hypersensitivity

Lymphocyte vs Monocyte

Cell	Nucleus	Cytoplasm
Small lymphocyte	Round, dense, no nucleolus visible	Thin rim
Monocyte	Indented, kidney-shaped	Abundant gray-blue

Recognition Logic

- **Pale biconcave disc, no nucleus** → RBC.
- **Small granular dot, in clusters, no nucleus** → platelet.
- **Multi-lobed nucleus + pale fine granules** → neutrophil.
- **Bilobed + bright red-orange granules** → eosinophil.
- **Bilobed obscured by dark blue granules** → basophil.
- **Round dense nucleus + thin cytoplasm** → small lymphocyte.
- **Indented nucleus + abundant gray-blue cytoplasm** → monocyte.

Clinical Correlation

Case — Bacterial Infection

A 7-year-old with fever and sore throat. CBC shows WBC 18,000 with 85% neutrophils and many band forms. The decisive feature is the left shift (bands) and neutrophilia.

Case — Parasitic Infection

A 25-year-old backpacker with eosinophilia. Stool ova and parasites show hookworm. The eosinophilia reflects IL-5-driven eosinophil release from marrow and recruitment to the parasite.

Case — Iron Deficiency Anaemia

A 30-year-old woman with menorrhagia has microcytic, hypochromic RBCs (smaller than normal, increased central pallor). Diagnosis: iron deficiency. The morphological signature is microcytosis + hypochromia.

Case — Acute Lymphoblastic Leukaemia (ALL)

A 7-year-old with fatigue, fever, bruising, bone pain. CBC: Hb 7, WBC 95,000 (80% blasts), platelets 30,000. Bone marrow shows monotonous blasts with high N:C ratio, fine chromatin, scant basophilic cytoplasm; TdT+, CD10+, CD19+ (B-ALL).

Lesson. ALL is the most common childhood malignancy; the decisive diagnostic feature is monotonous blast population replacing marrow.

Common Misconceptions

- Platelets are cells. They are anucleate cytoplasmic fragments.
- White blood cells are white in the body. "White" refers to the buffy coat on centrifugation.
- All granulocytes have multi-lobed nuclei. Eosinophils and basophils have bilobed nuclei.

- Lymphocytes are all the same. B, T, NK have distinct functions; morphology alone is insufficient.

High-Yield Knowledge

Must Know

- RBC: anucleate biconcave disc $\sim 7.5\ \mu\text{m}$; HbA ($2\alpha + 2\beta$); ~ 120 -day life span.
- Platelets: megakaryocyte fragments; primary haemostasis.
- Neutrophils: multi-lobed nucleus, neutral granules; bacterial phagocytosis.
- Eosinophils: bilobed, bright red-orange granules; parasites, allergy.
- Basophils: bilobed obscured, dark blue granules; histamine, immediate hypersensitivity.
- Lymphocytes: round dense nucleus, thin cytoplasm; adaptive immunity.
- Monocytes: indented nucleus, gray-blue cytoplasm; macrophage precursors.

Must Distinguish

- Three granulocytes by granule colour.
- Lymphocyte from monocyte.
- RBC pathological forms (spherocyte, sickle, target, schistocyte).

Must Understand

- Hypersegmented neutrophils $\rightarrow$ megaloblastic anaemia.
- Band neutrophils $\rightarrow$ left shift in infection.
- Eosinophilia $\rightarrow$ allergy or parasite.

Integrated Summary

Blood is a fluid connective tissue consisting of plasma and formed elements. RBCs are anucleate, biconcave discs packed with haemoglobin that transport O_2 and CO_2 and live ~ 120 days before splenic clearance. Platelets are megakaryocyte fragments that initiate primary haemostasis. The five leukocytes — neutrophils (multi-lobed, bacterial defence), eosinophils (bilobed, red-orange granules, parasites/allergy), basophils (bilobed obscured, dark blue granules, immediate hypersensitivity), lymphocytes (round dense nucleus, adaptive immunity), and monocytes (indented nucleus, macrophage precursors) — are recognised by nuclear morphology and granule colour.

Mastery Check

1. Compare the five leukocytes by nuclear morphology, granule colour, and function.
2. Why are mature RBCs anucleate, and how does this design optimise O_2 transport?
3. What are the stages of platelet plug formation?

4. Distinguish spherocytes, sickle cells, target cells, and schistocytes by appearance and disease.
5. Why is monocytosis associated with chronic infections?

Advanced Concepts

Neutrophil Detail

Neutrophils are the principal cells of acute inflammation. Key features:

- **Half-life** — 6–8 hours in blood; 1–4 days in tissue.
- **Granules:**
 - Primary (azurophilic) — myeloperoxidase (MPO), defensins, lysozyme, elastase, cathepsin G. MPO converts $\text{H}_2\text{O}_2 + \text{Cl}^- \rightarrow \text{HOCl}$ (bleach); deficiency → chronic granulomatous disease-like (actually MPO deficiency is milder).
 - Secondary (specific) — collagenase, lysozyme, lactoferrin, NADPH oxidase components.
 - Tertiary (gelatinase) — gelatinase, lysozyme.
- **Receptors** — FcγRIII (CD16), complement receptors (CR1, CR3), L-selectin, β2 integrins (LFA-1, Mac-1), TLRs, chemokine receptors.
- **Function** — phagocytosis, degranulation, NETosis (neutrophil extracellular traps = DNA + histones + granule proteins that trap bacteria).

Recruitment cascade:

1. **Rolling** — selectins (E-selectin on endothelium, P-selectin, L-selectin on leukocytes) bind sialyl-Lewis X.
2. **Activation** — chemokines (IL-8, CXCL8) activate integrins.
3. **Firm adhesion** — β2 integrins bind ICAM-1.
4. **Transmigration** — diapedesis between endothelial cells (paracellular) or through them (transcellular).
5. **Chemotaxis** — toward chemoattractants (C5a, LTB4, IL-8, bacterial peptides).

Neutrophilia (infection, inflammation, leukaemia); neutropenia (chemotherapy, viral infection, autoimmune); left shift (band neutrophils in circulation, immature).

Eosinophil Detail

Eosinophils mediate parasitic infections and allergy. Granules contain:

- Major basic protein (MBP) — toxic to parasites.
- Eosinophil cationic protein (ECP) — RNase activity.
- Eosinophil peroxidase — less potent than MPO.
- Eosinophil-derived neurotoxin (EDN) — RNase activity.

Eosinophil surface receptors: CCR3 (eotaxin receptor), FcεRI (IgE), CR1. Eosinophilia in allergy, parasitic infection, hypereosinophilic syndrome, certain leukaemias (chronic eosinophilic leukaemia). Curschmann spirals (extruded mucus casts) and Charcot-Leyden crystals (galectin-10, from eosinophil breakdown) are seen in sputum of asthmatics.

Lymphocyte Subsets (extended)

- **T cells** — mature in thymus. Subtypes:
 - CD4+ T helper (Th) — recognise MHC II; subtypes Th1 (IFN- γ , cell-mediated immunity), Th2 (IL-4, IL-5, IL-13, humoral/allergy), Th17 (IL-17, neutrophil recruitment), Tfh (B cell help in germinal centres), Treg (IL-10, TGF- β , immune tolerance).
 - CD8+ T cytotoxic (Tc) — recognise MHC I; kill via perforin/granzyme and FasL.
- **B cells** — mature in bone marrow; produce immunoglobulins; present antigen to CD4+ T cells; form germinal centres after activation.
- **NK cells** — innate lymphoid cells; kill via perforin/granzyme and ADCC; recognise missing-self (MHC I downregulation).
- **NKT cells** — express both T-cell receptor and NK markers; recognise CD1d-presented glycolipids.

Recirculation: naive T cells enter lymph nodes via HEVs; activated T cells up-regulate tissue-homing receptors (CLA for skin, $\alpha 4\beta 7$ integrin + CCR9 for gut).

Transition

Blood cells die in days to months. Where do they come from? The next chapter follows blood from the bone marrow, the body's most prolific haematopoietic factory.

Rapid Review

- Blood: plasma + formed elements (RBCs, WBCs, platelets).
- RBC: anucleate biconcave $\sim 7.5\ \mu\text{m}$; Hb; ~ 120 -day life span.
- WBCs:
 - Neutrophil: multi-lobed, neutral granules, bacteria.
 - Eosinophil: bilobed, bright red granules, parasites/allergy.
 - Basophil: bilobed obscured, dark blue granules, histamine/hypersensitivity.
- Lymphocyte: round dense nucleus, thin cytoplasm, adaptive immunity.
- Monocyte: indented nucleus, gray-blue cytoplasm, macrophage precursor.
- Platelets: megakaryocyte fragments; primary haemostasis.
- Haematocrit: % RBC volume ($\sim 45\%$ men).

Chapter 10 — Hemopoiesis

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 13

Opening Question

How does the bone marrow make 200 billion red cells a day without making a tumour — and how does it keep the lineages separated while still calling on a single stem-cell reserve?

Why This Matters

Haemopoiesis is essential for life. Marrow failure (aplastic anaemia, leukaemia, myelodysplastic syndrome) is rapidly fatal without transplant. Histology of the marrow and the recognition of immature precursors (blasts, promyelocytes, erythroblasts) is central to diagnosis.

Learning Objectives

1. Describe the hierarchy of haematopoietic stem and progenitor cells.
2. Explain the regulation of haemopoiesis by growth factors (EPO, TPO, G-CSF, GM-CSF, interleukins).
3. Identify the principal stages of erythroid, granulocytic, and megakaryocytic development.
4. Describe the morphology of erythroblasts, myeloblasts, and megakaryocytes.
5. Explain the difference between red and yellow marrow and the concept of red-marrow hyperplasia.
6. Recognise the histological signatures of marrow failure and leukaemia.

Big Picture

- **Sites** — yolk sac (early embryo) → liver and spleen (fetus) → bone marrow (from ~20 weeks; sole site in adult).
- **Hierarchy** — HSC (self-renew + multipotent) → MPP → CMP / CLP → committed progenitors → mature cells.
- **Regulation** — EPO (RBC), TPO (platelet), G-CSF (neutrophil), GM-CSF (granulocyte/macrophage), IL-3, IL-7.
- **Erythropoiesis** — proerythroblast → basophilic → polychromatophilic → orthochromatic → reticulocyte → RBC.
- **Granulopoiesis** — myeloblast → promyelocyte → myelocyte → metamyelocyte → band → segmented.
- **Megakaryopoiesis** — large polyploid megakaryocytes extend cytoplasmic projections through sinusoidal endothelium and fragment into platelets.

Core Concept

Haemopoiesis is the production of blood cells from a hierarchy of stem and progenitor cells in the bone marrow, regulated by colony-stimulating factors and other cytokines. Morphology reflects synthetic activity.

Structure

Sites

- **Yolk sac** (3–8 weeks): primitive haemopoiesis, nucleated RBCs in blood islands.
- **Liver** (6 weeks to birth): major fetal site; produces RBCs, granulocytes, platelets.
- **Spleen** (10–28 weeks): contributes.
- **Bone marrow** (from ~20 weeks): dominant from late gestation; the sole site of normal adult haemopoiesis.

In adults: red marrow (haematopoietically active) in vertebrae, sternum, ribs, pelvis, proximal femurs and humeri. Yellow marrow (adipose) replaces red marrow in long-bone diaphyses with age. In severe anaemia, yellow marrow converts back to red (red-marrow hyperplasia).

Stem Cell Hierarchy

- **HSCs** — rare (~1 in 10^4 – 10^5 marrow cells); CD34+, CD38–, lineage-negative; self-renew + multipotent.
- **MPPs** — committed to differentiation, no self-renew.
- **CMPs** (common myeloid) → GMPs (granulocyte-monocyte) + MEPs (megakaryocyte-erythroid).
- **CLPs** (common lymphoid) → B, T, NK precursors.
- **Precursors** → mature cells.

Growth Factors

- **EPO** (erythropoietin) — kidney (peritubular interstitial cells) in response to hypoxia; RBC production.
- **TPO** (thrombopoietin) — liver; platelet production.
- **G-CSF** — neutrophil production.
- **GM-CSF** — granulocyte and monocyte production.
- **IL-3** — multi-lineage.
- **IL-7** — T-cell development in thymus.

Growth Factor Detail and Clinical Use

EPO is the prototypical hypoxia-driven growth factor. The kidney senses arterial O_2 tension via peritubular interstitial cells; hypoxia → HIF-1 α stabilisation → EPO transcription → secretion → marrow stimulation of erythroid progenitors → increased reticulocyte release. Recombinant EPO is used to treat anaemia of chronic kidney disease and chemotherapy-induced anaemia; it is also misused in sports doping (increases O_2 -carrying capacity). Polycythaemia vera is an acquired JAK2 mutation that makes erythroid progenitors EPO-independent.

TPO binds the MPL receptor on megakaryocyte progenitors; constitutive hepatic production means platelet mass is mainly regulated by clearance of

TPO by platelets themselves. Thrombopoietin mimetics (eltrombopag, romiplostim) are used to treat immune thrombocytopenia.

G-CSF is released by macrophages, endothelial cells, and fibroblasts in response to bacterial products; it stimulates neutrophil release from the marrow (mature neutrophil pool shifts from marginating to circulating) and increases production. Recombinant G-CSF (filgrastim) is used to accelerate neutrophil recovery after chemotherapy.

GM-CSF stimulates both granulocyte and monocyte production; recombinant GM-CSF (sargramostim) is used after bone-marrow transplant.

IL-7 is produced by thymic epithelial cells and is essential for T-cell development. Therapeutic IL-7 is being investigated for immune reconstitution in HIV and after transplant.

Erythropoiesis

Morphological stages (in marrow):

Stage	Nucleus	Cytoplasm
Proerythroblast (rubriblast)	Large round, prominent nucleolus	Basophilic
Basophilic erythroblast (prorubricyte)	Smaller, basophilic	Basophilic
Polychromatophilic erythroblast (rubricyte)	Smaller	Gray-blue to pink (Hb synthesis)
Orthochromatic erythroblast (metarubricyte)	Pyknotic	Pink (Hb dominant)
Reticulocyte	Extruded nucleus	RNA visible with supravital stain
Mature erythrocyte	No nucleus	Hb only

Transit time ~7 days; reticulocytes mature in circulation within 1-2 days.

Granulopoiesis

Stage	Nucleus	Granules
Myeloblast	Large round, prominent nucleoli	None
Promyelocyte	Slightly indented	Azurophilic (primary, lysosomal)
Myelocyte	Last mitotic stage	Specific (lineage-specific)
Metamyelocyte	Indented (non-mitotic)	Both
Band cell	Band-shaped	Both
Mature granulocyte	Multi-lobed	Both

Megakaryopoiesis

Megakaryocytes are large (50–100 µm), polyploid (up to 64N) cells derived from MEPs. They extend cytoplasmic projections through the sinusoidal endothelium and fragment into platelets in the circulation.

Function

The marrow produces ~200 billion RBCs, ~10 billion neutrophils, and ~400 billion platelets daily in a normal adult. EPO from the kidney drives RBC production in response to tissue hypoxia. TPO from the liver drives platelet production. G-CSF drives neutrophil production, especially in infection.

Structure → Function

- **Basophilic cytoplasm of proerythroblast** = abundant RER synthesising globin.
- **Polychromatophilic cytoplasm** = globin synthesis dominating over ribosomal RNA → colour shifts blue → pink.
- **Pyknosis of late erythroblast** = nuclear condensation before extrusion.
- **Anucleate reticulocyte** = cytoplasmic maturity without nuclear baggage.
- **Azuropilic granules in promyelocyte** = lysosomal enzymes (the myeloperoxidase that defines myeloid lineage).
- **Specific granules in myelocyte** = lineage commitment (neutrophil, eosinophil, basophil).
- **Multi-lobed nucleus** = nuclear deformability for tissue migration.
- **Polyploid megakaryocyte** = single massive cell that can fragment into thousands of platelets.

Compare & Distinguish

Erythroid vs Granulocytic vs Megakaryocytic Lineage

Lineage	Earliest identifiable	Distinguishing feature
Erythroid	Proerythroblast	Basophilic → polychromatophilic → pink cytoplasm
Granulocytic	Myeloblast	Azuropilic granules in promyelocyte; specific granules in myelocyte
Megakaryocytic	Megakaryoblast	Large polyploid cell with multilobed nucleus

Red Marrow vs Yellow Marrow

Feature	Red	Yellow
Activity	Haematopoietically active	Replaced by adipose
Location	Vertebrae, sternum, ribs, pelvis, proximal long bones	Long-bone diaphyses
Response to anaemia	Hyperplasia	Converts back to red

Recognition Logic

- **Marrow with mixed haematopoietic precursors and megakaryocytes + fat spaces** → normal marrow.
- **Hypercellular marrow with reduced fat** → hyperplasia (anaemia, infection, leukaemia).
- **Hypocellular marrow with abundant fat** → aplastic anaemia.
- **Marrow replaced by monotonous blast population** → leukaemia.
- **Marrow with granulomas or fibrosis** → infiltrative disease.

Clinical Correlation

Case — Aplastic Anaemia

A 25-year-old presents with fatigue, epistaxis, and infections. CBC: pancytopenia. Bone marrow biopsy: markedly hypocellular marrow replaced by fat. Diagnosis: aplastic anaemia.

Lesson. Stem-cell niche failure → pancytopenia. Treatment is allogeneic stem-cell transplant.

Case — Acute Myeloid Leukaemia

A 60-year-old with fatigue, fever, bleeding. CBC: WBC 80,000 with 70% blasts; Hb 7; platelets 20,000. Marrow: monotonous myeloblasts with Auer rods (eosinophilic needle-like inclusions of fused azurophilic granules), myeloperoxidase positive.

Lesson. Auer rods are pathognomonic of AML; MPO positivity confirms myeloid lineage.

Case — Polycythaemia Vera

A 55-year-old with plethora, headache, splenomegaly. CBC: Hb 20, RBC 8 million, WBC and platelets elevated. JAK2 V617F mutation positive. Bone marrow hypercellular with panmyelosis.

Lesson. Acquired JAK2 mutation drives EPO-independent proliferation of all three lineages.

Common Misconceptions

- All marrow is haematopoietically active. Yellow marrow is fat; only red marrow makes blood cells.
- All blasts are leukaemic. Normal marrow contains a small percentage of blasts (<5%). The diagnosis of leukaemia requires >20% blasts (WHO).
- Erythropoiesis happens in blood. It happens in marrow; reticulocytes are released.

High-Yield Knowledge

Must Know

- HSC → MPP → CMP/CLP → precursors → mature cells.
- EPO (RBC), TPO (platelet), G-CSF (neutrophil), GM-CSF, IL-7 (T).

- Erythroid stages: proerythroblast → basophilic → polychromatophilic → orthochromatic → reticulocyte → RBC.
- Granulocytic stages: myeloblast → promyelocyte → myelocyte → metamyelocyte → band → segmented.
- Megakaryocytes: large polyploid cells that fragment into platelets.

Must Distinguish

- Red marrow vs yellow marrow.
- Erythroid lineage from granulocytic lineage by cytoplasm colour progression.
- Myeloblast from lymphoblast (cytoplasmic granules absent in both, but lymphoid markers CD19/TdT vs myeloid MPO/CD33).

Must Understand

- Red-marrow hyperplasia in chronic anaemia.
- Aplastic anaemia as niche failure.
- Leukaemia as clonal proliferation with maturation arrest.

Integrated Summary

Haemopoiesis is the production of blood cells from a hierarchy of HSCs in the marrow, regulated by colony-stimulating factors. The erythroid lineage progresses through proerythroblast, basophilic erythroblast, polychromatophilic erythroblast, orthochromatic erythroblast, reticulocyte, and mature erythrocyte over ~7 days; cytoplasm colour shifts from basophilic (RER) to pink (haemoglobin). The granulocytic lineage progresses through myeloblast, promyelocyte (azurophilic granules), myelocyte (specific granules), metamyelocyte, band, and mature segmented neutrophil. Megakaryocytes are large polyploid cells that fragment into platelets.

Mastery Check

1. Reconstruct the HSC hierarchy from stem cell to mature blood cell.
2. Trace erythropoiesis stage by stage with morphological features.
3. Trace granulopoiesis stage by stage with granule composition.
4. Why are megakaryocytes polyploid?
5. What is the difference between red and yellow marrow, and when does red-marrow hyperplasia occur?

Advanced Concepts

Stem Cell Niche Detail

Haematopoietic stem cells (HSCs) reside in a specific niche that regulates self-renewal and differentiation:

- **Endosteal niche** — near the endosteum; quiescent HSCs; osteoblasts produce CXCL12 (SDF-1) and SCF; HSCs adhere via CXCR4.

- **Vascular niche** — near sinusoidal endothelium; activated HSCs; CXCL12 also expressed by sinusoidal endothelial cells; mobilisation (G-CSF, chemotherapy) drives HSC egress into circulation.

HSC mobilisation: G-CSF → neutrophil activation → proteolytic cleavage of CXCL12 and VCAM-1 → HSC detachment → mobilisation into circulation. Mobilised HSCs can be harvested by leukapheresis for transplantation.

HSC homing to bone marrow after transplant: HSC CXCR4 binds CXCL12 → integrin activation → firm adhesion → transendothelial migration → niche.

Erythropoiesis Detail

Erythropoiesis proceeds through morphologically recognisable stages:

1. **Proerythroblast** — large, basophilic, euchromatic nucleus; first committed erythroid progenitor.
2. **Basophilic erythroblast** — smaller, deeply basophilic; high RER for globin synthesis.
3. **Polychromatophilic erythroblast** — intermediate staining (basophilic + eosinophilic); haemoglobin synthesis begins.
4. **Orthochromatophilic erythroblast** — small, pyknotic nucleus; cytoplasm strongly eosinophilic (haemoglobin dominant).
5. **Reticulocyte** — nucleus extruded; residual RNA and organelles; released into blood; matures to erythrocyte in 1-2 days. Supravital staining (new methylene blue) reveals residual RNA as reticular network → reticulocyte count.

Erythropoietin (EPO) acts on CFU-E and proerythroblast; iron is essential (Fe^{2+} incorporated into protoporphyrin IX by ferrochelatase); vitamin B₁₂ and folate are essential for DNA synthesis (deficiency → megaloblastic anaemia with nuclear-cytoplasmic asynchrony).

Reticulocyte count is the most useful index of effective erythropoiesis: increased in haemolysis or post-iron/B₁₂ replacement; decreased in aplastic anaemia, renal failure (EPO deficiency), marrow infiltration.

Granulopoiesis Detail

Granulopoiesis proceeds through myeloblast → promyelocyte → myelocyte → metamyelocyte → band → segmented (mature neutrophil).

- **Myeloblast** — large, euchromatic nucleus with prominent nucleoli; no granules; first recognisable granulocyte precursor.
- **Promyelocyte** — azurophilic (primary) granules appear (peroxidase, defensins).
- **Myelocyte** — specific (secondary) granules appear; last mitotic stage.
- **Metamyelocyte** — kidney-shaped nucleus; non-mitotic; nuclear indentation.
- **Band** — horseshoe-shaped nucleus; non-mitotic; immature circulating neutrophil.
- **Segmented** — multi-lobed nucleus (3-5 lobes); mature.

Specific granules determine cell identity (neutrophil, eosinophil, basophil). Granule content is packaged during the myelocyte stage; mutations

affecting granule formation cause specific granule deficiency.

G-CSF shortens the transit time of the granulocytic series and shifts the marginating pool to the circulating pool; used clinically to accelerate neutrophil recovery after chemotherapy.

Megakaryopoiesis and Platelet Production

Megakaryocytes are giant polyploid cells (up to 64N) that produce platelets by cytoplasmic fragmentation:

- 1. Megakaryoblast** — committed progenitor.
- 2. Promegakaryocyte** — endomitosis (DNA replication without cell division → polyploidy).
- 3. Megakaryocyte** — large polylobulated nucleus; abundant cytoplasm with platelet-specific granules; demarcation membrane system (invaginations of plasma membrane that partition the cytoplasm into future platelets).
- 4. Proplatelets** — cytoplasmic extensions that protrude through sinusoidal endothelium into the blood; platelets shear off from the proplatelet tips.

Platelets are anucleate cell fragments ~2-3 μm with α -granules (PDGF, TGF- β , vWF, fibrinogen, PF4) and δ -granules (dense bodies: ADP, ATP, serotonin, Ca^{2+}).

Transition

New lymphocytes produced in the marrow must be educated before they can defend. The next chapter enters the lymphoid organs — bone marrow, thymus, lymph nodes, spleen, MALT — where naive lymphocytes are selected and where they meet antigen.

Rapid Review

- Hierarchy: HSC → MPP → CMP/CLP → committed progenitors → mature cells.
- Growth factors: EPO, TPO, G-CSF, GM-CSF, IL-3, IL-7.
- Erythroid: proerythroblast → basophilic → polychromatophilic → orthochromatic → reticulocyte → RBC.
- Granulocytic: myeloblast → promyelocyte → myelocyte → metamyelocyte → band → segmented.
- Megakaryocyte: large polyploid cell; cytoplasmic fragmentation → platelets.
- Red marrow in axial skeleton; yellow marrow in long bones.

Chapter 11 — The Immune System and Lymphoid Organs

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 14

Opening Question

Why does losing one gland in the newborn chest leave a child defenseless against viruses for life — and how does the body separate the sites where lymphocytes are taught from the sites where they are tested?

Why This Matters

The immune system defends against infection, surveils for malignancy, and is the basis of vaccination. Its dysfunction causes immunodeficiency (HIV, SCID), autoimmunity (lupus, RA), and lymphoid malignancy (lymphoma, leukaemia). Histology of the lymphoid organs is central to diagnosis.

Learning Objectives

1. Distinguish primary (central) from secondary (peripheral) lymphoid organs.
2. Describe the structure and function of the thymus.
3. Describe the structure and function of the lymph node.
4. Describe the structure and function of the spleen.
5. Describe the organisation of mucosa-associated lymphoid tissue (MALT).
6. Distinguish the lymphoid organs histologically.

Big Picture

- **Primary** — bone marrow (B maturation), thymus (T maturation). Site of lymphocyte production and education.
- **Secondary** — lymph node (filters lymph), spleen (filters blood), MALT (mucosal defence). Site where lymphocytes meet antigen.
- **T zones** — paracortex (node), periarteriolar lymphoid sheath PALS (spleen).
- **B zones** — follicles with germinal centres (cortex of node, white pulp of spleen).
- **Conduits** — lymph node has afferent/efferent lymphatics and HEVs; spleen has central arteriole, no afferent lymphatics.

Core Concept

The lymphoid system separates education (primary organs, where self-reactive cells are deleted and self-MHC restriction is imposed) from execution (secondary organs, where naive cells meet antigen and mount immune responses).

Structure

Primary Lymphoid Organs

Bone Marrow (also secondary site for B cells)

Site of B-cell maturation from pro-B → pre-B → immature B → mature naive B. Also site of haematopoiesis (Chapter 10). Provides the progenitors for T cells that migrate to the thymus.

Thymus

Located in the mediastinum. Two lobes, each divided by septa into lobules. Each lobule has a cortex and a medulla.

- **Cortex** — densely packed immature thymocytes; dark-staining. Site of positive selection (T cells whose TCRs recognise self-MHC survive; the rest die by apoptosis).
- **Medulla** — paler; mature thymocytes, epithelial reticular cells, Hassall's corpuscles (concentric whorls of keratinised epithelial reticular cells). Site of negative selection (T cells whose TCRs strongly recognise self-antigen are deleted).
- **Epithelial reticular cells** — form a meshwork; cortical and medullary subtypes have different functions.
- **Macrophages** — phagocytose apoptotic thymocytes.
- **Dendritic cells** — contribute to negative selection.
- **Blood-thymus barrier** — continuous endothelium + basal lamina + epithelial reticular cells; prevents premature antigen exposure of developing T cells.

Thymic involution begins at puberty; by old age, the thymus is largely fat. Consequence: reduced naive T-cell output.

Secondary Lymphoid Organs

Lymph Node

Bean-shaped organ along lymphatic vessels. Capsule of dense CT.

- **Cortex** — B-cell follicles: primary (small, quiescent) and secondary (larger, with pale germinal centre where B cells proliferate, undergo somatic hypermutation and class-switch recombination, and are selected for high-affinity receptors). Surrounding darker mantle zone of small naive B cells.
- **Paracortex** — T-cell zone; contains T cells and dendritic cells. High-endothelial venules (HEVs) — the entry point for circulating lymphocytes into the node.
- **Medulla** — medullary cords (B cells, plasma cells, macrophages) and medullary sinuses (irregular vascular channels carrying lymph).
- **Hilum** — efferent lymphatic exit; arteries and veins.

Lymph flow: afferent lymphatics → subcapsular sinus → cortical sinuses → medullary sinuses → efferent lymphatic at hilum. Antigens captured by macrophages and dendritic cells and presented to lymphocytes in cortex and paracortex.

Lymph Node Detail

Germinal centres are not single cell types; they are microanatomical structures with a dark zone (rapidly proliferating centroblasts, AID-expressing, undergoing somatic hypermutation and class-switch recombination), a light zone (centrocytes + follicular dendritic cells presenting antigen + T follicular helper cells), and tingible-body macrophages (phagocytosing apoptotic B cells that failed selection). The dark zone is "dark" because the densely packed large basophilic blasts; the light zone is "light" because of larger centrocytes and the FDC network.

Follicular dendritic cells (FDCs) are not derived from haematopoietic stem cells; they are stromal cells of mesenchymal origin that retain antigen-antibody complexes on their surface via Fc receptors for long periods, presenting antigen to B cells during affinity maturation.

High-endothelial venules (HEVs) are specialised post-capillary venules with cuboidal endothelium expressing addressins (PNAd) that bind L-selectin on circulating lymphocytes, enabling lymphocyte homing into the node. The HEV is the gateway for adaptive immune surveillance.

Medullary sinuses contain macrophages that phagocytose debris from the lymph and antigens that bypassed the cortex. Medullary cords contain plasma cells producing antibodies that enter the efferent lymph.

Spleen

Largest lymphoid organ; filters blood.

- **Capsule** of dense CT with smooth muscle; trabeculae extend into parenchyma, dividing into lobules.
- **White pulp** — lymphoid tissue around central arterioles. Contains PALS (T cells) and follicles (B cells, with germinal centres when active).
- **Red pulp** — bulk of spleen; sinusoids (large fenestrated vessels) and cords of Billroth (splenic cords of reticular tissue and macrophages).

Blood flow: splenic artery → central arteriole (white pulp) → penicillar arteriole → capillary → either directly into splenic sinuses (closed circulation) or into cords of Billroth (open circulation, where blood percolates through and is squeezed through the slits between sinusoidal endothelial cells — a mechanical test that removes abnormal or aged RBCs) → splenic sinuses → splenic vein.

No afferent lymphatics. Spleen is not essential for life; splenectomy increases risk of overwhelming infection by encapsulated bacteria (*S. pneumoniae*, *H. influenzae*, *N. meningitidis*), especially in children.

MALT (Mucosa-Associated Lymphoid Tissue)

Largest lymphoid organ by mass. Includes:

- **Tonsils** (pharyngeal, palatine, lingual): aggregates with crypts.
- **Peyer's patches** in the ileum: lymphoid follicles with overlying M cells that sample luminal antigens.
- **Appendix**: lymphoid tissue in cecal wall.
- **BALT** (bronchus-associated) and lymphoid tissue in the urogenital tract.

Unencapsulated, in mucosal walls.

Function

Bone marrow produces all blood cells and matures B cells. Thymus educates T cells (positive and negative selection). Lymph node filters lymph and initiates immune responses to lymph-borne antigens. Spleen filters blood, recycles aged RBCs, stores platelets, and initiates immune responses to blood-borne antigens. MALT defends mucosal surfaces.

Structure → Function

- **Thymus cortex densely packed + medulla with Hassall's** = site of T-cell selection and the diagnostic landmark.
- **Germinal centres in secondary follicles** = B-cell proliferation, hypermutation, selection; the centre is pale (large euchromatic blasts), the mantle is dark (small resting B cells).
- **HEVs in paracortex** = the gateway for circulating lymphocytes to enter the node.
- **Splenic open circulation** = the mechanical test that removes abnormal/aged RBCs.
- **M cells overlying Peyer's patches** = the gateway for luminal antigen to enter mucosal immune surveillance.

Compare & Distinguish

Thymus vs Lymph Node

Feature	Thymus	Lymph node
Capsule	Yes	Yes
Cortex/medulla	Cortex = immature T; medulla = mature T + Hassall's	Cortex = B follicles; paracortex = T; medulla = cords + sinuses
Germinal centres	No	Yes
Afferent lymphatics	No	Yes
Function	T education	Lymph filtration

Lymph Node vs Spleen

Feature	Lymph node	Spleen
Filters	Lymph	Blood
Afferent lymphatics	Yes	No
T zone	Paracortex	PALS
B zone	Cortex follicles	White pulp follicles
Sinusoids	Medullary	Red pulp

Recognition Logic

- **Lobules with cortex + medulla + Hassall's corpuscles** → thymus.

- **Bean-shaped organ with cortex (follicles) + paracortex + medulla (cords/sinuses) → lymph node.**
- **Organ with white pulp (around central arterioles) + red pulp (sinusoids + cords) → spleen.**
- **Lymphoid tissue in mucosa, no capsule, often with follicles → MALT.**

Clinical Correlation

Case — DiGeorge Syndrome

A newborn with hypocalcaemic tetany and recurrent viral infections. Chest X-ray shows absent thymic shadow. Diagnosis: 22q11 deletion → failure of 3rd and 4th pharyngeal pouch development → thymic aplasia + parathyroid aplasia.

Lesson. T-cell deficiency + hypocalcaemia (parathyroid) is the diagnostic pair.

Case — Follicular Lymphoma

A 50-year-old with painless lymphadenopathy. Biopsy shows effaced architecture by monotonous CD20+, BCL6+, BCL2+ lymphoid cells.

Lesson. BCL2 overexpression (t(14;18)) prevents apoptosis of germinal-centre B cells; the neoplastic cells accumulate as tumour. The decisive marker is BCL2 in germinal centres (normal germinal centres are BCL2-negative).

Case — Splenectomy and Howell-Jolly Bodies

After splenectomy, RBCs show Howell-Jolly bodies (nuclear remnants). Normally the spleen removes these; their presence indicates asplenia or hyposplenism.

Lesson. The spleen removes nuclear remnants from RBCs; without it, the remnants persist.

Common Misconceptions

- All lymphocytes are made in the marrow. B cells mature in marrow; T cells mature in thymus.
- The thymus is essential throughout life. It involutes after puberty; naive T-cell output declines with age.
- The spleen is essential for life. It can be removed, but splenectomy increases risk of encapsulated-bacteria infection.
- Germinal centres are a single cell type. They have a dark zone (proliferating B cells), a light zone (selected B cells + FDCs), and tingible-body macrophages.

High-Yield Knowledge

Must Know

- Primary organs: marrow (B) and thymus (T); site of lymphocyte production and education.
- Secondary organs: lymph node (filters lymph), spleen (filters blood), MALT (mucosal defence).
- Thymus: cortex (positive selection) + medulla (negative selection, Hassall's corpuscles); involutes after puberty.
- Lymph node: cortex B follicles + germinal centres; paracortex T + HEVs; medulla cords + sinuses.
- Spleen: white pulp (PALS T + follicles B) + red pulp (sinusoids + cords of Billroth); open circulation filters aged RBCs.

Must Distinguish

- Thymus vs lymph node (no germinal centres vs germinal centres).
- Lymph node vs spleen (filters lymph vs blood; afferent lymphatics vs none).

Must Understand

- T zone = paracortex (node) or PALS (spleen).
- HEVs are the lymphocyte entry gate in the node.
- Splenectomy increases risk of encapsulated bacteria.

Integrated Summary

The lymphoid system separates primary (education) from secondary (execution) organs. Bone marrow matures B cells; thymus matures T cells with positive (cortex) and negative (medulla, Hassall's) selection. Lymph node filters lymph through cortex B follicles, paracortex T zone with HEVs, and medullary cords and sinuses. Spleen filters blood through white pulp (PALS + follicles) and red pulp (sinusoids + cords). MALT is unencapsulated mucosal lymphoid tissue including tonsils, Peyer's patches (with M cells), appendix, and BALT.

Mastery Check

1. Trace the flow of lymph through a lymph node.
2. Compare the open and closed circulations of the spleen.
3. Why is the thymus essential in childhood but dispensable in adulthood?
4. What is the role of M cells in Peyer's patches?
5. Why does splenectomy increase the risk of overwhelming infection by encapsulated bacteria?
6. Why is BCL2 expression in germinal centres a marker of follicular lymphoma?

Advanced Concepts

Lymphocyte Trafficking

Lymphocytes continuously recirculate between blood, lymphoid tissue, and lymph — a process essential for immune surveillance. Trafficking is governed by addressins (homing receptors) on endothelium and selectins/integrins on lymphocytes:

- **Naïve T cells** enter lymph nodes via HEVs using L-selectin (CD62L) → PNAd (peripheral node addressin) binding + CCR7 → CCL19/21 chemokine signalling.
- **Activated T cells** in tissues up-regulate tissue-specific homing receptors:
 - Skin-homing — CLA (cutaneous lymphocyte antigen) binds E-selectin on dermal endothelium.
 - Gut-homing — $\alpha 4\beta 7$ integrin binds MAdCAM-1 on gut endothelium; CCR9 binds CCL25.
 - Lung-homing — CCR4 binds CCL17.
- **Memory T cells** — central memory (CCR7+, in lymph nodes) vs effector memory (CCR7-, in tissues).

This compartmentalisation explains the regional specialisation of mucosal immunity (MALT: GALT in gut, BALT in bronchus, NALT in nasopharynx).

Antigen Presentation and MHC

- **MHC class I** — on all nucleated cells; presents endogenous peptides (8-10 aa) to CD8+ T cells; recognised after viral infection, tumour, intracellular pathogen. $\beta 2$ -microglobulin is the invariant light chain.
- **MHC class II** — on professional APCs (dendritic cells, macrophages, B cells); presents exogenous peptides (~15 aa) to CD4+ T cells.

Antigen processing:

- **Endogenous (cytosolic)** — proteasome cleaves proteins → TAP transporter to ER → MHC I binding → cell surface.
- **Exogenous** — endocytosis → lysosomal degradation → MHC II loaded in MHC II compartment → cell surface.

Dendritic cells (DCs) are the principal initiators of adaptive immune responses:

- **Conventional DC (cDC)** — antigen capture and presentation; migrate to lymph nodes after antigen uptake.
- **Plasmacytoid DC (pDC)** — produce massive type I IFN in viral infection.
- **Follicular DC (FDC)** — not from HSC; trap antigen-antibody complexes in B-cell follicles.
- **Langerhans cell** — epidermal dendritic cell; Birbeck granules (tennis racket-shaped, langerin/CD207+).

Hypersensitivity and Autoimmunity

Four classic hypersensitivity mechanisms:

- **Type I** (IgE-mediated, immediate) — allergens → IgE on mast cells → crosslinking → degranulation → histamine, leukotrienes; examples: allergic rhinitis, asthma, anaphylaxis, urticaria.

- **Type II** (antibody-mediated cytotoxic) — IgG/IgM against cell-surface antigens → complement-mediated lysis or ADCC; examples: autoimmune haemolytic anaemia, Goodpasture syndrome, myasthenia gravis, Graves disease (stimulating), transfusion reactions, Rh disease.
- **Type III** (immune complex) — antigen-antibody complexes deposit in tissues → complement activation → neutrophil recruitment → tissue damage; examples: serum sickness, post-streptococcal glomerulonephritis, SLE, Arthus reaction.
- **Type IV** (delayed-type, T-cell mediated) — CD4+ Th1 + macrophages + CD8+ Tc; 48–72 hours; examples: contact dermatitis (poison ivy), tuberculin (PPD) test, granulomatous inflammation (TB, sarcoid), transplant rejection, type 1 diabetes.

Autoimmunity arises from a combination of genetic susceptibility (HLA alleles, e.g., HLA-DR4 in RA, HLA-B27 in ankylosing spondylitis) and environmental triggers (infection, smoking, drugs). Central tolerance (negative selection in thymus) and peripheral tolerance (Treg, anergy, deletion) normally prevent autoreactive lymphocytes from causing disease; their failure → autoimmunity.

Transition

The immune system defends the body. The next chapter moves to the digestive tract, which is the largest interface with the outside world and uses all four tissue types — epithelium, connective tissue, muscle, nerve — to digest, absorb, and protect.

Rapid Review

- Primary: marrow (B), thymus (T).
- Thymus: cortex (positive selection, dense) + medulla (negative selection, Hassall's); involutes after puberty.
- Lymph node: capsule + cortex (B follicles + germinal centres) + paracortex (T + HEVs) + medulla (cords + sinuses); afferent + efferent lymphatics.
- Spleen: white pulp (PALS T + follicles B around central arteriole) + red pulp (sinusoids + cords of Billroth); no afferent lymphatics; open circulation filters aged RBCs.
- MALT: tonsils, Peyer's patches (M cells), appendix, BALT; unencapsulated.

Chapter 12 — The Digestive Tract

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 15

Opening Question

How does one tube — built from the same four tissue types — digest a steak, absorb fat, and protect itself from its own acid?

Why This Matters

The GI tract processes every nutrient that enters the body. Its dysfunctions (GERD, peptic ulcer, celiac disease, IBD, colorectal cancer) are among the most common human diseases. Most GI cancers are carcinomas of the mucosal epithelium.

Learning Objectives

1. Describe the four-layer organisation (mucosa, submucosa, muscularis externa, serosa/adventitia).
2. Identify the histological features of the esophagus, stomach, small intestine, and large intestine.
3. Explain the structural basis of digestion, absorption, and propulsion.
4. Describe the enteric nervous system (myenteric and submucosal plexuses).
5. Distinguish each region of the GI tract histologically.

Big Picture

- **Four layers** — mucosa (epithelium + lamina propria + muscularis mucosae), submucosa (CT + Meissner's plexus), muscularis externa (inner circular + outer longitudinal + Auerbach's plexus), serosa/adventitia.
- **Regional specialisations** — esophagus (stratified squamous, skeletal→smooth muscle transition), stomach (gastric pits, parietal HCl + chief pepsinogen), small intestine (villi + crypts; duodenum Brunner's, jejunum plicae, ileum Peyer's), large intestine (crypts, no villi, many goblet cells, taeniae coli).
- **Plexuses** — Meissner (submucosal, secretion), Auerbach (myenteric, motility). ENS is the largest subdivision of ANS.

Core Concept

The GI tract is a tube with a conserved four-layer organisation modified regionally for specialised function (ingestion → digestion → absorption → propulsion). The two enteric plexuses control secretion and motility independently of central input.

Structure

Four Layers

- **Mucosa** — epithelium (in contact with lumen) + lamina propria (loose CT with vessels, nerves, immune cells) + muscularis mucosae (1-3 smooth-muscle-cell layer that wrinkles the mucosa).
- **Submucosa** — dense CT with larger vessels, lymphatics, Meissner's (submucosal) plexus. Submucosal glands in some regions.
- **Muscularis externa** — typically inner circular + outer longitudinal smooth muscle; Auerbach's (myenteric) plexus between the layers.
- **Serosa or adventitia** — serosa (mesothelium + CT) for intraperitoneal regions; adventitia (dense CT) for retroperitoneal regions.

Enteric Nervous System

~200–600 million neurons (more than spinal cord). Two plexuses:

- **Myenteric (Auerbach's)** — between circular and longitudinal; controls motility.
- **Submucosal (Meissner's)** — in submucosa; controls secretion, local blood flow, mucosal sensation.

The ENS functions independently of CNS (can mediate peristalsis and secretion in response to luminal stimuli without central input).

Function

The GI tract processes nutrients: ingestion (mouth, esophagus), mechanical breakdown (stomach), chemical digestion (acid, pepsin, pancreatic enzymes, bile), absorption (small intestine), water reabsorption (colon), and storage/elimination (rectum). The ENS coordinates motility and secretion locally; the CNS modulates via parasympathetic (stimulatory) and sympathetic (inhibitory) input.

Structure → Function

- **Mucosa:** epithelium = barrier and transport; lamina propria = immune effector site; muscularis mucosae = mucosal movement (e.g., vomiting, villus contraction).
- **Submucosal plexus** modulates secretion and local blood flow.
- **Myenteric plexus** controls peristalsis (coordinated contraction of circular and longitudinal muscle).
- **Serosa vs adventitia:** serosa provides a smooth, low-friction surface within the peritoneal cavity; adventitia anchors the organ to retroperitoneal structures.

Regional Histology

Esophagus

- **Mucosa:** non-keratinised stratified squamous epithelium; mucus-secreting cardiac glands at upper and lower ends.
- **Submucosa:** esophageal glands proper (mucus-secreting, upper half).

- **Muscularis externa:** upper third skeletal; middle third mixed; lower third smooth. This transition is the key histological landmark.
- **Adventitia:** except for a short abdominal segment.

Esophagus Detail and Functional Anatomy

The esophagus is a muscular tube (~25 cm) that propels food by peristalsis. Its walls thicken from cranial to caudal: skeletal (upper third) → mixed (middle third) → smooth (lower third) as embryological origin (branchial arches) gives way to splanchnic mesoderm. The lower esophageal sphincter (LES) is a functional, not anatomic, sphincter — there is no visible thickening of the muscularis externa at the gastroesophageal junction. The LES is held closed by the sling fibres of the gastric cardia, the diaphragmatic crura, and the angle of His.

The stratified squamous non-keratinised epithelium is durable against abrasion but does not secrete mucus. The mucus for lubrication comes from submucosal mucous glands (proper mucous glands, distinct from the cardiac glands near the GE junction). Barrett's esophagus is the metaplastic change from stratified squamous to columnar with goblet cells (intestinal metaplasia) in response to chronic acid reflux; it is a precursor to esophageal adenocarcinoma. Esophageal varices are dilated submucosal veins in the lower esophagus from portal hypertension (cirrhosis); they are a major source of upper GI bleeding.

Auerbach's plexus (myenteric) lies between the longitudinal and circular layers of the muscularis externa and is required for peristalsis; loss of these neurons in Chagas disease causes megaesophagus. Meissner's plexus (submucosal) controls secretion and local blood flow.

Stomach

Four regions: cardia, fundus, body, pylorus.

- **Mucosa:** simple columnar epithelium of surface mucous cells. Gastric pits (foveolae) into which gastric glands open.
- **Fundic (gastric) glands** contain:
 - Surface and neck mucous cells — mucus protection.
 - Parietal (oxyntic) cells — eosinophilic (abundant mitochondria); secrete HCl and intrinsic factor (essential for B₁₂ absorption in ileum).
 - Chief (zymogenic) cells — basophilic (abundant RER); secrete pepsinogen and gastric lipase.
 - Enteroendocrine cells — G cells (gastrin), D cells (somatostatin).
 - Stem cells in the neck region; turnover ~4-7 days.
- **Muscularis externa:** three layers (inner oblique, middle circular — thickens into pyloric sphincter, outer longitudinal).
- **Acid secretion:** H⁺ from carbonic anhydrase, pumped by H⁺/K⁺-ATPase (target of proton-pump inhibitors like omeprazole); Cl⁻ via Cl⁻ channel; HCO₃⁻ exchanged for Cl⁻ across basolateral membrane.

Small Intestine

Three regions: duodenum (short, retroperitoneal), jejunum (longest free segment, intraperitoneal), ileum (longest, joins colon at ileocecal valve).

Mucosal specialisations (amplify surface area $\sim 600\times$):

- **Plicae circulares** (Kerckring's folds) — large circular folds of mucosa + submucosa, most prominent in jejunum; absent in lower ileum and colon.
- **Villi** — finger-like mucosal projections with lamina propria core, lacteal (lymphatic), capillaries, and simple columnar absorptive epithelium with brush border.
- **Microvilli** — apical microvilli forming the brush border.

Epithelium:

- Enterocytes — tall columnar, apical microvilli, basal nuclei; absorb nutrients; brush-border disaccharidases and peptidases.
- Goblet cells — mucus; more numerous distally.
- Enteroendocrine cells — CCK, secretin, GIP, somatostatin, motilin.
- Paneth cells — at base of crypts; eosinophilic apical granules containing lysozyme, defensins; mucosal defence.
- M cells — over Peyer's patches; sample luminal antigens.
- Stem cells — in crypts; turnover $\sim 3\text{--}5$ days.

Submucosa: Brunner's glands (duodenum only) — alkaline (HCO_3^- -rich) mucus to neutralise gastric acid.

Large Intestine

Four regions: cecum (with appendix), ascending colon, transverse colon, descending colon, sigmoid colon, rectum, anal canal.

- **Mucosa:** flat (no villi, no plicae); straight crypts of Lieberkühn with abundant goblet cells, absorptive columnar cells, stem cells, enteroendocrine cells.
- **Submucosa:** submucosal plexus.
- **Muscularis externa:** inner circular + outer longitudinal condensed into three teniae coli (visible grossly as three bands); teniae shorten the colon relative to the other layers $\rightarrow$ haustra (sacculations).
- **Appendix:** mucosa similar to colon (crypts + goblet cells), but densely infiltrated by lymphoid follicles (continuous ring of lymphoid tissue).

Anal Canal

Rectal columnar $\rightarrow$ transitional $\rightarrow$ non-keratinised stratified squamous at pectinate line. Above pectinate: visceral innervation (stretch, not pain); below: somatic innervation (pain, temperature, touch).

Compare & Distinguish

Esophagus vs Stomach vs Small Intestine vs Colon

Feature	Esophagus	Stomach	Small intestine	Colon
Surface epithelium	Stratified squamous non-keratinised	Simple columnar with gastric pits	Simple columnar with villi + crypts	Simple columnar with crypts (no villi)

Feature	Esophagus	Stomach	Small intestine	Colon
Glands in mucosa	None	Fundic glands (parietal + chief)	Crypts (Paneth at base)	Crypts (many goblet cells)
Submucosal glands	Esophageal glands proper (upper half)	None	Brunner's (duodenum only)	None
Muscularis externa	Skeletal→smooth transition	3 layers (oblique + circular + longitudinal)	2 layers	2 layers (taeniae coli)
Special	—	Pyloric sphincter	Plicae, villi, Peyer's patches	Taeniae coli, haustra

Duodenum vs Jejunum vs Ileum

Region	Submucosal glands	Villi	Peyer's patches
Duodenum	Brunner's	Tall, leaf-shaped	Few
Jejunum	None	Tall, finger-like	Few
Ileum	None	Shorter	Numerous

Recognition Logic

- **Stratified squamous non-keratinised + skeletal→smooth muscle transition** → esophagus.
- **Simple columnar with gastric pits + fundic glands with eosinophilic parietal cells** → stomach.
- **Simple columnar with villi + Brunner's glands in submucosa** → duodenum.
- **Simple columnar with villi + Peyer's patches (no Brunner's)** → ileum.
- **Simple columnar with crypts (no villi) + many goblet cells + taeniae coli** → colon.
- **Crypts + dense lymphoid tissue in muscular wall** → appendix.

Clinical Correlation

Case — Barrett's Esophagus

A 35-year-old with chronic GERD. Biopsy of distal esophagus shows columnar epithelium with goblet cells where stratified squamous should be. Diagnosis: Barrett's esophagus.

Lesson. Chronic acid injury → squamous-to-columnar (intestinal) metaplasia. The goblet cells specifically (not just any columnar epithelium) define Barrett's. Increased risk of esophageal adenocarcinoma.

Case — Peptic Ulcer

A 50-year-old man with epigastric pain. Endoscopy shows a gastric ulcer with clean base. Biopsy of the ulcer edge shows destruction of the mucosa and inflammation; *H. pylori* organisms visible on Giemsa stain.

Lesson. *H. pylori* infection + acid hypersecretion → mucosal ulceration. Treatment is PPI + antibiotics.

Case — Celiac Disease

A 30-year-old with chronic diarrhea, weight loss, and bloating. Duodenal biopsy shows villous atrophy, crypt hyperplasia, increased intraepithelial lymphocytes, and plasma-cell infiltrate in lamina propria. Anti-tissue transglutaminase antibodies positive.

Lesson. Immune reaction to gluten → enterocyte damage → malabsorption. Treatment is lifelong gluten-free diet; histology shows recovery on repeat biopsy.

Case — Hirschsprung Disease

A newborn with failure to pass meconium. Rectal biopsy shows absence of ganglion cells in submucosal and myenteric plexuses, with hypertrophied nerve bundles.

Lesson. Failure of neural crest migration → aganglionic segment cannot relax → functional obstruction. The decisive histological feature is absence of ganglion cells.

Common Misconceptions

- The GI tract has the same histology throughout. Four layers are conserved, but each region has distinctive specialisations.
- The stomach absorbs nutrients. Mostly water, alcohol, some drugs; small intestine is the principal absorption site.
- The colon absorbs nutrients. Water and electrolytes, not nutrients.
- Goblet cells are the same throughout the GI tract. They differ in abundance (most in colon) and mucin type.
- The ENS is purely autonomic. The ENS can function independently of the CNS; it is the "second brain."

High-Yield Knowledge

Must Know

- Four layers: mucosa, submucosa, muscularis externa, serosa/adventitia.
- ENS: myenteric (Auerbach's, motility) + submucosal (Meissner's, secretion).
- Esophagus: stratified squamous non-keratinised + skeletal→smooth muscle transition.
- Stomach: simple columnar with gastric pits; fundic glands with parietal HCl + chief pepsinogen.
- Small intestine: villi + crypts + microvilli (600× surface area amplification).
- Duodenum: Brunner's glands (alkaline mucus). Jejunum: plicae. Ileum: Peyer's patches.
- Colon: crypts, no villi, many goblet cells, taeniae coli.

Must Distinguish

- Esophagus vs stomach vs intestine by epithelium.
- Duodenum vs jejunum vs ileum by submucosal glands and Peyer's patches.
- Small intestine vs colon (villi vs no villi).

Must Understand

- Acid secretion: H^+/K^+ -ATPase is target of PPIs.
- Celiac: villous atrophy + intraepithelial lymphocytes + anti-tTG.
- Barrett: columnar with goblet cells in distal esophagus.

Integrated Summary

The GI tract is a tube with a conserved four-layer organisation (mucosa, submucosa, muscularis externa, serosa/adventitia). The mucosa is the variable layer: stratified squamous in esophagus, simple columnar with gastric pits in stomach, simple columnar with villi in small intestine, simple columnar with crypts in colon. Submucosal glands are characteristic of esophagus (esophageal glands proper) and duodenum (Brunner's glands). The ENS (myenteric + submucosal plexuses) controls motility and secretion independently of CNS input.

Mastery Check

1. Compare the four layers of the GI tract by component and function.
2. Reconstruct the cell types of the fundic gland and their secretions.
3. Why are PPIs effective for acid-related disease?
4. Compare duodenum, jejunum, ileum, and colon histologically.
5. Explain the pathogenesis of celiac disease and its histological features.
6. What is Hirschsprung disease and what is the histological finding?

Advanced Concepts

Gastric Acid Secretion Mechanism

The parietal cell is the most sophisticated acid-secreting cell in the body. Resting parietal cells have tubulovesicles containing H^+/K^+ -ATPase in the cytoplasm. On stimulation:

1. **Histamine** (from ECL cells) → H_2 receptor → G_s → cAMP → PKA.
2. **Gastrin** (from G cells) → CCK-B receptor → G_q → PLC → IP_3/Ca^{2+} .
3. **ACh** (from vagus) → M_3 receptor → G_q → PLC → IP_3/Ca^{2+} .

All three converge on PKA/Ca^{2+} → tubulovesicles fuse with apical membrane → H^+/K^+ -ATPase inserted → acid secretion.

Acid generation:

- $CO_2 + H_2O \rightarrow H_2CO_3$ (carbonic anhydrase) → $H^+ + HCO_3^-$.
- H^+ pumped into lumen by H^+/K^+ -ATPase (proton pump; target of PPIs).
- Cl^- follows via $ClC-2$ channel.

- HCO_3^- exchanged for Cl^- across basolateral membrane (anion exchanger AE2).

Parietal cells also secrete **intrinsic factor** (essential for B_{12} absorption in the terminal ileum); autoimmune gastritis → anti-parietal cell + anti-intrinsic factor antibodies → pernicious anaemia (B_{12} deficiency).

PPI pharmacokinetics: PPIs are prodrugs activated by acid; they irreversibly inhibit H^+/K^+ -ATPase (covalent cysteine modification), so acid secretion resumes only after new pump synthesis (~24 hours). Best taken 30–60 min before meals when parietal cells are most active.

Pancreatic Exocrine Activation Cascade

Pancreatic enzymes are secreted as inactive zymogens to prevent autodigestion. Activation is a controlled cascade in the duodenum:

1. **Trypsinogen** → **trypsin** by enterokinase (enteropeptidase) on the duodenal brush border. Trypsin is the master activator.
2. Trypsin activates:
 - Chymotrypsinogen → chymotrypsin.
 - Proelastase → elastase.
 - Procarboxypeptidase → carboxypeptidase.
3. Trypsin also activates phospholipase A2 and procolipase (essential for lipid digestion).

Failure of activation: trypsinogen gene mutations (cationic trypsinogen PRSS1) → premature intra-pancreatic activation → hereditary pancreatitis.

Premature activation within the pancreas (trypsin autodigestion) → acute pancreatitis; common causes: gallstones (ampullary obstruction), alcohol, hypertriglyceridaemia, hypercalcaemia, drugs, ERCP, trauma. Trypsin also activates complement and kinins → systemic inflammation, ARDS, DIC.

Intestinal Absorption Mechanisms

- **Glucose/galactose** — SGLT1 (Na^+ -coupled cotransporter, apical); GLUT2 (basolateral).
- **Fructose** — GLUT5 (apical facilitated diffusion); GLUT2 (basolateral).
- **Amino acids** — various apical transporters (B^0 , X^-AG , b^0 , +); Na^+ -coupled.
- **Di- and tripeptides** — PepT1 (H^+ -coupled); broken down to amino acids in cytosol.
- **Long-chain fatty acids and monoglycerides** — diffuse across apical membrane; re-esterified to TG in ER; packaged with apolipoproteins (ApoB-48, ApoA-IV) into chylomicrons; secreted by exocytosis into lacteals (lymphatics); reach blood via thoracic duct.
- **Short-chain fatty acids** — absorbed directly into portal blood.
- **Iron** — DMT1 (apical, Fe^{2+} uptake); ferroportin (basolateral export); hephaestin (oxidises Fe^{2+} → Fe^{3+} for binding to transferrin).
- **Vitamin B_{12}** — intrinsic factor- B_{12} complex absorbed in terminal ileum via cubilin receptor.
- **Calcium** — transcellular (duodenum, vitamin D-dependent); paracellular (jejunum, ileum, concentration-dependent).

GI Hormones

- **Gastrin** (G cells, antrum) — stimulates acid secretion; growth of gastric mucosa.
- **CCK** (I cells, duodenum, jejunum) — stimulates pancreatic enzyme secretion, gallbladder contraction, relaxes sphincter of Oddi; satiety signal.
- **Secretin** (S cells, duodenum) — stimulates pancreatic bicarbonate secretion.
- **GIP** (K cells, duodenum, jejunum) — glucose-dependent insulinotropic peptide; insulin release in response to oral glucose (incretin effect).
- **GLP-1** (L cells, ileum, colon) — incretin; insulin release, glucagon suppression, satiety, gastric emptying delay. GLP-1 agonists (exenatide, liraglutide, semaglutide) used in type 2 diabetes and obesity.
- **Somatostatin** (D cells) — inhibits gastrin, CCK, secretin, gastric acid, pancreatic secretion.
- **Motilin** (M cells, duodenum) — stimulates gastric motility; target of erythromycin (motilin agonist).

Transition

The GI tract processes the food. The next chapter turns to the accessory digestive organs — salivary glands, pancreas, and liver — that produce saliva, pancreatic juice, and bile, and to the structural organisation of their secretions.

Rapid Review

- Four layers: mucosa, submucosa, muscularis externa, serosa/adventitia.
- ENS: myenteric (Auerbach's, motility) + submucosal (Meissner's, secretion).
- Esophagus: stratified squamous non-keratinised + skeletal→smooth muscle transition.
- Stomach: simple columnar with gastric pits; fundic glands (parietal HCl + intrinsic factor, chief pepsinogen).
- Small intestine: villi + crypts + microvilli (600×); Brunner duodenum, plicae jejunum, Peyer's ileum.
- Colon: crypts, no villi, many goblet cells, taeniae coli.

Chapter 13 — Organs Associated with the Digestive Tract

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 16

Opening Question

How does the liver do 500 jobs with only a few cell types — and why does its blood flow make it both central to metabolism and uniquely vulnerable to injury?

Why This Matters

The salivary glands, pancreas, and liver are essential organs. Liver pathology in particular is enormous: hepatitis, cirrhosis, hepatocellular carcinoma, and metabolic liver disease are among the most common causes of morbidity and mortality worldwide.

Learning Objectives

1. Compare the three major salivary glands (parotid, submandibular, sublingual).
2. Describe the histological organisation of the exocrine and endocrine pancreas.
3. Explain the structure of the liver lobule and the hepatic acinus.
4. Describe the structure and function of hepatocytes, sinusoidal cells, and bile-duct epithelium.
5. Describe the structure and function of the gallbladder.
6. Identify each digestive gland histologically.

Big Picture

- **Salivary glands** — parotid (serous), submandibular (mixed serous-predominant), sublingual (mucous-predominant); myoepithelial cells; intercalated → striated → excretory ducts.
- **Pancreas** — exocrine serous acini (digestive enzymes as inactive zymogens) + endocrine islets (α glucagon, β insulin, δ somatostatin).
- **Liver** — hexagonal lobule (central vein + portal tracts at corners); acinus (zones 1 periportal → 3 centrilobular) for metabolic zonation; hepatocytes (most functions), Kupffer cells (macrophages), stellate cells (vitamin A + fibrosis).
- **Gallbladder** — simple columnar epithelium with apical microvilli; no muscularis mucosae or submucosa; rugae when empty.

Core Concept

The accessory digestive organs produce saliva (lubrication, amylase digestion, antimicrobial protection), pancreatic juice (digestive enzymes),

and bile (emulsification), and the liver combines hundreds of metabolic functions in a single hepatocyte.

Structure

Salivary Glands

General architecture: acinus (secretory unit) → intercalated duct (low cuboidal) → striated duct (columnar with basal striations representing mitochondria for ion transport) → excretory duct (larger, stratified or pseudostratified epithelium) → opening into oral cavity.

Myoepithelial cells — contractile cells with processes wrapping acini and intercalated ducts; squeeze secretions on sympathetic stimulation.

Parotid — almost entirely serous (watery, enzyme-rich, amylase). Mumps primarily affects the parotid.

Submandibular — mixed, predominantly serous with some mucous acini. Serous demilunes (crescent-shaped serous caps) at the ends of mucous tubules.

Sublingual — mixed, predominantly mucous with some serous.

Histological distinction:

- Serous acini: round basal nuclei, apical eosinophilic granular cytoplasm (zymogen granules).
- Mucous acini: flattened basal nuclei, pale foamy bluish cytoplasm (mucinogen).

Salivary Gland Detail

The intercalated duct is the short, narrow, low cuboidal segment that drains the acinus. It has a basal lamina, no striations, and serves primarily as a passage for secretion. It is the site of origin of the salivary gland tumours that recapitulate this segment (intercalated duct cells in pleomorphic adenoma, intercalated duct-like cells in Warthin tumour of the parotid). It is also where most of the IgA secretion from plasma cells is added (secretory component added to dimeric IgA).

The striated duct is the principal site of salivary electrolyte modification. Tall columnar cells with extensive basolateral membrane infoldings and mitochondria (basal striations visible in H&E) actively reabsorb Na^+ and Cl^- and secrete K^+ and HCO_3^- , producing a hypotonic final saliva. Striated duct cells express carbonic anhydrase VI (gustin) — secreted as a zinc-binding protein involved in taste perception.

Myoepithelial cells wrap acini and the proximal intercalated ducts; they have contractile processes (actin + myosin) and squeeze secretion on autonomic stimulation. They are difficult to see in routine H&E but are highlighted by p63, calponin, or smooth muscle actin IHC.

Sjögren syndrome is a chronic autoimmune disease with lymphocytic infiltration of salivary and lacrimal glands, leading to xerostomia (dry mouth) and keratoconjunctivitis sicca (dry eyes). The histology shows focal lymphocytic sialadenitis with focal lymphocytic aggregates adjacent to

ducts. Anti-Ro/SSA and anti-La/SSB antibodies are characteristic.

Pancreas

Compound tubuloacinar gland with serous acini.

Exocrine — pyramidal acinar cells around small central lumen; basal basophilia (RER) and apical eosinophilia (zymogen granules). Centroacinar cells (within the acinus, distinctive of pancreas) → intercalated ducts → intralobular → interlobular → main pancreatic duct → duodenal ampulla.

Secretions: trypsinogen, chymotrypsinogen, proelastase, procarboxypeptidase (proteases, secreted as inactive zymogens); α -amylase, lipase, colipase, phospholipase (carbohydrate/lipid digestion); RNase, DNase.

Activation: in duodenum, enterokinase converts trypsinogen → trypsin; trypsin activates the other zymogens. Premature intra-pancreatic activation → pancreatitis (autodigestion).

Endocrine — islets of Langerhans: pale-staining clusters within exocrine tissue.

- β cells (~70%) — insulin; central in islet.
- α cells (~20%) — glucagon; peripheral.
- δ cells (~5-10%) — somatostatin; peripheral.
- PP cells — pancreatic polypeptide.
- ϵ cells — ghrelin.

Distinguishable by IHC (insulin, glucagon antibodies) but not in routine H&E. Highly vascularised; hormones released into fenestrated capillaries.

Liver

Two complementary organising concepts:

The classical lobule: hexagonal mass of hepatocytes with a central vein in the middle and portal tracts (portal vein + hepatic artery + bile duct + lymphatics + nerves) at the six corners. Hepatocytes arranged in cords (1-2 cells thick) radiating from central vein to periphery. Sinusoids (large, fenestrated capillaries) between the cords carry mixed blood (portal vein + hepatic artery) toward the central vein.

The hepatic acinus: functional unit defined by blood supply — region of hepatocytes supplied by a single terminal portal vein/hepatic artery branch, between two adjacent central veins. Divided into three zones based on proximity to portal tract:

- Zone 1 (periportal) — most oxygenated blood; oxidative metabolism, gluconeogenesis, urea synthesis.
- Zone 2 (midzonal) — intermediate.
- Zone 3 (centrilobular) — least oxygenated; glycolysis, lipogenesis, detoxification (cytochrome P450). Most susceptible to ischemic injury and centrilobular necrosis (acetaminophen toxicity).

Hepatocytes — large, polygonal, central round nucleus (often polyploid, prominent nucleolus), abundant eosinophilic cytoplasm (mitochondria, smooth ER, rough ER, Golgi, peroxisomes). Glycogen granules (PAS+,

diastase-sensitive); small lipid droplets.

Functions: bile production, plasma protein synthesis (albumin, clotting factors), gluconeogenesis and glycogen storage, urea synthesis, cholesterol/lipoprotein metabolism, drug detoxification, vitamin storage (A, D, B₁₂).

Hepatocyte Detail

A hepatocyte is ~25 µm in diameter, with a polygonal shape defined by its neighbours in the hepatic cord. The nucleus is large, round, and often polyploid (binucleate hepatocytes are common — up to 25% of adult human hepatocytes). Polyploidy is associated with metabolic stress and regenerative capacity. The cytoplasm has abundant mitochondria (~1000-2000 per cell — the highest density of any cell in the body) to support oxidative metabolism.

Ultrastructurally, hepatocyte cytoplasm contains:

- **Rough ER** — clusters (basophilic bodies) for plasma protein synthesis (albumin, clotting factors, transferrin).
- **Smooth ER** — extensive; for bile salt synthesis, cholesterol synthesis, drug detoxification (cytochrome P450 induction → barbiturates cause SER proliferation visible as eosinophilic ground-glass hepatocytes in biopsy).
- **Golgi** — multiple regions, for protein glycosylation and bile acid secretion.
- **Lysosomes** — abundant, for turnover of aged proteins and organelles.
- **Peroxisomes** — for fatty acid β-oxidation and plasmalogen synthesis.
- **Glycogen** — rosettes of glycogen granules (PAS+, diastase-sensitive). Fasting depletes glycogen within 24 hours.
- **Lipid droplets** — small in normal hepatocytes; large in steatosis (NAFLD, alcoholic fatty liver).

Hepatocytes have two distinct basolateral surfaces: the sinusoidal surface (with microvilli extending into the space of Disse, increasing exchange area ~6×) and the canalicular surface (forming the bile canaliculus between adjacent hepatocytes, sealed by tight junctions — this is the structural basis of the blood-bile barrier).

Sinusoidal cells — fenestrated endothelium + Kupffer cells (resident macrophages, derived from monocytes) + hepatic stellate (Ito) cells in the space of Disse between endothelium and hepatocytes. Stellate cells store vitamin A in lipid droplets; in chronic injury, they activate into myofibroblast-like cells producing collagen → fibrosis → cirrhosis.

Bile — produced by hepatocytes, secreted into bile canaliculi (small grooves between adjacent hepatocytes, sealed by tight junctions). Bile flows centrifugally (opposite to blood flow) toward the portal tracts, then through canals of Hering (lined partly by hepatocytes, partly by bile-duct epithelium) → bile ductules → interlobular bile ducts → larger ducts → common bile duct → gallbladder (storage) or duodenum.

Gallbladder

Mucosa: simple columnar epithelium with apical microvilli (water absorption); mucosa thrown into folds (rugae) when empty. Rokitansky-Aschoff sinuses (epithelial invaginations into muscularis) are common.

No muscularis mucosae, no submucosa — distinctive. Muscularis externa of irregular smooth muscle. Serosa on free surface, adventitia on hepatic surface.

Function

Salivary glands produce saliva for lubrication, amylase digestion, and antimicrobial protection (lysozyme, lactoferrin, IgA). Pancreatic exocrine produces digestive enzymes (zymogens) and bicarbonate; endocrine produces insulin, glucagon, somatostatin. Liver produces bile, plasma proteins, glycogen, urea, and many other molecules; Kupffer cells are macrophages; stellate cells store vitamin A and contribute to fibrosis. Gallbladder stores and concentrates bile.

Structure → Function

- Serous acini (parotid) → watery enzyme-rich saliva.
- Mucous acini (sublingual) → viscous lubricating saliva.
- Centroacinar cells → bicarbonate-rich alkaline pancreatic juice.
- Pancreatic islets with β cells central and α cells peripheral → rapid response to blood glucose changes.
- Sinusoidal fenestrations → high permeability for protein exchange.
- Acinar zone 1 → oxidative metabolism; zone 3 → CYP450 detoxification.
- Bile canaliculi sealed by tight junctions → blood-bile barrier.
- Gallbladder microvilli → water absorption concentrates bile $\sim 10\times$.

Compare & Distinguish

Salivary Glands

Feature	Parotid	Submandibular	Sublingual
Type	Serous	Mixed (serous-pred)	Mixed (mucous-pred)
Adipose infiltration	Common	Less	Less

Serous vs Mucous Acini

Feature	Serous	Mucous
Nucleus	Round, basal	Flattened, basal
Cytoplasm	Eosinophilic granular	Pale foamy bluish

Pancreatic Acinar Cells vs Islet Cells

Feature	Acinar	Islet
Appearance	Basophilic basal + eosinophilic apical (zymogen)	Pale, less basophilic
IHC	Trypsin, amylase	Insulin, glucagon, somatostatin

Liver Lobule vs Acinus

Feature	Lobule	Acinus
Shape	Hexagon	Diamond/lozenge
Centre	Central vein	Portal tract
Zones	—	1 (periportal) → 2 → 3 (centrilobular)
Use	Anatomical	Metabolic

Recognition Logic

- **Serous acini only + adipose infiltration** → parotid.
- **Mixed acini with serous demilunes** → submandibular.
- **Mucous-predominant acini** → sublingual.
- **Serous acini with basal basophilia + apical eosinophilia + pale islets** → pancreas.
- **Hexagonal lobule with central vein + portal tracts + cords of hepatocytes + sinusoids** → liver.
- **Mucosa with rugae, no muscularis mucosae or submucosa** → gallbladder.

Clinical Correlation

Case — Cirrhosis

A 55-year-old man with chronic alcohol use presents with jaundice, ascites, and confusion. Liver biopsy shows nodules of regenerating hepatocytes surrounded by fibrous bands, with distortion of the normal lobular architecture.

Lesson. Alcohol injury → stellate cell activation → collagen deposition → fibrosis. Continued injury and regeneration → nodules and bands. Portal hypertension (ascites) and impaired hepatic function (jaundice, hyperammonemia, encephalopathy) result.

Case — Type 1 Diabetes

A 12-year-old with polydipsia, polyuria, weight loss. Fasting glucose 350; HbA1c 12%; anti-GAD antibodies positive. Pancreas shows insulinitis (lymphocytic infiltrate of islets) and β -cell loss.

Lesson. Autoimmune β -cell destruction → absolute insulin deficiency.

Case — Sialolithiasis

A 40-year-old with recurrent parotid swelling. Imaging shows a stone in the submandibular duct.

Lesson. Salivary stones (sialoliths) obstruct ductal outflow; the submandibular gland is most commonly affected because its secretion is more mucous (more viscous) and its duct has an uphill course.

Common Misconceptions

- Salivary glands only produce saliva. They also produce antimicrobial peptides (lysozyme, lactoferrin) and IgA.
- Pancreas is primarily exocrine. ~85% exocrine, ~2% endocrine (islets); islet hormones are essential.
- Liver has multiple cell types. Many cell types (hepatocytes, Kupffer cells, stellate cells, sinusoidal endothelium, bile-duct epithelium), but hepatocyte is the only cell that performs most metabolic functions.
- Bile is made by the gallbladder. Bile is made by the liver; the gallbladder stores and concentrates it.
- All zones of the liver lobule are the same. Zone 1 vs Zone 3 have different metabolic specialisations and injury susceptibilities.

High-Yield Knowledge

Must Know

- Salivary: parotid (serous), submandibular (mixed serous-pred), sublingual (mucous-pred).
- Pancreas: serous acini (zymogens) + islets (β insulin central, α glucagon peripheral, δ somatostatin).
- Liver lobule: central vein + portal tracts (portal vein + hepatic artery + bile duct); sinusoids; cords of hepatocytes.
- Acinus zones: 1 periportal (oxidative) → 3 centrilobular (CYP450, glycolysis).
- Hepatocytes (functions), Kupffer cells (macrophages), stellate cells (vitamin A, fibrosis).
- Gallbladder: simple columnar with apical microvilli; rugae; no muscularis mucosae or submucosa.

Must Distinguish

- Salivary glands by acinar type.
- Pancreatic acinar cells (basophilic basal + eosinophilic apical) from islet cells (pale).
- Liver lobule from acinus.

Must Understand

- Zymogens activated in duodenum (trypsinogen → trypsin).
- Bile flows opposite to blood (canaliculi → portal tracts).
- Zone 3 most susceptible to acetaminophen toxicity.

Integrated Summary

The salivary glands (parotid serous, submandibular mixed, sublingual mucous-predominant) produce saliva with amylase and antimicrobial peptides. The pancreas has serous acini (zymogens, activated in duodenum) and islets (β insulin central, α glucagon peripheral, δ somatostatin). The liver is organised into hexagonal lobules (central vein + portal tracts) and acini with three metabolic zones (1 periportal oxidative, 3 centrilobular CYP450); hepatocytes perform most functions, Kupffer cells are macrophages, stellate cells store vitamin A and drive fibrosis. The gallbladder stores and concentrates bile.

Mastery Check

1. Compare the three salivary glands by acinar type.
2. Why are pancreatic enzymes secreted as zymogens, and what happens with premature activation?
3. Compare the hepatic lobule and the hepatic acinus; which is more useful for metabolic zonation?
4. Why is zone 3 of the hepatic acinus most susceptible to acetaminophen toxicity?
5. What are the histological features of cirrhosis?

Advanced Concepts

Bile Acid Metabolism

Bile acids are synthesised from cholesterol in hepatocytes (~500 mg/day). Two pathways:

- Classical (neutral) — cholesterol 7 α -hydroxylase (CYP7A1) is rate-limiting; produces cholic acid (CA) and chenodeoxycholic acid (CDCA).
- Alternative (acidic) — sterol 27-hydroxylase (CYP27A1); produces CDCA.

Primary bile acids (CA, CDCA) are conjugated with glycine or taurine (increased solubility, decreased passive reabsorption), secreted into bile via BSEP (bile salt export pump), stored in gallbladder, released into duodenum after a meal. In the ileum, bacteria deconjugate and 7 α -dehydroxylate primary bile acids → secondary bile acids (deoxycholic acid, lithocholic acid). ~95% of bile acids are reabsorbed in the ileum (enterohepatic circulation) and returned to the liver via portal blood; only ~5% lost in faeces (compensated by new synthesis).

Bile acid functions:

- Micelle formation — emulsify dietary lipids and fat-soluble vitamins; critical for lipid absorption.
- Bile flow — choleric; primary driving force for bile secretion.
- Signalling — bind FXR (farnesoid X receptor) and TGR5; regulate glucose, lipid, and energy homeostasis. FXR agonists (obeticholic acid) used in primary biliary cholangitis and NASH.

Gallstones: most are cholesterol stones (supersaturation of bile with cholesterol, nucleation, growth); risk factors are "5 F's" (Female, Fat, Forty, Fertile, Fair). Pigment stones (bilirubin): chronic haemolysis, cirrhosis.

Pancreatic Islet Cell Organisation

Islet cell distribution is non-random:

- **Mouse/rat** — β cells central, α cells peripheral (mantle).
- **Human** — β cells ~55–70%, α cells ~20–35%, δ cells ~5–10%, scattered throughout islet; α cells more peripheral than in rodents; PP cells peripheral.

Vascular supply is rich and fenestrated; islets receive $\sim 10\times$ the blood flow of surrounding exocrine tissue. Insulin (β) and glucagon (α) flow into the portal vein $\rightarrow$ liver (first-pass effect; insulin acts directly on hepatocytes).

β -cell stimulus-secretion coupling: glucose $\rightarrow$ GLUT2 $\rightarrow$ glycolysis $\rightarrow$ ATP/ADP ratio rises $\rightarrow$ ATP-sensitive K^+ channel closes $\rightarrow$ depolarisation $\rightarrow$ voltage-gated Ca^{2+} channel opens $\rightarrow$ Ca^{2+} entry $\rightarrow$ insulin granule exocytosis. Sulfonylureas close the K^+ channel directly $\rightarrow$ insulin release (independent of glucose); risk of hypoglycaemia.

Liver Regeneration Detail

Liver regeneration after partial hepatectomy is the classical model of regenerative biology:

- 70% hepatectomy in rat $\rightarrow$ full mass restoration in ~ 2 weeks.
- Hepatocytes are the principal proliferating cell; oval cells (Hepatic stem cells) participate in severe injury when hepatocyte proliferation is blocked.
- Regeneration is driven by cytokines (TNF, IL-6) and growth factors (HGF, EGF, TGF- α); inhibited by TGF- β once appropriate size is reached.
- Architecture is preserved — original lobular pattern restored.

Liver disease severity:

- **Steatosis** — fat accumulation; reversible.
- **Steatohepatitis** (NASH/ASH) — fat + inflammation + ballooning degeneration; reversible if cause removed.
- **Fibrosis** — collagen deposition (Ito cell activation); potentially reversible if cause removed (new antifibrotic therapies).
- **Cirrhosis** — nodules + fibrous bands; irreversible; portal hypertension, synthetic failure, risk of HCC.

Liver Sinusoidal Cell Functions

- **Hepatocytes** — most functions.
- **Sinusoidal endothelial cells** — fenestrated (~ 100 nm pores), no diaphragm; large surface area for plasma exchange; clear macromolecules (hyaluronic acid) from blood; express L-SIGN and SR-B1 (hepatitis B entry).

- **Kupffer cells** — resident macrophages; phagocytose aged RBCs, bacteria from portal blood (first-pass defence against gut-derived pathogens), tumour cells, debris; produce cytokines (TNF, IL-6).
- **Hepatic stellate (Ito) cells** — store vitamin A in lipid droplets (largest reservoir of vitamin A in body); quiescent in normal liver; activated in chronic injury to myofibroblast-like cells producing type I and III collagen → fibrosis.
- **Pit cells** — liver-specific NK cells; tumour surveillance.

Liver Pathology Recognition

Pattern	Likely Cause
Centrilobular necrosis	Acetaminophen, ischaemia, CCl ₄
Midzonal necrosis	Yellow phosphorus
Periportal necrosis	Hepatitis B, autoimmune hepatitis
Microvesicular steatosis	Reye syndrome, valproate, acute fatty liver of pregnancy
Macrovesicular steatosis	NAFLD, alcohol, obesity
Canalicular cholestasis	Drugs, sepsis
Ductular cholestasis	Primary biliary cholangitis, primary sclerosing cholangitis
Mallory-Denk bodies	Alcohol, NASH
Ground-glass hepatocytes	Chronic hepatitis B (HBsAg accumulation), drug-induced SER proliferation

Transition

The digestive system processes nutrients. The next chapter turns to the system that delivers oxygen from the atmosphere and removes carbon dioxide — the respiratory system.

Rapid Review

- Salivary glands: parotid (serous), submandibular (mixed, serous-pred), sublingual (mucous-pred). Myoepithelial cells. Striated ducts.
- Pancreas: exocrine (serous acini with zymogen granules, basal basophilia, apical eosinophilia) + endocrine (islets: β insulin, α glucagon, δ somatostatin).
- Liver: hexagonal lobule (central vein + portal tracts); acinus with zones 1–3. Hepatocytes (most metabolic functions), Kupffer cells (macrophages), stellate cells (vitamin A, fibrosis). Bile flows hepatocyte → canaliculi → bile duct → duodenum.
- Gallbladder: rugae, simple columnar epithelium, no muscularis mucosae or submucosa.

Chapter 14 — The Respiratory System

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 17

Opening Question

How does an organ that must stay open also clean itself — and how does a ~0.5 μm barrier let oxygen through but keep blood in?

Why This Matters

The respiratory system is essential for life. Asthma, COPD, pneumonia, lung cancer, and pulmonary fibrosis are among the most common human diseases. Lung cancer is the leading cause of cancer death in many countries.

Learning Objectives

1. Describe the organisation into conducting and respiratory zones.
2. Identify the histological features of each segment of the respiratory tree.
3. Explain the structural basis of gas exchange at the alveolus.
4. Describe the alveolar epithelium (type I, type II pneumocytes) and the blood-air barrier.
5. Distinguish the regions of the respiratory system histologically.
6. Recognise the cell types of the alveolus.

Big Picture

- **Conducting zone** (nasal cavity → terminal bronchioles; no gas exchange) — warms, moistens, filters air; pseudostratified ciliated columnar with goblet cells; cartilage decreases; smooth muscle increases.
- **Respiratory zone** (respiratory bronchioles → alveoli; gas exchange) — progressively more alveoli; simple squamous → simple cuboidal → type I/II pneumocytes.
- **Alveolar wall** — type I pneumocytes (~95% surface, gas exchange), type II pneumocytes (~60% of cells, surfactant + stem cell), alveolar macrophages (dust cells), capillary endothelium.
- **Blood-air barrier** — type I pneumocyte + fused basal laminae + capillary endothelium, ~0.5 μm thin.

Core Concept

The respiratory tree is a conducting zone (air conditioning) that transitions into a respiratory zone (gas exchange). The alveolar wall is the thinnest of all physiological barriers — a fused basal lamina sandwich between type I pneumocyte and capillary endothelium — optimised for O_2 and CO_2 diffusion.

Structure

Nasal Cavity

- **Vestibule** — most anterior; keratinised stratified squamous epithelium with vibrissae (coarse hairs that filter large particles).
- **Respiratory region** — respiratory epithelium (pseudostratified ciliated columnar with goblet cells); lamina propria rich in venous plexuses (warm inspired air); submucosal glands (mucus).
- **Olfactory region** — olfactory epithelium: pseudostratified columnar with olfactory receptor neurons (bipolar, apical cilia with odorant receptors, basal axons through cribriform plate to olfactory bulb), supporting (sustentacular) cells, and basal cells (stem cells for continuous olfactory neuron replacement — unique among neurons).

Paranasal Sinuses

Air-filled cavities in the frontal, maxillary, ethmoid, and sphenoid bones, lined by respiratory epithelium. Mucus moves by ciliary action into the nasal cavity.

Nasopharynx

Respiratory epithelium.

Larynx

Connects pharynx to trachea. Contains vocal cords. Respiratory epithelium above the vocal cords, stratified squamous over the vocal cords (mechanical abrasion during phonation). Cartilaginous framework (thyroid, cricoid, arytenoid, corniculate, cuneiform). Mixed seromucous glands in submucosa.

Trachea

- **Mucosa:** respiratory epithelium on basement membrane; lamina propria with elastic fibers.
- **Submucosa:** dense CT with submucosal glands (mixed seromucous) producing mucus for the mucociliary escalator.
- **Cartilaginous layer:** 16–20 C-shaped hyaline cartilage rings (open part of C faces esophagus, allowing esophageal expansion during swallowing); connected by annular ligament.
- **Adventitia:** loose CT.
- **Trachealis muscle:** smooth muscle bridging the open part of cartilage rings posteriorly.

Bronchi

Primary bronchi → secondary (lobar) → tertiary (segmental) → smaller. Same wall structure as trachea but with progressively:

- Smaller cartilage plates (irregular).
- More smooth muscle in the muscularis.
- Fewer submucosal glands.

The wall shows: respiratory epithelium → lamina propria → smooth muscle (muscularis; target of bronchodilators) → submucosal glands → cartilage plates → adventitia. The prominent wavy internal elastic lamina helps distinguish muscular arteries from bronchi when both are adjacent.

Bronchioles

Lumen < 1 mm; **no cartilage, no submucosal glands**. Wall:

- Simple ciliated columnar (initially) → simple ciliated cuboidal (smaller).
- Few or no goblet cells.
- Smooth muscle in relatively thick layer.
- **Clara (club) cells** — dome-shaped; secrete Clara cell secretory protein (CCSP/CC10) and surfactant-like substances; serve as stem cells.

Absence of cartilage means caliber depends on smooth muscle tone. Bronchospasm = asthma.

Terminal bronchioles — smallest conducting airways; last segment of conducting zone. → Respiratory bronchioles — first segment of respiratory zone, with occasional alveoli in the wall.

Respiratory Zone

Respiratory Bronchioles

Bronchioles with occasional alveoli opening from the wall. Lined by simple cuboidal epithelium with Clara cells, interrupted by alveoli.

Alveolar Ducts

Elongated airways whose walls are entirely alveolar openings. Wall between adjacent alveoli is a septum with smooth muscle knobs at alveolar openings, elastic fibers, and capillaries.

Alveolar Sacs

Clusters of alveoli at the terminations of alveolar ducts.

Alveoli

Terminal air spaces; polygonal, ~200–300 µm diameter, extremely thin walls. Total number ~300–500 million; total surface area ~70 m².

Alveolar wall (septum):

- **Alveolar epithelium:** type I and type II pneumocytes.
- **Capillary endothelium** (continuous, attenuated).
- **Fused basement membranes** (air-blood barrier).
- **Interstitium** with elastic and reticular fibers, fibroblasts, immune cells.

Alveolar epithelium:

- **Type I pneumocytes** — very thin, flattened; cover ~95% of alveolar surface; principal site of gas exchange.
- **Type II pneumocytes** — cuboidal, apical microvilli, abundant RER/Golgi, characteristic lamellar bodies (membrane-bound organelles)

- containing pulmonary surfactant). Produce surfactant (reduces surface tension, prevents alveolar collapse); serve as stem cells.

Type II Pneumocyte Detail

Type II pneumocytes are the defenders and regenerators of the alveolus. Each cell has apical microvilli, abundant rough ER, a large Golgi, and characteristic lamellar bodies — membrane-bound concentrically arranged organelles containing surfactant. Surfactant is synthesised in the ER, modified in the Golgi, packaged in lamellar bodies, and secreted by exocytosis into the alveolar lumen where it spreads as a monolayer at the air-liquid interface. The monolayer reduces surface tension (which would otherwise collapse small alveoli and over-expand large ones) and provides alveolar stability across the breathing cycle.

Surfactant proteins:

- **SP-A, SP-D** — collectins; innate immune functions (opsonisation, bacterial aggregation, modulation of inflammation).
- **SP-B, SP-C** — hydrophobic proteins that organise the phospholipid monolayer and are essential for surface activity. Mutations in SP-B are lethal.

Type II pneumocytes also serve as the stem cells of the alveolar epithelium. After injury (pneumonia, ARDS), type II cells proliferate, then differentiate into type I cells to re-establish the alveolar lining. Hyperplasia of type II pneumocytes is a histological hallmark of alveolar injury and regeneration.

Neonatal respiratory distress syndrome (NRDS) in premature infants results from immature type II pneumocytes that cannot produce adequate surfactant. Without surfactant, alveoli collapse on each breath (atelectasis), work of breathing rises, and gas exchange fails. Treatment: antenatal corticosteroids (accelerate type II maturation) + exogenous surfactant (instilled via endotracheal tube).

Surfactant — complex of phospholipids (mainly dipalmitoylphosphatidylcholine), proteins (SP-A, SP-B, SP-C, SP-D), and other components. SP-B and SP-C essential for surface activity; SP-A and SP-D are collectins involved in innate immunity. Neonatal respiratory distress syndrome = surfactant deficiency (premature type II pneumocytes).

Alveolar macrophages (dust cells) — abundant in alveolar lumen and septum; phagocytose inhaled particles; in smokers contain anthracotic (carbon) pigment.

Blood-Air Barrier

Type I pneumocyte cytoplasm → fused basal laminae → capillary endothelial cytoplasm. Total thickness often <1 µm — the thinnest physiological barrier. Site of O₂ and CO₂ diffusion by simple diffusion.

Function

Conducting zone warms, moistens, filters air; mucociliary escalator clears debris. Respiratory zone exchanges O₂ and CO₂. Alveolar macrophages clear inhaled particles. Surfactant reduces surface tension and prevents alveolar collapse. Type II pneumocytes regenerate the alveolar epithelium after injury.

Structure → Function

- **C-shaped tracheal cartilage** allows esophageal expansion during swallowing.
- **Pseudostratified ciliated epithelium + mucus** = mucociliary escalator that traps and propels debris.
- **Clara cells** secrete surfactant-like substances into bronchioles (anti-adhesive), CCSP (anti-inflammatory), and serve as stem cells.
- **Type I pneumocytes** are extremely thin → minimal diffusion distance.
- **Type II pneumocytes** secrete surfactant → reduced surface tension → alveolar stability.
- **Fused basal laminae** create the air-blood barrier.
- **Capillary network** in the septum provides a large surface area for gas exchange.
- **Smooth muscle** in bronchioles regulates airflow; bronchoconstriction → asthma.

Compare & Distinguish

Bronchus vs Bronchiole

Feature	Bronchus	Bronchiole
Cartilage	Yes	No
Submucosal glands	Yes	No
Epithelium	Pseudostratified ciliated columnar	Simple ciliated columnar/cuboidal
Goblet cells	Yes	Few
Clara cells	No	Yes
Smooth muscle	Present	Relatively thicker

Type I vs Type II Pneumocyte

Feature	Type I	Type II
Shape	Flat, attenuated	Cuboidal
Coverage	~95% of surface	~60% of cells (smaller total area)
Function	Gas exchange	Surfactant, stem cell
Diagnostic	EM only (thin squamous lining)	Lamellar bodies on EM

Recognition Logic

- **Pseudostratified ciliated columnar + goblet + hyaline cartilage + submucosal glands** → trachea or bronchus.
- **Simple ciliated columnar/cuboidal + Clara cells + no cartilage + no glands** → bronchiole.
- **Simple cuboidal with alveoli in wall** → respiratory bronchiole.
- **Wall entirely alveolar openings** → alveolar duct.
- **Polygonal air spaces with thin septa (capillaries, type I, type II)** → alveoli.
- **Pseudostratified columnar with olfactory receptor neurons + supporting + basal cells** → olfactory epithelium.

Clinical Correlation

Case — Squamous Cell Carcinoma of the Lung

A 70-year-old man with 50-pack-year smoking history, cough, weight loss, lung mass. Biopsy shows nests of large cells with keratin pearls.

Lesson. Smoking is the principal risk factor for squamous-cell lung cancer. Keratin pearls are pathognomonic of squamous differentiation.

Case — Asthma

A 25-year-old with episodic wheezing and dyspnea. Lung biopsy (rarely done) would show smooth muscle hyperplasia, goblet-cell hyperplasia, thickened basement membrane, and eosinophilic infiltrate.

Lesson. Bronchiolar smooth muscle hypertrophy + mucus hypersecretion → airway obstruction.

Case — Neonatal Respiratory Distress Syndrome

A 30-week premature infant with tachypnea, grunting, cyanosis. Chest X-ray shows diffuse ground-glass opacities. Diagnosis: surfactant deficiency (type II pneumocytes immature).

Lesson. Premature infants lack adequate surfactant; alveolar collapse → atelectasis → impaired gas exchange. Treatment is antenatal corticosteroids + postnatal surfactant.

Common Misconceptions

- The trachea is one long tube. Branches at the carina into primary bronchi, then secondary, tertiary, smaller bronchi, eventually bronchioles.
- All airways have goblet cells. Goblet cells are in trachea and bronchi; absent in bronchioles (Clara cells produce protective secretions).
- Alveoli are empty sacs. Contain alveolar macrophages that clear debris; smokers' macrophages contain anthracotic pigment.
- The air-blood barrier is one membrane. Fused basal laminae sandwiched between type I pneumocyte and capillary endothelium.
- Surfactant is one molecule. Complex of phospholipids + proteins (SP-A, SP-B, SP-C, SP-D).

High-Yield Knowledge

Must Know

- Conducting zone: nasal → terminal bronchioles (no gas exchange); respiratory epithelium, cartilage (trachea, bronchi), submucosal glands.
- Respiratory zone: respiratory bronchioles → alveoli; thin alveolar walls.
- Type I pneumocytes (~95% surface, gas exchange), Type II (surfactant + stem cell, lamellar bodies).
- Surfactant: dipalmitoylphosphatidylcholine + proteins; reduces surface tension.
- Blood-air barrier: type I + fused BL + capillary endothelium, ~0.5 μm .

Must Distinguish

- Bronchus vs bronchiole (cartilage, glands, Clara).
- Type I vs Type II pneumocyte.
- Terminal bronchiole (no alveoli) vs respiratory bronchiole (with alveoli).

Must Understand

- Asthma = bronchiolar smooth muscle hypertrophy + goblet hyperplasia.
- RDS = surfactant deficiency in premature infants.
- Smoking → squamous-cell carcinoma (keratin pearls) + anthracotic pigment in macrophages.

Integrated Summary

The respiratory system is organised into a conducting zone (nasal cavity → terminal bronchioles; warms, moistens, filters air; pseudostratified ciliated epithelium + cartilage + submucosal glands) and a respiratory zone (respiratory bronchioles → alveoli; gas exchange). The alveolar wall is the thinnest physiological barrier: type I pneumocyte + fused basal laminae + capillary endothelium, ~0.5 μm . Type II pneumocytes produce surfactant (dipalmitoylphosphatidylcholine + proteins) and serve as stem cells. Alveolar macrophages clear inhaled debris.

Mastery Check

1. Compare conducting and respiratory zones by function and histology.
2. Compare bronchus and bronchiole histologically.
3. Compare Type I and Type II pneumocytes.
4. What is the blood-air barrier, and what are its layers?
5. Why does surfactant deficiency cause neonatal respiratory distress syndrome?

Advanced Concepts

Mucociliary Escalator Detail

The mucociliary escalator is the principal defence of the conducting zone. Components:

- **Mucus layer** — two sublayers: a low-viscosity periciliary layer ($\sim 7\ \mu\text{m}$, surrounds cilia) and a high-viscosity gel layer ($\sim 5\text{--}10\ \mu\text{m}$, on top). Mucus traps inhaled particles (dust, bacteria, allergens).
- **Cilia** — ~ 200 per cell, $\sim 5\text{--}7\ \mu\text{m}$ long, $\sim 0.3\ \mu\text{m}$ diameter; beat at $\sim 10\text{--}20\ \text{Hz}$ with metachronal rhythm (recovery stroke in the periciliary layer, effective stroke in the gel layer). Velocity of mucus transport: $\sim 5\ \text{mm/min}$ in small airways, up to $\sim 20\ \text{mm/min}$ in trachea.

Ciliary structure:

- 9+2 axoneme — 9 outer doublet microtubules + 2 central singlets.
- Dynein arms (inner and outer) on outer doublets — motor proteins that generate sliding force.
- Nexin links connect adjacent doublets; radial spokes to central pair; basal body anchors cilium to cell.

Ciliary dysfunction:

- Primary ciliary dyskinesia (Kartagener syndrome) — dynein arm defects $\rightarrow$ immotile cilia $\rightarrow$ recurrent sinopulmonary infections, bronchiectasis, infertility (sperm flagella), situs inversus ($\sim 50\%$).
- Smoking — paralyzes cilia; metaplasia (squamous instead of pseudostratified columnar); goblet cell hyperplasia $\rightarrow$ chronic bronchitis.
- CF — dehydrated mucus $\rightarrow$ impaired ciliary transport $\rightarrow$ chronic infection.

Pulmonary Vascular Bed

The lung has a dual blood supply:

- **Pulmonary circulation** — low-pressure ($\sim 25/8\ \text{mmHg}$), high-flow, thin-walled; carries deoxygenated blood from RV for gas exchange; arteries accompany bronchi.
- **Bronchial circulation** — high-pressure (systemic pressure), low-flow; supplies the conducting airways down to respiratory bronchioles; bronchial veins drain systemic venous blood.

Pulmonary capillaries form a dense sheet in the alveolar septum; each capillary is shared between adjacent alveoli. The capillary endothelium is continuous and extremely thin ($\sim 0.2\ \mu\text{m}$), apposed to the type I pneumocyte across a fused BM (air-blood barrier).

Pulmonary vascular responses:

- **Hypoxic pulmonary vasoconstriction** — unique to the lung (systemic vessels dilate in hypoxia); diverts blood from poorly ventilated regions to better-ventilated regions; mediated by smooth muscle Ca^{2+} entry. Failure $\rightarrow$ V/Q mismatch.
- **V/Q ratio** — normal = 0.8 (4 L/min ventilation / 5 L/min perfusion); apex has high V/Q (~ 3.3) $\rightarrow$ wasted ventilation; base has low V/Q (~ 0.6) $\rightarrow$ physiological shunt.

Lung Pathophysiology Correlates

Disease	Pathology	Functional Consequence
Asthma	Bronchiolar smooth muscle hypertrophy, mucus plugging, eosinophilic inflammation, subepithelial fibrosis	Reversible airflow obstruction (FEV ₁ /FVC ↓)
COPD	Centriacinar emphysema (smoking), panacinar (α_1 -antitrypsin), chronic bronchitis (Reid index ↑)	Irreversible airflow obstruction, air trapping
Pulmonary fibrosis	Interstitial fibrosis, fibroblast foci, honeycombing	Restrictive pattern, ↓ DLCO
Pulmonary embolism	Thrombus in pulmonary artery, often from DVT	V/Q mismatch, RV strain
ARDS	Diffuse alveolar damage, hyaline membranes	Severe hypoxaemia, bilateral infiltrates
Bronchogenic carcinoma	Most common cancer worldwide; smoking; adenocarcinoma most common type, then squamous, then small-cell	Mass effect, paraneoplastic syndromes
Tuberculosis	Caseating granulomas with Langhans giant cells, AFB stain	Cavitary disease, miliary spread, meningitis

Transition

The respiratory system exchanges gases. The next chapter turns to the body's barrier against the external environment — the skin.

Rapid Review

- Conducting zone: nasal → terminal bronchioles. Respiratory epithelium (pseudostratified ciliated columnar + goblet); cartilage (trachea, bronchi); submucosal glands; Clara cells (bronchioles).
- Respiratory zone: respiratory bronchioles → alveolar ducts → alveolar sacs → alveoli.
- Alveolar epithelium: Type I (~95% surface, gas exchange), Type II (surfactant + stem cell, lamellar bodies).
- Alveolar macrophages (dust cells): phagocytosis, anthracotic pigment in smokers.
- Surfactant: DPPC + SP-A/B/C/D; reduces surface tension.
- Blood-air barrier: Type I + fused BL + capillary endothelium.

Chapter 15 — Skin

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 18

Opening Question

How does the body build a tissue that is simultaneously tough, waterproof, self-renewing, and interactive — containing immune cells, sensory receptors, and pigment-producing cells?

Why This Matters

Skin is the body's largest organ and first line of defence. Skin diseases — acne, eczema, psoriasis, melanoma, non-melanoma skin cancer — are among the most common. Burns, wound healing, and cosmetic dermatology depend on understanding skin histology.

Learning Objectives

1. Describe the three layers of the skin: epidermis, dermis, hypodermis.
2. Identify the five layers of the epidermis and the cells of each.
3. Explain the process of keratinisation.
4. Describe the epidermal appendages: hair, nails, sebaceous glands, sweat glands.
5. Identify the cells of the epidermis and their functions.
6. Distinguish thick from thin skin histologically.
7. Recognise the histological signatures of common skin pathologies.

Big Picture

- **Epidermis** — keratinised stratified squamous epithelium with five layers in thick skin (basale → spinosum → granulosum → lucidum → corneum).
- **Dermis** — dense irregular CT with papillary (loose) and reticular (dense) layers.
- **Hypodermis** — loose CT with adipose.
- **Cells** — keratinocytes (~85%), melanocytes (~5%, neural-crest-derived, melanin), Langerhans cells (~3%, antigen-presenting), Merkel cells (~1%, mechanosensory).
- **Appendages** — hair follicles, sebaceous glands (holocrine, sebum), eccrine sweat (merocrine, thermoregulation), apocrine sweat (axilla/areola/perineum, odoriferous).

Core Concept

Skin is a renewing keratinised surface supported by dense connective tissue, containing pigment, immune, and sensory cells. The epidermis turns over every 4–6 weeks from basal stem cells.

Structure

Epidermal Cells

- **Keratinocytes** (~85%) — principal cells; produce keratin; form the stratified squamous architecture.
- **Melanocytes** (~5%) — neural-crest-derived; in basal layer; produce melanin in melanosomes; transfer melanosomes to keratinocytes via dendritic processes.
- **Langerhans cells** (~3%) — antigen-presenting dendritic cells from bone marrow; capture antigens penetrating skin.
- **Merkel cells** (~1%) — mechanoreceptors in basal layer; associated with Merkel discs (expanded nerve endings) for touch.

Five Epidermal Layers (thick skin, deep to superficial)

- **Stratum basale** — single layer of cuboidal/columnar keratinocytes on BM; stem cells; melanocytes; Merkel cells; anchored by hemidesmosomes.
- **Stratum spinosum** — several layers of polygonal keratinocytes connected by desmosomes; "spiny" appearance from desmosomes (cells shrink during fixation); keratins K1, K10 produced.
- **Stratum granulosum** — 3–5 layers of flattened keratinocytes with keratohyalin granules (profilaggrin + loricrin) and lamellar bodies (lipids + enzymes). Filaggrin aggregates keratin filaments. Lamellar bodies release lipid into intercellular space → lipid barrier that waterproofs skin.
- **Stratum lucidum** — only in thick skin (palms, soles); thin, clear layer of dead, flattened keratinocytes with densely packed keratin, no nuclei or organelles.
- **Stratum corneum** — multiple layers of dead, flattened, anucleate, keratin-filled squames in the lipid matrix; principal barrier against water loss and microbial invasion; surface cells continuously shed (desquamation).

Epidermal Barrier Detail

The epidermal barrier has two arms: a mechanical arm (keratin filaments and desmosomes) and a chemical arm (lipid matrix). In the stratum granulosum, lamellar bodies (membrane-bound secretory organelles) bud from the Golgi, fuse with the apical membrane, and release their lipid-rich content (ceramides, cholesterol, free fatty acids, cholesterol sulfate) into the intercellular space. Profilaggrin in the keratohyalin granules is proteolytically cleaved into filaggrin, which aggregates keratin filaments into tight bundles (macrofibrils) and contributes to the cornified envelope.

In the stratum corneum, the squames are embedded in the lipid matrix like bricks in mortar. Disruption of this barrier (filaggrin mutations, lipid synthesis defects, desmosomal defects) compromises barrier function. Loss-of-function mutations in FLG (filaggrin) cause ichthyosis vulgaris and are the strongest known genetic risk factor for atopic dermatitis. Lipid defects (ABCA12 mutations) cause harlequin ichthyosis.

Desquamation is controlled by desmosomal degradation (desmoglein-1 cleaved by kallikreins) and corneodesmosin degradation. Retained squames form the keratin plugs seen in retention hyperkeratosis (e.g., lichen planus).

Keratinisation

Terminal differentiation of keratinocytes. Begins in stratum basale, progresses through the layers:

- **Basal:** K5/K14 (basal keratins); mitotically active.
- **Spinosum:** K1/K10 (differentiation-specific keratins); cells flatten.
- **Granulosum:** keratohyalin + lamellar bodies; filaggrin aggregates keratin; lipid barrier forms.
- **Corneum:** dead, anucleate, keratin-dense; lipid barrier mature.

Transit time basal → desquamation: ~4 weeks.

Dermis

Connective-tissue layer supporting the epidermis:

- **Papillary layer** — superficial; loose CT, thin collagen/elastic fibers, capillaries, sensory nerve endings; forms dermal papillae that interdigitate with epidermal rete ridges.
- **Reticular layer** — deep; dense irregular CT, thick collagen/elastic, larger vessels, skin appendages (hair follicles, sweat glands, sebaceous glands).

Principal site of dermal collagen (type I, some III). Aging skin loses collagen/elastin → wrinkling.

Hypodermis (Subcutis)

Loose CT beneath dermis with abundant adipose (white adipose), larger vessels, nerves, deep portions of appendages. Insulation, cushioning, energy storage.

Appendages

Hair

Produced by hair follicles. Components:

- **Hair bulb** — expanded base with hair matrix (mitotically active keratinocytes producing the hair) and dermal papilla (CT core with capillaries).
- **Hair shaft** — keratinised filament protruding from surface.
- **Inner root sheath** — surrounds lower shaft; degenerates above sebaceous gland.
- **Outer root sheath** — continuous with epidermis.
- **Arrector pili muscle** — smooth muscle; pulls follicle upright ("goose bumps").
- **Sebaceous gland** — usually associated with follicle.

Hair growth is cyclic: anagen (growth, 2–7 yr) → catagen (regression, 2–3 wk) → telogen (rest, 3 mo).

Hair color: eumelanin (brown/black), pheomelanin (red/yellow), grey/white from loss of melanocyte function.

Nails

Plates of hard keratin (more disulfide bonds than soft keratin of hair) produced by the nail matrix (specialised epidermis beneath the nail root). The lunula is the white crescent at the base, marking the distal nail matrix.

Sebaceous Glands

Holocrine glands secreting sebum (triglycerides, wax esters, squalene) into the hair follicle (or directly onto surface in some regions). Most numerous on face and scalp. Sebum lubricates skin and hair and has antimicrobial properties.

Sweat Glands

- **Eccrine (merocrine)** — most numerous; distributed over entire body surface; simple coiled tubular; watery hypotonic secretion (water, NaCl, urea, ammonia); thermoregulation (evaporative cooling).
- **Apocrine** — axilla, areola, perineum; larger; viscous odorless fluid into hair follicle; becomes odorous when modified by skin bacteria; stimulated by adrenergic stimuli.

Thin vs Thick Skin

- **Thick** (palms, soles): all five layers (including stratum lucidum); thick stratum corneum; no hair follicles; no sebaceous glands.
- **Thin** (most of body): thinner stratum corneum; no stratum lucidum; hair follicles, sebaceous glands, apocrine glands in some regions.

Function

Epidermis provides barrier (chemical, microbial, UV, mechanical), waterproofing, and renewal. Dermis provides support, elasticity, and houses appendages. Hypodermis provides insulation, cushioning, energy storage. Melanocytes protect against UV. Langerhans cells provide immune surveillance. Merkel cells mediate touch. Sebaceous glands lubricate and protect. Eccrine sweat glands regulate temperature. Apocrine glands contribute to body odour.

Structure → Function

- **Keratin + lipid matrix of stratum corneum** → waterproof barrier.
- **Hemidesmosomes in basale + desmosomes in spinosum** → mechanical resistance to shearing.
- **Melanin in supranuclear cap of keratinocytes** → UV absorption.
- **Lamellar body lipids in granulosum** → water barrier.
- **Anucleate squames in corneum** → barrier against microbial invasion.
- **Hard keratin in nails** → mechanical resistance.

- **Myoepithelial cells around sweat glands** → squeeze secretion on sympathetic stimulation.

Compare & Distinguish

Thick vs Thin Skin

Feature	Thick	Thin
Layers	All 5 including lucidum	No lucidum
Corneum	Thick	Thin
Hair follicles	No	Yes
Sebaceous glands	No	Yes
Location	Palms, soles	Most body

Eccrine vs Apocrine Sweat Gland

Feature	Eccrine	Apocrine
Distribution	Entire body	Axilla, areola, perineum
Secretion	Watery, hypotonic	Viscous, odorless until modified
Mode	Merocrine	Apocrine
Function	Thermoregulation	Odor
Stimulation	Heat, sympathetic	Stress, adrenergic

Recognition Logic

- **Stratified squamous keratinised with all 5 layers, no hair, no sebaceous glands** → thick skin.
- **Stratified squamous keratinised with thin corneum, hair follicles + sebaceous glands** → thin skin.
- **Pale cell in basal layer with clear halo** → melanocyte (cytoplasm shrinks during fixation).
- **Dendritic IHC+ cell in spinosum** (CD1a, S100) → Langerhans cell.
- **Basal cell associated with nerve ending** → Merkel cell.
- **Sebaceous gland** = holocrine, foamy cells with central duct.
- **Eccrine sweat** = simple coiled tubular with cuboidal/columnar cells and small lumens.

Clinical Correlation

Case — Malignant Melanoma

A 60-year-old with a pigmented lesion on the back that has grown and changed color. Biopsy shows atypical melanocytes at the dermoepidermal junction with some cells invading the dermis. S100+, HMB-45+, Melan-A+.

Lesson. UV-induced melanocyte transformation. Dermal invasion = invasive melanoma (vs in situ melanoma, which is confined to the epidermis).

Case — Basal Cell Carcinoma

An elderly patient with a pearly nodule on the face. Biopsy shows nests of basaloid cells with peripheral palisading and clefting artefact from the surrounding stroma.

Lesson. The most common skin cancer; from stratum basale keratinocytes; locally invasive but rarely metastatic. UV is the major risk factor.

Case — Pemphigus Vulgaris

A 40-year-old with flaccid blisters. Biopsy shows intraepidermal split (acantholysis) with tombstone appearance of basal cells. Direct immunofluorescence shows IgG deposition in a net-like pattern around keratinocytes (anti-desmoglein).

Lesson. Autoantibodies against desmoglein disrupt desmosomes → intraepidermal blister.

Common Misconceptions

- The epidermis is just dead cells. Four cell types; only stratum corneum is dead.
- Skin is impermeable. Relatively impermeable to water and most solutes; percutaneous absorption of small, lipophilic molecules (basis of transdermal drug delivery).
- Hair grows continuously. Cyclic (anagen, catagen, telogen).
- Skin color depends only on melanin. Melanin + hemoglobin (dermal capillaries) + carotenoids (stratum corneum and adipose).

High-Yield Knowledge

Must Know

- Three layers: epidermis, dermis, hypodermis.
- Epidermal cells: keratinocytes, melanocytes, Langerhans cells, Merkel cells.
- Five epidermal layers: basale, spinosum, granulosum, lucidum, corneum.
- Appendages: hair, nails, sebaceous (holocrine), eccrine (merocrine, thermoregulation), apocrine (axilla).
- Thick skin: palms, soles; no hair; stratum lucidum.

Must Distinguish

- Thick vs thin skin (lucidum, hair).
- Sebaceous vs sweat gland (holocrine, sebum vs merocrine, sweat).
- Melanocyte (basal, melanin) from Langerhans (dendritic, immune) from Merkel (touch).

Must Understand

- Keratinisation: K5/K14 basal → K1/K10 spinous → filaggrin/lamellar in granulosum → anucleate corneum.
- Desmosomes vs hemidesmosomes (cell-cell vs cell-BM).
- UV → melanin cap → DNA protection.

Integrated Summary

Skin is the body's largest organ and principal barrier. Three layers: epidermis (keratinised stratified squamous epithelium with five layers in thick skin), dermis (papillary + reticular dense irregular CT), hypodermis (loose CT with adipose). Epidermal cells: keratinocytes (barrier, keratin), melanocytes (UV protection), Langerhans cells (immune), Merkel cells (touch). Appendages: hair (follicle + arrector pili + sebaceous), nails (hard keratin), sebaceous (holocrine), eccrine (merocrine thermoregulation), apocrine (axilla). Skin turns over every 4–6 weeks.

Mastery Check

1. Compare the four epidermal cells by origin, location, and function.
2. Compare the five epidermal layers by cell morphology and function.
3. Compare thick and thin skin histologically.
4. Compare eccrine and apocrine sweat glands.
5. What are the cells of origin of basal-cell carcinoma, squamous-cell carcinoma, and melanoma?
6. Reconstruct keratinisation from basal layer to stratum corneum.

Advanced Concepts

Melanocyte Biology Extended

Melanocytes are dendritic, neural-crest-derived cells that reside in the stratum basale (~1 melanocyte per 5 basal keratinocytes) and transfer melanosomes to surrounding keratinocytes via their dendritic processes.

Melanin synthesis (melanogenesis):

1. Tyrosine → DOPA → dopaquinone (tyrosinase; rate-limiting; Cu^{2+} cofactor).
2. Dopaquinone → DOPA-chrome → DHICA → eumelanin (brown/black) or dopaquinone + cysteine → pheomelanin (red/yellow).
3. Melanin polymerises on a tyrosinase scaffold within melanosomes (lysosome-related organelles).
4. Mature melanosomes are transported along melanocyte dendrites to tips, then transferred to keratinocytes (cytocrine secretion).
5. Within keratinocytes, melanosomes form a supranuclear cap that absorbs UV before it can damage nuclear DNA.

Pigmentation variation:

- Dark skin — large, numerous, non-degraded melanosomes in keratinocytes.

- Light skin — smaller, fewer melanosomes, more rapid degradation in keratinocytes.
- Red hair — predominance of pheomelanin (high cysteine); pheomelanin is photoprotective but also generates ROS upon UV exposure.
- Albinism — tyrosinase mutations (oculocutaneous albinism type 1) → no melanin → severe sunburn risk, nystagmus, photophobia.
- Vitiligo — autoimmune destruction of melanocytes; well-demarcated depigmented patches.
- Freckles (ephelides) — increased melanin production without increase in melanocyte number; UV-induced.
- Lentigo — increased melanocytes along the basal layer; benign; solar lentigo = age spot.
- Melasma — hyperpigmentation on face; pregnancy, OCPs, sun exposure.

Melanoma staging (Breslow depth, Clark level, ulceration, mitotic rate) is the principal prognostic indicator. ABCDE rule for clinical suspicion: Asymmetry, Border irregularity, Colour variation, Diameter >6 mm, Evolution.

Hair Growth Cycle Extended

Hair cycles through anagen (growth, 2-7 years), catagen (regression, 2-3 weeks), telogen (rest, 3 months), then back to anagen. At any time, ~85% of scalp hairs are in anagen, ~1-2% in catagen, ~10-15% in telogen.

- Alopecia areata — autoimmune destruction of anagen hair follicles; exclamation-point hairs; T-cell mediated.
- Androgenetic alopecia — dihydrotestosterone (DHT, from testosterone via 5 α -reductase) miniaturises follicles in genetically susceptible individuals; vertex and frontal scalp in men (Hamilton-Norwood pattern); diffuse thinning in women (Ludwig pattern).
- Telogen effluvium — premature shift of anagen follicles to telogen after stress (illness, childbirth, surgery); diffuse shedding ~3 months later.

Sweat Gland Detail

Eccrine sweat glands:

- Simple coiled tubular; ~3-4 million in the body; highest density on palms, soles, forehead.
- Secretory coil (cuboidal/columnar, clear cells + dark cells); reabsorptive duct (two layers of cuboidal cells, no myoepithelium) — primary sweat is isotonic; duct reabsorbs Na⁺ (via ENaC) and Cl⁻ (CFTR); final sweat is hypotonic. In CF, CFTR defective → impaired Cl⁻ reabsorption → salty sweat (basis of the pilocarpine iontophoresis sweat test).
- Stimulated by sympathetic cholinergic fibers (unusual: sympathetic innervation but ACh neurotransmitter); thermoregulation.

Apocrine sweat glands:

- Larger; coiled tubular; secrete by decapitation (apical cytoplasm pinched off).

- Viscous, protein-rich, odourless secretion; bacterial metabolism in the axilla produces characteristic body odour.
- Stimulated by sympathetic adrenergic fibers; become active at puberty (androgen-dependent).

Wound Healing Detail

Wound healing proceeds in three overlapping phases:

1. Inflammatory phase (0–3 days) — haemostasis (platelet plug, fibrin clot); neutrophils (24–48 h); macrophages (day 2–3) phagocytose debris and release cytokines (TGF- β , PDGF, FGF, VEGF).

2. Proliferative phase (3 days to 3 weeks) — fibroblasts migrate into wound along fibrin scaffold (fibronectin-mediated); proliferate; produce type III collagen (granulation tissue). Myofibroblasts (TGF- β -driven) contract the wound. Angiogenesis (VEGF-driven) restores blood supply. Re-epithelialisation from wound edges.

3. Remodelling phase (3 weeks to 1–2 years) — type III collagen replaced by type I; wound contracts; tensile strength increases to ~80% of unwounded tissue. Matrix metalloproteinases (MMPs) remodel; TIMPs regulate MMPs.

Primary vs secondary intention:

- Primary — wound edges approximated (sutured); minimal granulation tissue; minimal scarring.
- Secondary — wound left open; abundant granulation tissue; more scarring.

Keloid: scar extends beyond original wound boundaries; exuberant collagen synthesis; more common in dark-skinned individuals; high recurrence after excision. Hypertrophic scar: raised but within wound boundaries.

Transition

Skin is the body's external barrier. The next chapter turns to the body's internal signalling system — the endocrine glands.

Rapid Review

- Three layers: epidermis (stratified squamous keratinised), dermis (papillary + reticular), hypodermis (adipose).
- Epidermal cells: keratinocytes, melanocytes, Langerhans cells, Merkel cells.
- Five epidermal layers: basale, spinosum, granulosum, lucidum, corneum.
- Appendages: hair follicles (with sebaceous, arrector pili), nails, eccrine sweat glands (thermoregulation), apocrine sweat glands (axilla, odor).
- Thick skin: palms, soles; no hair; stratum lucidum.

Chapter 16 — The Endocrine System

Corresponds to Junqueira's Basic Histology, 17th Edition, Chapter 20

Opening Question

How does the body coordinate the slow, sustained regulation of metabolism, growth, reproduction, and stress response through chemical messengers in the bloodstream — and how is this coordination organised in histologically distinct glands, each with its own characteristic cell types?

Why This Matters

The endocrine system regulates virtually every aspect of physiology. Its dysfunctions — diabetes, hypothyroidism, hyperthyroidism, adrenal insufficiency, Cushing syndrome, pituitary adenoma — are common clinical problems.

Learning Objectives

1. Describe the general principles of endocrine histology: hormone-secreting cells, fenestrated capillaries, no ducts.
2. Identify and describe the histology of the anterior pituitary, posterior pituitary, thyroid, parathyroid, adrenal cortex, and adrenal medulla.
3. Identify the cell types of each endocrine gland and the hormones they produce.
4. Distinguish the endocrine glands histologically.
5. Recognise the histological signatures of common endocrine pathologies.

Big Picture

- **General** — ductless glands; hormone-secreting cells; fenestrated capillaries; rich vascularisation.
- **Peptide-secreting cells** — abundant RER, prominent Golgi, storage granules (anterior pituitary, pancreatic islets, parathyroid).
- **Steroid-secreting cells** — abundant smooth ER, lipid droplets, mitochondria with tubular cristae (adrenal cortex, Leydig cells of testis).
- **Amine-secreting cells** — dense-core granules (adrenal medulla chromaffin, thyroid C cells).

Core Concept

The endocrine system is a collection of ductless glands that secrete hormones into the bloodstream. Each gland has characteristic cell types and architecture that reflect the hormones it produces.

Structure

Pituitary (Hypophysis)

Located in sella turcica. Two divisions from different embryological origins:

- **Adenohypophysis (anterior pituitary)** — from oral ectoderm (Rathke's pouch).
- **Neurohypophysis (posterior pituitary)** — from neuroectoderm (downgrowth of diencephalon).

Anterior Pituitary — Pars Distalis

Three cell types by tinctorial properties (historical classification; modern classification is by immunohistochemistry):

- **Acidophils** (eosinophilic, ~50% of cells):
- Somatotrophs — GH (linear growth in children, IGF-1 production in liver).
- Lactotrophs — prolactin (milk production).
- **Basophils** (basophilic, ~10% of cells):
- Corticotrophs — ACTH (adrenal cortex stimulation).
- Thyrotrophs — TSH (thyroid stimulation).
- Gonadotrophs — FSH + LH (gonadal function).
- **Chromophobes** — pale, poorly staining; degranulated secretory cells or stem cells.

B-FLAT mnemonic for basophils: **B** = FSH/LH (gonadotroph), **F** = FSH/LH, **L** = LH, **A** = ACTH (corticotroph), **T** = TSH (thyrotroph). Mnemonic aid; modern classification is by IHC.

Anterior Pituitary Detail

The anterior pituitary is the classical example of a peptide-secreting endocrine gland. Each cell type has a characteristic ultrastructural profile and a characteristic secretory granule size that correlates with H&E staining:

- **Somatotrophs** are the most numerous (~50% of all adenohypophyseal cells), have densely granulated cytoplasm (large, 300–400 nm granules), and stain intensely eosinophilic. They produce GH (a 191-amino-acid polypeptide) in response to hypothalamic GHRH (stimulatory) and somatostatin (inhibitory). GH acts directly and indirectly via hepatic IGF-1. Pituitary GH adenoma in children causes gigantism; in adults (after epiphyseal closure) causes acromegaly.
- **Lactotrophs** have sparse, small granules (~200 nm). The cells are typically chromophobic in routine H&E but are PAS- and IHC-positive. Dopamine from the hypothalamus tonically inhibits prolactin release; dopamine antagonists (antipsychotics, antiemetics) cause hyperprolactinaemia and galactorrhoea. During pregnancy, lactotrophs undergo hyperplasia (the anterior pituitary enlarges ~30%) and accounts for the increased pituitary size on MRI in pregnancy (which can mimic adenoma).

- **Corticotrophs** contain large, basophilic, PAS-positive granules. They synthesise proopiomelanocortin (POMC), which is cleaved into ACTH, MSH, β -lipotropin, and other peptides. Corticotroph hyperplasia/adenoma → Cushing disease (ACTH-dependent hypercortisolism). Loss of cortisol feedback in primary adrenal insufficiency (Addison disease) → reactive corticotroph hyperplasia (Crooke hyaline change in corticotrophs reflects sustained suppression by excess cortisol).
- **Thyrotrophs** and **gonadotrophs** are basophilic and PAS-positive. Gonadotrophs are typically bihormonal (FSH + LH in the same cell). All adenohypophyseal cell types are ultimately identified by IHC (anti-GH, anti-prolactin, anti-ACTH, anti-TSH, anti-FSH/LH) on pituitary biopsy specimens.

Posterior Pituitary — Neurohypophysis

Composed of:

- Axons of hypothalamic neurons (cell bodies in supraoptic and paraventricular nuclei). These axons carry oxytocin (paraventricular) and ADH/vasopressin (supraoptic) to the posterior pituitary, where they are stored and released.
- Herring bodies — dilated axon terminals with neurosecretory granules.
- Pituicytes — specialised glia supporting axons.

The posterior pituitary is a storage depot, not a synthesis site. In H&E: pale tissue with unmyelinated axons and scattered nuclei of pituicytes.

Thyroid

Two cell types:

- **Follicular cells** — cuboidal (low in inactive gland; tall columnar in active). Arranged in follicles (spherical structures) around central colloid containing thyroglobulin (iodinated precursor of thyroid hormones). Follicular cells produce T4 and T3.
- **Parafollicular cells (C cells)** — scattered between follicles; produce calcitonin (lowers blood calcium).

Thyroid hormone synthesis:

1. Synthesise thyroglobulin and secrete into colloid.
2. Take up iodide via sodium-iodide symporter; oxidise via thyroid peroxidase.
3. Iodinate tyrosine residues on thyroglobulin (organification) → MIT, DIT.
4. Couple MIT + DIT → T3; DIT + DIT → T4 (still bound to thyroglobulin).
5. Endocytose iodinated thyroglobulin back into cell.
6. Hydrolyse in lysosomes → release T3 and T4 into bloodstream.

TSH stimulates follicular cells.

Parathyroid

Typically four glands on posterior surface of thyroid. Two cell types:

- **Chief (principal) cells** — small polygonal with pale cytoplasm; produce PTH (raises blood calcium by stimulating osteoclasts indirectly via osteoblasts, increasing renal calcium reabsorption, and vitamin D activation).
- **Oxyphil cells** — larger, eosinophilic (abundant mitochondria); function unclear; appear after puberty and increase with age.

Adrenal Gland

Two functionally distinct regions:

- **Adrenal cortex** — mesoderm-derived; produces steroid hormones.
- **Adrenal medulla** — neural-crest-derived; produces catecholamines.

Adrenal Cortex — Three Concentric Zones

- **Zona glomerulosa** (outermost) — clusters of small cells; produces aldosterone (mineralocorticoid); regulated by renin-angiotensin and serum K^+ .
- **Zona fasciculata** (middle, largest) — cords of large, lipid-laden spongiocytes with abundant smooth ER, mitochondria with tubular cristae, lipid droplets (cholesterol esters as steroid substrate); produces cortisol (glucocorticoid) and some androgens; regulated by ACTH.
- **Zona reticularis** (innermost) — cords of smaller, more basophilic cells with less lipid and more lipofuscin pigment; produces DHEA and other androgens.

Steroid-secreting cells have abundant smooth ER and mitochondria with tubular cristae — features distinguishing them from peptide-secreting cells.

Adrenal Medulla

Composed of chromaffin cells (modified postganglionic sympathetic neurons). Secrete epinephrine (~80%) and norepinephrine (~20%).

- Dense-core granules with catecholamines.
- Chromaffin reaction — chromium salts oxidise catecholamines to brown pigments.
- Innervated by preganglionic sympathetic fibers (release ACh → stimulate catecholamine release).
- Also contains ganglion cells (sympathetic ganglion cells that innervate chromaffin cells).

Function

Anterior pituitary produces GH, prolactin, ACTH, TSH, FSH/LH. Posterior pituitary stores oxytocin and ADH (synthesised in hypothalamus). Thyroid produces T3/T4 (metabolism) and calcitonin (calcium). Parathyroid produces PTH (calcium). Adrenal cortex produces aldosterone (sodium), cortisol (glucose), androgens. Adrenal medulla produces epinephrine/norepinephrine (stress).

Structure → Function

- Acidophils (somatotrophs, lactotrophs) → GH, prolactin → euchromatic nuclei + basophilic cytoplasm with RER.
- Basophils (corticotrophs, thyrotrophs, gonadotrophs) → ACTH, TSH, FSH/LH → euchromatic + more RER.
- Posterior pituitary Herring bodies → oxytocin/ADH storage.
- Thyroid follicles → colloid = thyroglobulin storage.
- C cells → calcitonin (neural-crest-derived, between follicles).
- Adrenal cortex spongiocytes → cholesterol ester droplets for steroid synthesis.
- Chromaffin cells → catecholamine granules.

Compare & Distinguish

Anterior vs Posterior Pituitary

Feature	Anterior	Posterior
Embryology	Oral ectoderm	Neuroectoderm
Cell types	Acidophils, basophils, chromophobes	Axons, pituicytes, Herring bodies
Hormones	Synthesised in situ	Stored (synthesised in hypothalamus)

Thyroid vs Parathyroid

Feature	Thyroid	Parathyroid
Colloid	Yes (follicles)	No
Cell types	Follicular cells + C cells	Chief cells + oxyphil cells
Hormone	T3/T4 + calcitonin	PTH

Adrenal Cortex vs Medulla

Feature	Cortex	Medulla
Origin	Mesoderm	Neural crest
Hormone	Steroid	Catecholamine
Cells	Spongiocytes, etc.	Chromaffin cells

Recognition Logic

- **Anterior pituitary with acidophils + basophils + chromophobes in cords** → anterior pituitary.
- **Pale tissue with axons + Herring bodies + pituicytes** → posterior pituitary.
- **Follicles of pink colloid + cuboidal cells + scattered C cells** → thyroid.
- **Sheets of small chief cells with interspersed oxyphil cells** → parathyroid.

- **Three zones (glomerulosa + fasciculata + reticularis) + central medulla** → adrenal.
- **Chromaffin reaction positive** → adrenal medulla.

Clinical Correlation

Case — Prolactinoma

A 30-year-old woman with galactorrhea and amenorrhea. MRI shows a pituitary mass. Biopsy shows sheets of acidophilic cells with IHC positive for prolactin.

Lesson. Lactotroph adenoma → excess prolactin. The clinical syndrome includes galactorrhea, amenorrhea, and infertility. Mass effect at the optic chiasm produces bitemporal hemianopia.

Case — Hashimoto Thyroiditis

A 50-year-old woman with fatigue, weight gain, and a goiter. Thyroid biopsy shows lymphocytic infiltrate with germinal centres, Hürthle cells (oxyphilic change), and fibrosis.

Lesson. Autoimmune thyroiditis with anti-TPO and anti-thyroglobulin antibodies. Histological signature: lymphocytic infiltrate with germinal centres + Hürthle cell change.

Case — Cushing Disease

A 40-year-old with weight gain, moon facies, central obesity, hypertension. 24-hour urine cortisol elevated. MRI shows pituitary mass. Biopsy: basophilic cells with ACTH IHC+.

Lesson. Corticotroph adenoma → excess ACTH → adrenal hyperplasia → excess cortisol. Distinguish from ectopic ACTH (small-cell lung cancer) and adrenal adenoma (primary cortisol excess with suppressed ACTH).

Case — Pheochromocytoma

A 35-year-old with episodic hypertension, headaches, palpitations. Plasma metanephrines elevated. Adrenalectomy shows a mass; histology shows nests of chromaffin cells (Zellballen pattern) with chromaffin reaction positive.

Lesson. Adrenal medulla tumour producing catecholamines. The classic triad is episodic hypertension, headaches, palpitations.

Common Misconceptions

- The anterior pituitary stores hormones. It synthesises and releases; posterior pituitary stores.
- The thyroid stores hormones. It stores iodinated thyroglobulin in the colloid; T3/T4 are released from this storage form.
- All endocrine cells look the same. Peptide cells have abundant RER + Golgi + granules; steroid cells have smooth ER + tubular mitochondrial cristae + lipid droplets; amine cells have dense-core granules.

- The adrenal cortex and medulla are the same tissue. Different origins and functions.
- Calcitonin is essential for calcium regulation in adults. Plays a minor role; PTH is the principal regulator.

High-Yield Knowledge

Must Know

- Anterior pituitary: acidophils (GH, prolactin), basophils (B-FLAT: FSH/LH, ACTH, TSH), chromophobes.
- Posterior pituitary: stores oxytocin and ADH produced in hypothalamus; pituicytes + Herring bodies.
- Thyroid: follicles (T3/T4) + C cells (calcitonin); colloid is thyroglobulin storage.
- Parathyroid: chief cells (PTH) + oxyphil cells.
- Adrenal cortex: zona glomerulosa (aldosterone), fasciculata (cortisol, spongiocytes), reticularis (androgens).
- Adrenal medulla: chromaffin cells (epinephrine/norepinephrine).

Must Distinguish

- Anterior vs posterior pituitary.
- Thyroid (colloid) vs parathyroid (no colloid).
- Adrenal cortex vs medulla.

Must Understand

- Peptide vs steroid vs amine cells (organelle profile).
- B-FLAT mnemonic for basophils (aid, not definition).
- PTH raises calcium; calcitonin lowers it (minor role in adults).

Integrated Summary

The endocrine system is a collection of ductless glands that secrete hormones into the bloodstream. The anterior pituitary produces GH, prolactin, ACTH, TSH, FSH/LH; the posterior pituitary stores oxytocin and ADH from the hypothalamus. The thyroid has follicles (T3/T4 from follicular cells) and C cells (calcitonin); the parathyroid has chief cells (PTH) and oxyphil cells. The adrenal cortex has three zones (glomerulosa aldosterone, fasciculata cortisol as spongiocytes, reticularis androgens); the adrenal medulla has chromaffin cells (catecholamines). Steroid-secreting cells have abundant smooth ER and tubular mitochondrial cristae; peptide-secreting cells have abundant RER and storage granules.

Mastery Check

1. List the hormones of the anterior pituitary and their cell types.
2. Reconstruct the synthesis and release of thyroid hormones.
3. Compare the three zones of the adrenal cortex by hormone and histology.

4. Why is the zona fasciculata called spongiocytes?
5. Compare chromaffin cells and ordinary sympathetic neurons.
6. Why does the posterior pituitary have axons but the anterior pituitary has glandular cells?

Advanced Concepts

Steroid Hormone Synthesis

Steroid hormones are synthesised from cholesterol through a series of cytochrome P450 enzymes and hydroxysteroid dehydrogenases. The pathway:

- **Adrenal cortex zones:**
- Glomerulosa — aldosterone synthase (CYP11B2) at end of pathway → aldosterone (mineralocorticoid).
- Fasciculata — 17 α -hydroxylase (CYP17) converts pregnenolone to 17 α -hydroxypregnenolone → cortisol; no aldosterone synthase → no aldosterone.
- Reticularis — CYP17 activity + 17,20-lyase activity → DHEA → androstenedione.
- 21-hydroxylase (CYP21) — required for cortisol and aldosterone synthesis; most common cause of congenital adrenal hyperplasia (CAH) (95%); 21-hydroxylase deficiency → impaired cortisol/aldosterone synthesis → shunting into androgen pathway → virilisation of female fetuses (ambiguous genitalia at birth), salt wasting (hypotension, hyperkalaemia) in ~75%.
- 11 β -hydroxylase (CYP11B1) — converts 11-deoxycortisol to cortisol; deficiency → CAH with hypertension (excess 11-deoxycorticosterone has mineralocorticoid activity).
- 17 α -hydroxylase (CYP17) — converts pregnenolone/progesterone to 17 α -hydroxy forms; deficiency → CAH with hypertension (excess mineralocorticoids), sexual infantilism (no sex steroids).

Steroid hormones bind intracellular receptors → heat-shock protein dissociation → receptor-ligand complex translocates to nucleus → binds hormone-response elements → modulates gene transcription (slow onset, hours to days).

Thyroid Hormone Action

T3 binds nuclear thyroid hormone receptors (TR α , TR β) → recruit co-activators or co-repressors → regulate gene transcription. Effects of thyroid hormone:

- Basal metabolic rate — $\uparrow$ Na⁺/K⁺-ATPase, mitochondrial biogenesis, O₂ consumption, heat production.
- Cardiovascular — $\uparrow$ β 1-adrenergic receptors → $\uparrow$ heart rate, contractility.
- CNS — development (myelination, synaptogenesis); deficiency in childhood → cretinism (intellectual disability, short stature, deafness).
- Growth — synergises with GH.

- Bone — ↑ bone turnover (osteoclast and osteoblast activity).

Hyperthyroidism: weight loss despite appetite, heat intolerance, tachycardia, tremor, anxiety, exophthalmos (Graves-specific), pretibial myxoedema.

Hypothyroidism: weight gain, cold intolerance, bradycardia, fatigue, dry skin, myxoedema, depression. Hashimoto thyroiditis (most common cause in iodine-sufficient regions); iodine deficiency (most common cause worldwide).

Calcium Homeostasis

PTH (chief cells, parathyroid) — raises blood Ca^{2+} :

- Bone — stimulates osteoblasts to express RANKL → activates osteoclast precursors → bone resorption (Ca^{2+} + phosphate release).
- Kidney — increases distal tubule Ca^{2+} reabsorption; decreases proximal tubule phosphate reabsorption (phosphaturia); stimulates 1α -hydroxylase → $1,25\text{-(OH)}_2\text{-vitamin D}$ (calcitriol).
- Intestine (indirect) — via calcitriol → increases Ca^{2+} and phosphate absorption.

Calcitonin (C cells, thyroid) — lowers blood Ca^{2+} (minor role in adults):

- Inhibits osteoclast activity; promotes renal Ca^{2+} excretion.

Vitamin D — increases blood Ca^{2+} (and phosphate):

- Skin: 7-dehydrocholesterol → cholecalciferol (vitamin D_3) by UV.
- Liver: cholecalciferol → 25-hydroxyvitamin D (calcidiol) by 25-hydroxylase.
- Kidney: 25-OH-D → $1,25\text{-(OH)}_2\text{-vitamin D}$ (calcitriol) by 1α -hydroxylase (stimulated by PTH, inhibited by FGF23 and high phosphate).
- Calcitriol binds VDR → ↑ intestinal Ca^{2+} absorption; ↑ bone mineralisation; feedback inhibition of PTH.

FGF23 (osteocytes) — lowers blood phosphate:

- Inhibits 1α -hydroxylase, stimulates 24-hydroxylase → ↓ calcitriol → ↓ phosphate absorption.
- Causes phosphaturia.

Clinical correlations:

- Primary hyperparathyroidism (parathyroid adenoma) → hypercalcaemia (stones, bones, abdominal groans, psychiatric overtones); PTH elevated, phosphate low.
- Hypoparathyroidism (post-thyroidectomy, DiGeorge) → hypocalcaemia (tetany, Chvostek and Trousseau signs, prolonged QT); PTH low.
- Vitamin D deficiency → rickets (children), osteomalacia (adults); hypocalcaemia, secondary hyperparathyroidism.
- Paget disease of bone (out of core scope but illustrative) — disordered bone remodelling with abnormal osteoclasts (paramyxovirus-like inclusions suspected); mosaic pattern of lamellar bone.

Transition

This completes the four-tissue foundation and the sixteen-chapter core syllabus of Essential Histology. The student who has mastered this foundation can read any tissue section, predict function from structure, recognise the most common diseases, and prepare for clinical practice and advanced study.

Rapid Review

- Pituitary: anterior (acidophils GH/prolactin; basophils B-FLAT: FSH/LH, ACTH, TSH) + posterior (oxytocin, ADH stored; pituicytes, Herring bodies).
- Thyroid: follicles (T3/T4) + C cells (calcitonin); colloid = thyroglobulin.
- Parathyroid: chief (PTH) + oxyphil.
- Adrenal cortex: glomerulosa (aldosterone), fasciculata (cortisol, spongiocytes), reticularis (androgens).
- Adrenal medulla: chromaffin cells (catecholamines).
- Peptide cells (RER + Golgi + granules) vs steroid cells (SER + tubular mitochondria + lipid droplets).